# PY - QUEST: DEVELOPMENT OF AN ANDROID BASED GAME APPLICATION FOR PYTHON PROGRAMMING FUNDAMENTALS

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**MICHAEL D. BLANCO**
**MAYVEL T. LANDICHO**
**MIRIAM FAITH D. MEDINA**
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**CAVITE STATE UNIVERSITY**
**Don Severino delas Alas Campus**
Indang, Cavite

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**

**Department of Information Technology**

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Authors: **MAYVEL T. LANDICHO**
**MIRIAM FAITH D. MEDINA**
**MICHAEL D. BLANCO**


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| | | | |
|---|---|---|---|
| **ANABELLE J. ALMAREZ**<br>Adviser | 12-12-23<br>Date | **JAY- ARR C. BUHAIN**<br>Technical Critic | 12-12-23<br>Date |
| **GLADYS G. PEREY**<br>Unit Research Coordinator | 1-3-24<br>Date | **CHARLOTTE B. CARANDANG**<br>Department Chairperson | 1-3-24<br>Date |
| **EDWIN R. ARBOLEDA, DEng**<br>College Research Coordinator | 1/9/24<br>Date | **WILLIE C. BUCLATIN, PhD**<br>Dean | Date |

**MIRIAM D. BALTAZAR, PhD**
Director for Research

Date

*Not valid without University seal*

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**MICHAEL D. BLANCO**

**MAYVEL T. LANDICHO**

**MIRIAM FAITH D. MEDINA**

## ABSTRACT

**BLANCO, MICHAEL D., LANDICHO, MAYVEL T., and MEDINA, MIRIAM FAITH D. Development of Py – Quest: An Android – Based Game Application for Python Programming Fundamentals.** Undergraduate Capstone Project. Bachelor of Science in Information Technology. Cavite State University, Indang, Cavite. April 2025. Adviser: Prof. Anabelle J. Almarez.

The capstone project, “Py-Quest: An Android-Based Game Application for Python Programming Fundamentals," was conducted at Cavite State University from March 2022 to June 2024. It aimed to develop an interactive mobile game for Python programming fundamentals to ICT senior high school students. The project was designed to address the challenges of accessing and understanding Python, especially when offline resources are limited. By offering an engaging and user-friendly platform accessible anytime and anywhere, Py-Quest aimed to make learning Python both enjoyable and effective.

The development process involved analyzing student needs through surveys to identify the specific difficulties in learning Python. The game features a user-friendly interface and covers a wide range of topics from basic concepts to loops, with various engaging question types. It was built using Java, XML, SQLite, and Android Studio, ensuring compatibility with Android versions 8 to 13. The game includes two main modules: the book of knowledge for lessons and the Gameplay for assessments.

The application underwent thorough testing and evaluation based on ISO 25010 software standards. It received very good remarks in functional suitability, performance efficiency, usability, and portability from both technical instructors and users. Specifically, the system's reliability was marked as "excellent," with a mean score of 4.48 and a standard deviation of 0.59. Its usability also rated as an "excellent”, with an average score of 4.72 and a standard deviation of 0.48. The maintainability was rated excellent, achieving a mean score of 4.48 and a standard deviation of 0.59. Additionally, the system's portability was rated as "excellent," with a mean score of

4.73 and a standard deviation of 0.43. The performance efficiency was rated as "very good," with a mean score of 4.37 and a standard deviation of 0.70. The successful evaluation demonstrated the application's effectiveness as an educational tool, highlighting its potential for broader implementation in educational settings and providing an excellent solution for teaching Python programming to a wider audience.

## TABLE OF CONTENTS

## LIST OF TABLES

## LIST OF FIGURES

## LIST OF APPENDIX TABLES

## LIST OF APPENDIX FIGURES

# LIST OF APPENDICES

**PY - QUEST: DEVELOPMENT OF AN ANDROID BASED GAME APPLICATION FOR PYTHON PROGRAMMING FUNDAMENTALS**

**Michael D. Blanco**
**Mayvel T. Landicho**
**Miriam Faith D. Medina**

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## INTRODUCTION

The gaming industry experienced a revolutionary transformation with the advent of smartphones. Android, being an open-source platform, provided developers with the tools and flexibility needed to unleash their creativity and build captivating games that catered to diverse audiences. The popularity of Android games could be attributed to their ease of accessibility, smooth performance, and compatibility with a wide range of devices. Additionally, gaming offered a dynamic environment that motivated learners to develop skills and form emotional connections with the subject matter. By integrating games as instructional strategies, learners could become more self-assured and independent individuals (Dadheech, 2021).

The survey, conducted with the help of Mr. Edizon Feranil from Tanza National Trade School (See Appendix 1) reveals that students struggle with programming due to difficulties in understanding lessons, memorizing syntax, and dealing with distractions leading to poor retention of essential knowledge (See Appendix Figure 1).

The traditional programming education methods face several issues, including high textbook costs, outdated materials, and ineffective teaching approaches.

Overcrowded classrooms and inadequate facilities further complicate the learning environment, making it difficult for students to engage with and understand programming concepts effectively (See Appendix Figure 2).

In senior high school, students typically learn programming languages like Java. However, college introductory courses usually focus on Python, creating a significant challenge for students from schools where Python is not part of the curriculum. This lack of preparation leaves them unready for college-level programming demands, compounded by limited resources and outdated materials (See Appendix Figure 3).

To address these challenges, an innovative solution is needed. A proposed mobile-based game aims to provide an interactive and engaging platform for learning Python, helping students build essential skills and confidence. This game will support students in bridging the gap between senior high school and college programming education, making programming more accessible and enjoyable for all learners.

**Objectives of the Study**

The general objective of the study was to create an engaging mobile game that teaches Python programming fundamentals to ICT senior high school students, making learning accessible and fun to enhance their coding skills for future education and career opportunities.

Specifically, the study aimed to:

1. identify and analyze the problem of ICT senior high-school students encountered in terms of having difficulty accessing knowledge about python programming language.
2. design an Android-based- game application that has the following features:
    a) user-friendly mobile game interface that appeals to ICT students.
    b) topics ranging from basic concepts to loops in Python fundamentals. It starts with introductory lessons on variables, data types, and basic syntax, ensuring that players understand the foundation of Python

programming. As players progress, they move on to more advanced topics such as conditional statements and loops.

c) accessible anytime and anywhere.

d) different types of engaging questions such as debugging, completing, and guessing the output.

e) built-in scoreboard where players can easily track their progress. It shows all the questions they've completed along with their scores, helping users monitor how well they're doing as they play.

3. develop an Android-based game application using Java, XML, SQLite and Android Studio.
4. test the application's functionality, usability, and compatibility.
5. evaluate the game application based on ISO 25010 software standards.
6. create an implementation plan for its deployment.

**Purpose and Description**

This project aimed to support and empower ICT students from senior high-school by teaching them the fundamentals of Python programming through a mobile game. The goal was to make learning Python fun and accessible. Once deployed, the application will students with access to valuable learning materials about Python programming fundamentals. The mobile game will include interactive lessons, quizzes, and tasks designed to build coding skills and knowledge step-by-step.

The mobile game Py-Quest: An Android-Based Game Application for Python Programming Fundamentals possesses several key features:

1. list of topics including introduction, comments, variables, data types, numbers, casting, strings, booleans, operators, lists, tuples, sets, dictionaries, if...else statements, while loops, and for loops. Mastering these topics will give users a strong foundation in Python and improve their coding skills.

2. locked topics to ensure that users thoroughly understand and master each one before advancing to the next. This method ensures a strong knowledge foundation, enhancing learning and preventing gaps in understanding as user’s progress to more advanced material.
3. the game has audio features presentation in every lesson that provides an alternative mode of content delivery that caters to different learning styles.
4. quiz that contains a variety of questions. Users were given tasks such as debugging code, completing code snippets, and guessing the output of given code.
5. scoreboard that serves as a measure of their understanding and mastery of the content. Users can track their progress through this.
6. offline mode to guarantee that users can access it anywhere and anytime, whether the users are at home, traveling, or on a break at work, the game is readily accessible on their mobile device. This flexibility ensures that players can engage with the game and make progress at their convenience, without being constrained by specific times or places.

This study addressed the challenges faced by the ICT students, particularly their lack of knowledge in programming languages. By implementing the game application, the project aimed to bring the following benefits:

1. ICT senior high-school students. The game provided a valuable opportunity for students to learn the fundamentals and terminology in basic Python to prepare for introduction to programming I in college. It also encouraged them to learn actively, presenting a self-challenge and enhancing their skills
2. Researcher. As student researchers involved in programming, this project allowed them to share their knowledge and learning experiences with ICT students. It displayed their proficiency in programming and system development.

3. Future researchers. The study would serve as a valuable resource for future researchers, providing additional insights and opportunities for further improvements to the mobile game application.

Overall, the project aimed to bridge the knowledge gap for ICT students in programming and contribute to their educational growth and development.

**Time and Place of the Study**

The study was conducted at Tanza National Trade School, Cavite, from July 2022, to July 2024.

**Scope and Limitation of the Study**

Py- Quest promotes awareness to those who are unfamiliar with the Python programming language. The study is an android-based mobile application that is accessed by a user in their smartphone. It was developed using java and consists of a gameplay module, scoreboard module and book of knowledge module.

The gameplay module is responsible for the game mechanics, rules, levels, and progression. The system allows the user to play and learn python programming language through a level-driven game. When the player begins the game, they'll need to choose from available topics such as Python introduction, syntax, comments, variables, data types, numbers, casting, strings, booleans, operators, lists, tuples, sets, dictionaries, if-else, while loops, and for loops. Once a topic is selected, the user must pick a question related to that topic. To initiate the process, the user starts the game. Upon starting, the game system presents the home page, featuring various navigation buttons such as start, scoreboard, avatar, and settings. Additionally, it highlights the available topics and questions that users can engage and play with. The user must then finish the lesson, which is kept inside the book of knowledge to start answering the quiz; these lessons correspond to the topic selected by the user. The user will then have to provide the answers to each question, after answering; the game system will display the result of the attempt to the user.

Avatars appear randomly as a reward in their assigning tasks like completing code. Users fill in missing parts with accurate answers. Multiple-choice questions provide answer options, and users aim to pick the correct one. In debugging codes, users fix wrong code to get the given output. Identification questions have no options and users only rely on the book of knowledge. True or false statements are presented for users to confirm if the statement is true or false. A button representing the book of knowledge is available for assistance. The difficulty varies based on the chosen topic.

The scoreboard module contains the score summary of all the finished topics or levels. This module could be viewed upon clicking the specific topic, allowing users to review their previous attempt results and progress.

The book of knowledge module contains the lesson for the selected topic. It allows users to have more ideas about the questions. This module activates when the designated button is pressed which then displays the lesson for the selected topic. This module also allows users to choose whether to play the lesson with audio by tapping the music icon or by simply reading the lesson itself.

The game is specifically created for single-player gameplay and was not intended for multiplayer functionality. Recognizing that not all users may have constant internet access, it offered offline mode, enabling learners to continue their Python programming journey anytime, anywhere using their Android mobile devices.

**Conceptual Framework of the Study**

Figure 1 displays the conceptual framework of the study. The input consists of knowledge, hardware, and software requirements needed to create the game. It required the researchers to know the Python programming language as well as how to use Java. Hardware requirements refer to personal computers/laptops and mobile phones.

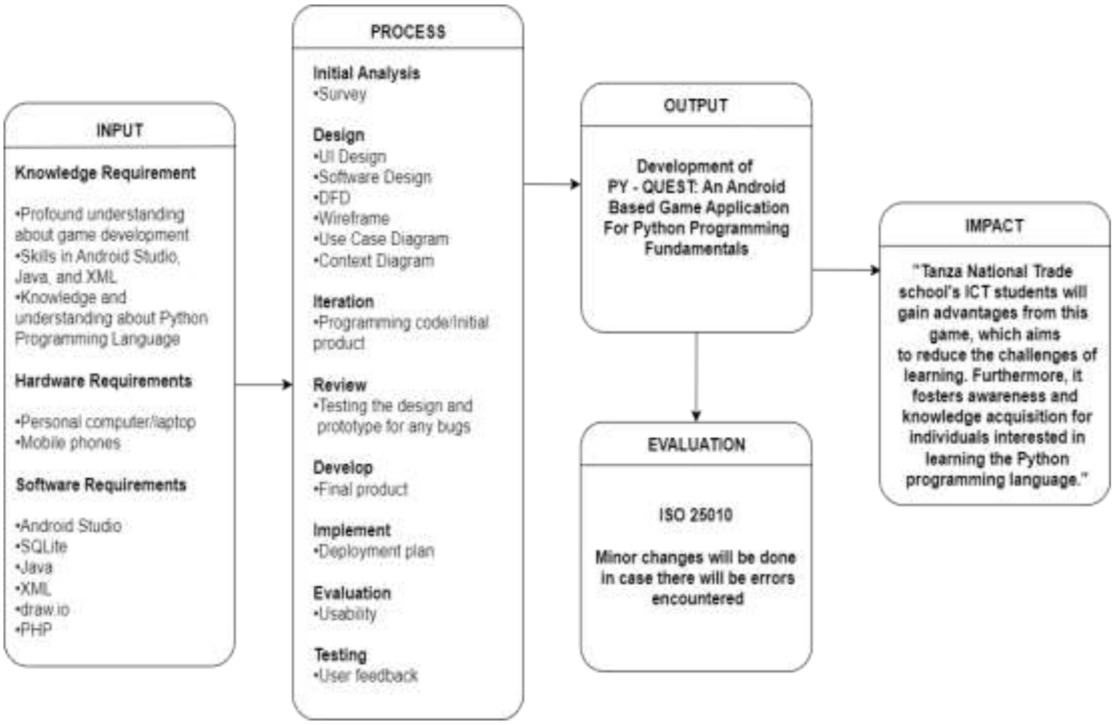



Figure 1. Conceptual framework of Py-Quest: An Android-Based Game Application for Python Programming Fundamentals

Next, it showed the process where the Rapid Prototyping Model approach was used, consisting of the initial analysis, design, iteration, review, development, implementation, and evaluation phases of creating the system. In the initial analysis, the researchers analyzed the data they collected during data gathering. The data or information that they got from it helped them analyze what kind of system their beneficiaries needed. Design analysis refers to creating a system design through UI design, data flow diagrams, use case diagrams, wireframes, prototypes, and system architecture, all of which contribute to the overall design of the system. Iteration/Prototype was a process where researchers started writing, coding to develop, form system specifications, and functionality.

The review was a process where researchers needed to test a prototype for any bugs or errors. Development was the most important part of the model since this phase was where the concept and design were being coded.

The final product would be implemented for the intended beneficiaries in the implementation stage. During this stage, the researchers would implement the game and train the target users on how to use or play the game. The users would be instructed and guided properly, and researchers would provide the necessary manuals and other materials required for implementation. Lastly, the evaluation phase, which included the evaluation of the system according to ISO 25010, also included technical and non-technical evaluation for recommendations and feedback to improve the system.

The output section contained a developed system of a researcher. The impact section referred to how the target beneficiaries benefited from the system. The system aimed to help ICT students to learn about the Python programming language for preparation to college. The evaluation phase is a test of the system according to ISO 25010. It also determined if there were changes needed to improve the system.

**Definition of Terms**

To understand the study, the below terms are operationally defined:

**5G network** is the fifth generation of wireless communication technology, which provides significantly faster internet speeds, reduced delays in data transmission, and the ability to connect a large number of devices simultaneously.

**Android-based application** is designed to run on the Android operating system, allowing developers to freely modify and create applications for smartphones, tablets, and smartwatches.

**Android Studio** is an IDE (Integrated Development Environment) that is used to create applications for the Android platform.

**Book of knowledge module** is a feature that provides players with the lesson about the topic they selected.

**Code completion** is a task in the game where players are required to fill in missing parts of Python code to make it functional or complete.

**Code debugging** is a task where players are presented with faulty Python code and must identify and correct the errors to make the code work as intended.

**Game-based learning** is an active technique that enhances student learning by using games to promote critical thinking and problem-solving skills.

**Game mechanics** are the rules, processes, and interactions that govern how the game works, including the logic behind scoring, level progression, and user interactions.

**Integrated Development Environment (IDE)** is a comprehensive software application that provides a user-friendly environment for computer programmers and developers to write, edit, compile, run, and debug their software code efficiently.

**Mobile games** are digitally based games that are mostly played on mobile devices such as cell phones, personal digital assistants, smartphones, and tablets.

**Offline mode** is an operating state of the game that allows users to access and use the application without an active internet connection, enabling them to continue learning Python programming at any time and place.

**Output guessing** is a task that challenges players to predict the output of a given piece of Python code without actually running it.

**Programming language** is a type of computer language used by programmers to create software programmes, scripts, or other sets of commands for computers to perform.

**Py-Quest** is an Android-based game application to learn python fundamentals.

**Python** is a high-level programming language with dynamic semantics, making it suitable for rapid and scripting. Its easy syntax minimizes maintenance costs and promotes modularity.

**Python syntax** refers to the set of rules and conventions governing the structure and composition of programs written in the Python programming language.

**Scoreboard** module is where users can view their scores in their completed topics.

**Single player gameplay** is a mode where one player engages with the game's challenges and content independently.

# REVIEW OF RELATED LITERATURE

This chapter presents the review of related literature and studies underlying the framework of the study.

**System Technical Background**

To establish the technical foundation of this project, this section provides an overview of the key technologies, tools, and programming languages employed in the development of the proposed study. This background information will serve as a reference point for understanding the system's architecture and functionality.

**Android Operating System.** The Android operating system presents numerous advantages to its users. One of its primary strengths lies in the extensive availability of a myriad of apps, which are organized into categories and can be obtained from various sources, including third-party websites. Android's notification system is designed with user-friendliness in mind, presenting notifications in an organized manner and providing convenient access. Additionally, it facilitates internet sharing through mobile hotspots, potentially leading to cost savings on separate internet packages within a household. Android exhibits compatibility with a diverse range of mobile devices, spanning from economical options to pre. Moreover, Android benefits from a substantial and dynamic community of developers and users who contribute support and verify app authenticity. It consistently introduces new features while phasing out outdated ones and operates as an open-source system, permitting customization and code modifications by developers. Android seamlessly integrates with cloud storage, providing users with 15 GB of complimentary storage and facilitating synchronization across multiple devices. The platform's popularity creates a robust demand for Android developers, and it incorporates features for app restoration and data backup, ensuring the resilience of apps in the face of issues or updates (Rehman 2021).

The history of Android is a fascinating journey that spans several decades, characterized by innovation, collaboration, and evolution. From its humble beginnings, when it was competing with Nokia's mobile OS Symbian, the Windows Phone OS, and Blackberry OS, to its dominant position in the mobile operating system market, Android has significantly influenced how we interact with technology (Basumallick, 2024).

**Android Studio.** Android Studio is an integrated development environment (IDE) that is strong and offers an extensive array of tools and functionalities designed specifically for developing Android applications. Its user-friendly interface facilitates the seamless creation, testing, and debugging of Android apps, along with the optimization of code performance and stability. Android Studio encompasses essential features and tools, such as the manifest file, libraries, Git integration, debugging, and App Inspection. The manifest file serves to delineate crucial features and prerequisites of an Android application, while libraries enable the incorporation of external code and functionality into the project (GeeksforGeeks, 2024).

**Java.** Java is a class-based, object-oriented programming language designed to minimize implementation dependencies. It allows developers to write code once and run it anywhere (WORA) without needing recompilation on different platforms that support Java. First released in 1995, Java is widely used for developing applications across desktop, web, and mobile platforms. Its simplicity, and strong security features make it a preferred language for enterprise-level applications (GeeksforGeeks, 2024).

**Python.** Python is a general-purpose, object-oriented programming language that has several implications across software, web development, data science, and automation environments. The language's dynamic semantics, high-level built-in data structures, dynamic typing, and dynamic binding make it one of the most useful languages for rapid application development (Grant, 2022).

**Extensible Markup Language (XML).** The rules and protocols for defining, storing, and sharing data between computer systems can be found in the Extensible Markup Language (XML), a markup language. These rules simplify data exchange between websites, applications, and databases on any network. Owing to their human and machine readability, the parties involved in data sharing can easily read and decipher the data hidden within XML files (Kanade, 2023).

**Related Literature**

To provide a comprehensive understanding of the development of educational game applications, this review explores existing literature on game-based learning, mobile app development, and the use of technology in education.

**Mobile Technology.** Abou-Samra (2020) claims that mobile technologies are changing how individuals discover, receive, and share information. It is a subset of technology that is mostly utilized in cellular communication and other associated aspects. Individuals are using their phones further as innovative applications extend the range of uses for mobile technology. Mobile technology has advanced from a simple device used for making and receiving calls to a multi-tasking device capable of Global Positioning System tracking or navigation, internet browsing, mobile gaming, online communications, mobile banking, and other functions.

**Mobile Games.** Mobile games are sometimes characterized as digital games played on smartphones or other mobile devices such as iPads and tablet PCs (IGI Global, 2018). According to Newzoo (2023), the rise of smartphones has marked a new era in gaming, leading to significant growth in mobile games over recent years. By harnessing the advanced hardware and software features of smartphones, these games offer immersive and engaging experiences that were previously reserved for traditional gaming consoles.

Mobile gaming has surpassed both console and PC gaming as the most popular type of gaming in the world. One of the reasons for the success of mobile gaming is its ease of access since almost everyone owns a smartphone capable of playing a game (AppLovin, 2022). The games are not only for entertainment; there is also called “game-based learning". Dadheech (2021) believes that game-based learning (GBL) may be used to improve both learning and teaching. The rising integration of digital games and applied sciences into the learning environment affects both educator instruction and student learning. In addition, all games are learning experiences, whether they be for entertainment or education. Moreover, Games developed specifically with the intent to educate children may greatly encourage self-learning and problem-solving skills. Game-based learning is simply the incorporation of subject knowledge into games.

Gaming offers a dynamic that can motivate learners to develop skills and develop emotional connections to learning and subject matter. Games may be used as instructional strategies, assisting learners to become more self-assured and independent individuals (Dadheech 2021).

**Educational Games.** Chimpz (2019) claims that educational games are essential to attaining new learning goals and objectives interactively. In addition, integrating educational learning objectives into the curriculum takes learning to a whole new level.

Chueng and Yin Ng (2021) conducted a study about the Application of the Educational Game to Enhance Student Learning, and they believed that an educational game that allows students to "play" while still studying conveniently and effectively may be more advantageous to students, especially in our fast-changing modern society in which individuals spend a majority of their time on their mobile phones. In their study, they mention the game called “PaGaMo”, which is an online gamification-learning platform that enables participants to learn and compete with one another using a game. Individuals must properly answer questions in the game to effectively create, occupy, and conquer territories. Because of the game's growing popularity, it is now frequently utilized for a variety of reasons, including but not limited to language acquisition, licensure examinations, and other professional training. Additionally, one of the benefits of the mentioned educational game is that it is accessible by computer, mobile phone, and tablet applications. Learners cannot only evaluate their knowledge before academic examinations, but they can also utilize educational games as study tools in their leisure or free time. Furthermore, the result of the study concludes that because of the simplicity of the game, playing educational games on mobile devices is more popular than utilizing PCs and tablet devices. In addition, it improves learners' educational motivation and learning success.

Cuadra and Tejero (2019) conducted a study about the 2D educational entertainment Android game, 'QRY,' targeting the enhancement of user skills and education in Structured Query Language (SQL). The research methodology involves conducting interviews and surveys to identify potential problems, subsequently analyzed using a fishbone diagram. The gameplay encompasses multiple stages and chapters, allowing progression only upon successful completion of each chapter. This structure challenges the player's continuity, comprising eight chapters, each comprising three stages. Enhancement of End-User Skills: QRY aims to enhance users' proficiency in SQL. The study adds insights into Android game design and development for the researchers. Offers guidance for future students or researchers entering Android game development, potentially inspiring improvements in subsequent studies. The researchers employed a structured methodology in the game's development process. The evaluation relied on the standard, assessing software quality across functionality, reliability, usability, efficiency, maintainability, and portability.

**Adobe Photoshop.** Adobe Photoshop, a popular graphics editing software application, was created and published by Adobe Inc., led by John and Thomas Knoll in 1987. This software application, designed for image editing and photo retouching, is available for both Windows and MacOS computers. Photoshop provides users with the tools to create, enhance, and edit images, artwork, and illustrations (Smith, 2022).

**Flaticon.** Flaticon is the byproduct of the Freepik, which specializes in offering incredible visual experiences to people worldwide. It holds the largest database of free customizable icons in all formats. (Freepik, 2021).

**Rapid Prototyping.** Rapid prototyping is an iterative design approach for apps or websites, where the design and functionality are quickly refined through regular updates and short cycles. This method helps identify and resolve common design issues early, saving time and money, and enabling faster market entry. The focus remains on addressing the needs of the end-user. In software development, rapid prototyping is often linked to the Rapid Application Development (RAD) methodology, though it can also be used with agile practices. The final outcome can be a new product or significant enhancements to existing software, such as improved user experience and a smoother user interface. (Merrill, M. 2023).

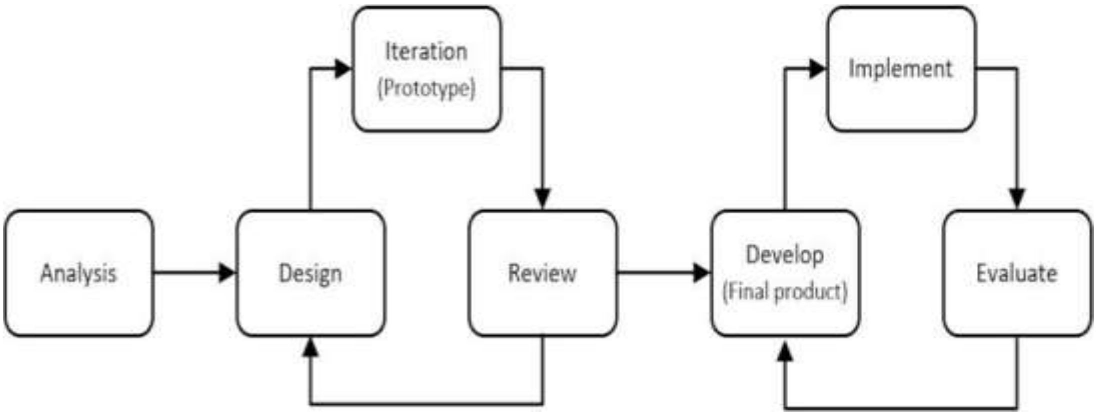



Figure 2. Rapid prototyping model (Merrill, M. 2023)

**System Testing.** Software testing is a crucial stage in the software development process, aimed at ensuring that software applications are free of bugs, adhere to technical specifications established during design and development, and effectively meet user requirements. This process assesses the application's capability to handle a variety of scenarios, including unusual circumstances, to guarantee a consistent and reliable user experience. By methodically identifying and resolving issues, software testing plays a pivotal role in delivering high-quality software that performs well under diverse conditions. In summary, it involves a systematic evaluation of a software program's functionality and operational performance, comparing the software's actual performance with expected criteria to detect and rectify errors, defects, or deviations. Through its focus on specifications, functionality, and overall performance, testing ensures that the software meets its intended goals, thereby enhancing its reliability and usability (GeeksforGeeks, 2024).

**Slovin's formula.** According to Zach Bobbit (2023), Slovin's formula is a widely used statistical tool that helps researchers determine an appropriate sample size when conducting surveys or experiments. The following statistical procedures are used to analyze the data to be gathered from the system evaluation questionnaire. The results of the statistical procedure determined the general perception of the respondent on the system.

Sample mean is the average score of a sample on a given variable.

Formula:

$$\underline{x} = \frac{\sum_{i=1}^{n} \quad X_i}{n}$$

Where:

x = mean

Xi = representation of each observation from respondents

n = total number of respondents

Sample standard deviation is a measure of the spread (variability) of the scores in the sample on a given variable.

Formula:

$$s = \sqrt{\frac{\sum_{i=1}^{n} \quad (X_i - \bar{x})^2}{n-1}}$$

Where:

x = mean

Xi = representation of each observation from respondents

n = number of respondents

s = sample standard deviation

Percentage determines the frequency counts and percentage distribution of personal related variables of the respondents.

Formula:

$$\% = \frac{f}{n} \times 100$$

Where:

% = percentage

f = frequency

n = total number of respondents

To determine the proper sample size of the evaluator who evaluated the system given a number of populations a Slovins' formula is used.

Formula:

$$n = \frac{N}{(1 + Ne^2)}$$

Where:

n = number of samples

N = total number of populations

e = error tolerance (level).

**ISO 25010.** ISO 25010, also known as "Systems and software engineering – Systems and software Quality Requirements and Evaluation (SQuaRE) – System and software quality models," serves as a comprehensive standard defining quality characteristics and evaluation models essential for assessing both software product quality and software quality in use. This standard outline detailed descriptions of various models encompassing characteristics and sub-characteristics crucial for evaluating software products. It also provides practical guidance on effectively applying these quality models in real-world scenarios. ISO 25010 comprises eight core product quality characteristics. Functional Suitability assesses the effectiveness of a product or system in meeting specified and implied functional requirements. Reliability assesses the performance of a system or product under specified conditions. Performance Efficiency measures performance relative to resources utilized. Usability examines how effectively and satisfactorily a product can be used to achieve goals. Security evaluates the protection of information and data from vulnerabilities. Compatibility assesses the ability to exchange information and perform functions within shared environments. Maintainability gauges how easily a product can be modified to improve or adapt to changes. Lastly, Portability assesses the ease with which a system or product can be transferred between different environments. These characteristics collectively provide a structured framework for organizations to ensure and enhance the quality of their software offerings throughout their lifecycle (Britton, 2021).

## Related Studies

To understand the context and significance of this study, it is important to review existing research on developing Android-based educational game applications. This section offers an overview of relevant studies, focusing on the theories, methods, findings, and implications that inform our understanding of mobile games as tools for programming education. By examining past work, we aim to identify knowledge gaps, build on existing research, and underscore the unique contribution of this study.

**Educational Games.** Educational games, often referred to as serious games, are interactive digital experiences designed to facilitate learning. Educational games are utilized in classrooms to boost learning, often serving as a supplement to traditional teaching methods. By making learning engaging and enjoyable, these games can enhance students' motivation, critical thinking skills, and overall learning outcomes (Study.com, 2023).

Koupritzioti and Xinagalos (2019) introduced Py Diophantus, an educational maze game prototype. This game serves as an additional tool to help students become familiar with arithmetic expressions and operator priorities. Py Diophantus is fully functional and comprises around 500 lines of code. It incorporates numerous programming concepts and constructs typically taught in introductory programming courses, along with various game programming principles. Furthermore, it's easily customizable to showcase additional game programming concepts, such as adding an animated sprite to navigate the maze as player’s advance. PyDiophantus offers multiple possibilities, including using it as a case study or a student-implemented project to teach programming concepts, Python, or fundamental game development processes. Moreover, it is in line with the notion that learning through game implementation is an efficient alternative to traditional instructor-centered programming instruction.

Bonza and Lopio (2023) developed the "Binary Wars" mobile game application to improve the logical thinking abilities of ICT students and educate them on binary and decimal conversion. Their development approach followed the Iterative development cycle, encompassing five phases: planning, requirements, analysis and design, implementation/coding, and evaluation. They employed various software tools in the process and designed the game with five modules: mode difficulty, game settings, game logic, score computation, and time management. This study underscores the potential of game-based learning in ICT education, emphasizing the significance of interactive and engaging tools to enhance student learning outcomes module.

“PROGDUNGEON” is an Android-based Python coding game designed to inspire and engage students in programming. The primary goal of this research is to encourage programming learning through block-based programming. The game employs a drag-and-drop mechanism, where players manipulate blocks containing Python syntax. This Android game is compatible with mobile phones and comprises four main modules: a user interface module, a block manager module, an interpreter module, and a game manager module. The researchers adopted an Iterative Model approach in their methodology. This approach begins with the initial implementation of a limited set of software requirements and iteratively refines the evolving versions until the entire system is fully developed and ready for deployment. The target audience for this game includes incoming first-year students and anyone interested in learning Python Programming (Dauz, Dela Cruz, & Pacaña, 2019).

Bagunu and Buan (2017) conducted a study titled “Silicon Valley: 2D Android Based Mobile Adventure Game for Computer Fundamentals” with the aim of assisting students in understanding certain aspects of Computer Fundamentals. The game is designed exclusively for Android- supported mobile phone and consist of modules such us menu selection, avatar selection, avatar profile, backpack, stage selection and gameplay

The development of the system followed the Spiral model, a risk-driven approach emphasizing the significance of risk analysis in project success. C# programming language and the Unity 3D platform were utilized for software development. This research explores the potential enhancement of the learning process for computer fundamentals through the integration of a mobile game. Moreover, the study highlights the dual benefit of providing both learning and entertainment simultaneously.

Table 1. Comparison table of related studies

| **MODULES** | **RELATED STUDIES** | | | | **PROPOSED STUDY** |
|---|---|---|---|---|---|
| | **S1** | **S2** | **S3** | **S4** | |
| Character Skin | | | | | ✓ |
| Menu/Settings | ✓ | ✓ | ✓ | ✓ | ✓ |
| Sounds | | ✓ | ✓ | ✓ | ✓ |
| Numerous Topics | ✓ | ✓ | ✓ | ✓ | ✓ |
| Level of Difficulty | | ✓ | ✓ | ✓ | ✓ |
| Scoreboard | | | | | ✓ |
| The Book of Knowledge | | | | | ✓ |
| Android-Based | | ✓ | ✓ | ✓ | ✓ |
| Methodology | | Iterative model | Iterative model | Spiral Model | Rapid Prototyping |

Legend

S1 – Diophantus Maze Game: Learning Mathematics by Playing or Learning Game Programming in Python
S2 – Binary wars: a mobile game development about binary and decimal conversion
S3 – Progdungeon: An android based python coding game

S4- Silicon Valley: 2D Android Based Mobile Adventure Game for Computer Fundamental

**Synthesis**

Educational games often referred to as serious games, have garnered significant attention in the field of education. They aim to inspire students, stimulate their interest in complex subjects, and make the learning experience more enjoyable.

Studies by Koupritzioti and Xinagalos (2019), Bonza and Lopio (2023), Dauz, Dela Cruz, and Pacaña (2019) have all employed game-based learning for their intended users and students. "PyDiophantus" (Koupritziot *et al*., 2019) is an educational maze game designed to help students grasp arithmetic expressions and programming concepts. "PROGDUNGEON" (Dauz *et al*., 2019) focuses on teaching programming through block-based coding using the Android platform. "Binary Wars" (Bonza *et al*., 2023) was developed to enhance logical thinking abilities related to binary and decimal conversion. The research paper (Bonza *et al*., 2019) employs the Iterative Model approach as their software development methodology, while the study by Bagunu *et al*. (2023) utilizes the Spiral model, combining learning and entertainment by offering modules for menu selection, avatar customization, and gameplay. These studies collectively demonstrate the effectiveness of using interactive and engaging game-based tools to enhance student learning outcomes across various educational domains.

Based on these findings, it has been shown that education in the form of mobile games is indeed an effective tool for educating individuals because they are entertaining, interesting, and capable of capturing people's full attention. Since this learning approach is not as conventional as most people believe, information technology may be utilized to simplify the delivery of education, making it easier for students to understand. Additionally, game-based learning is employed to engage, motivate, and retain students while also increasing their learning and understanding.

# METHODOLOGY

This chapter presents the materials, methods, and data gathering techniques, data analysis, and implementation plan.

**Requirement Analysis**

To collect the data, a letter request and a letter request to conduct data gathering was submitted by the researchers to Mr. Edizon Feranil, an ICT professor from Tanza National Trade School, Tanza-Trece, Cavite. Followed by approval, the researchers conducted a face-to-face survey with grade 12 students. The survey questions (see Appendix 7) are open-ended, with options for respondents. These survey questions are made up of 25 questions to cater to the respondents' difficulties.


Figure 3. Beneficiary’s location – Tanza National trade School (maps.google.com)

Based on the survey conducted with the ICT senior high-school students of Tanza National Trade School, Cavite, the following features for each module has been finalized by the researchers.

Table 2. Features of the proposed system

| MAIN FEATURES | DETAILED FEATURES |
|---|---|
| Topics | 1. The game provides a wide range of Python fundamental topics. In total, there are sixteen (16) available for users to explore. |
| Different types of questions | 1. The game provides different types of questions such as completing codes, guessing the output, and debugging codes. |
| The Book of Knowledge | 1. The application provides a specific button named "The Book of Knowledge" that contains lessons about the selected topic.<br>2. The main purpose of these lessons is to assist players in answering the given questions. |
| Audio Button | 1. The game provides an audio button for the users, when pressed; a narration of the lesson will start playing for the user to listen. |
| Settings | 1. The game provides a setting.<br>2. Users have the option to control the audio experience within the game. They can choose to mute or unmute background music and sound effects based on their preferences. |
| Avatar Skins | 1. The game offers a variety of avatar skins; these avatar skins will be given to the player as a reward for completing topics and are given randomly. |
| Scoreboard | 1. The game provides a scoreboard module where users can view their scores in their completed topics. |
| Attempt | 1. The game provides the user three (3) chances to answer the questions. Once the user consumes all the 'life' or they answered the question correctly, they will be directed to the next questions. |

The following table, however, presents the non-functional requirements of the proposed system.

Table 3. Non-functional requirements of the system

| QUALITY ATTRIBUTES | DESCRIPTION |
|---|---|
| Compatibility | The application feature is compatible with the android version 8 up to 11. |
| Usability | The application provides clear and easily readable contrast text and visual elements across all pages. |
| Functionality | Application allows users to select and answer questions based on selected categories. The navigation between buttons and pages is expected to function. |

Figure 4 illustrates the system's context diagram, depicting the flow of events. To initiate the process, the user starts the game. Upon starting, the game system presents the home page, featuring various navigation buttons such as the 'Start' that contains lessons and quizzes, the 'Scoreboard' where all the user's attempt results are recorded, the 'Avatar' where the user can choose their preferred avatar, and lastly the 'Settings' button which lets the user turn the music background on or off. Additionally, it highlights the available topics and questions that users can engage and play with.

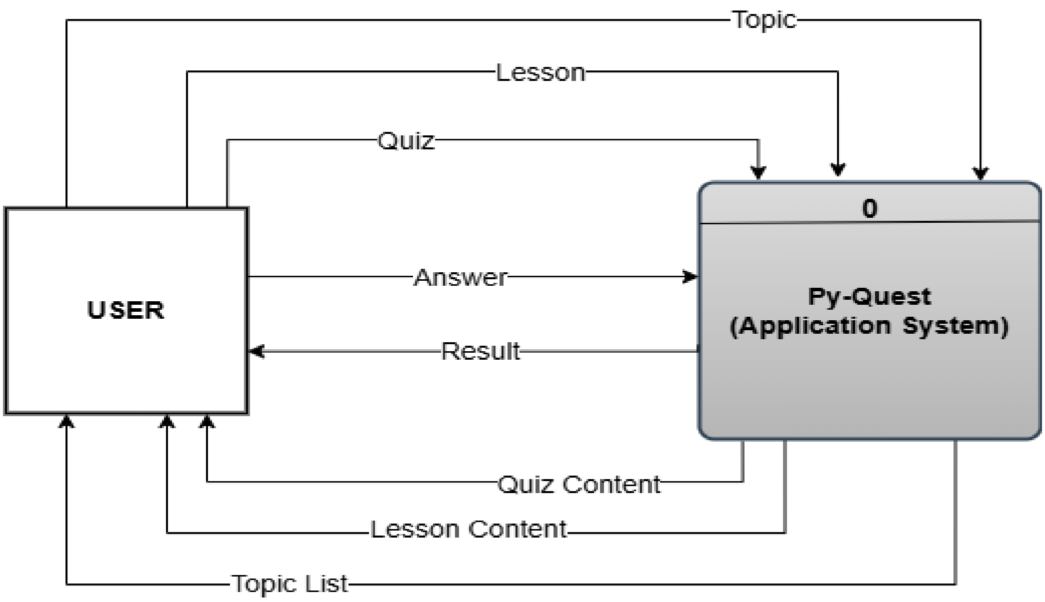



Figure 4. Context diagram of Py-Quest: An Android-Based Game Application for Python Programming Fundamentals

The user must then finish the lesson, which is kept inside the book of knowledge to start answering the quiz; these lessons correspond to the topic selected by the user. The user will then have to provide the answers to each question, after answering; the game system will display the result of the attempt to the user.

Figure 5 depicts the system architecture, comprising four modules: the gameplay module, the book of knowledge module, the achievement module, and reset module. The gameplay module will be responsible for the game's mechanics, rules, levels, and progression. The system allows the user to play and learn Python programming language through a level-driven game with different modes, which are identification, coding, and debugging. Within this module, users must answer all the questions on a specific topic to advance to the next level.

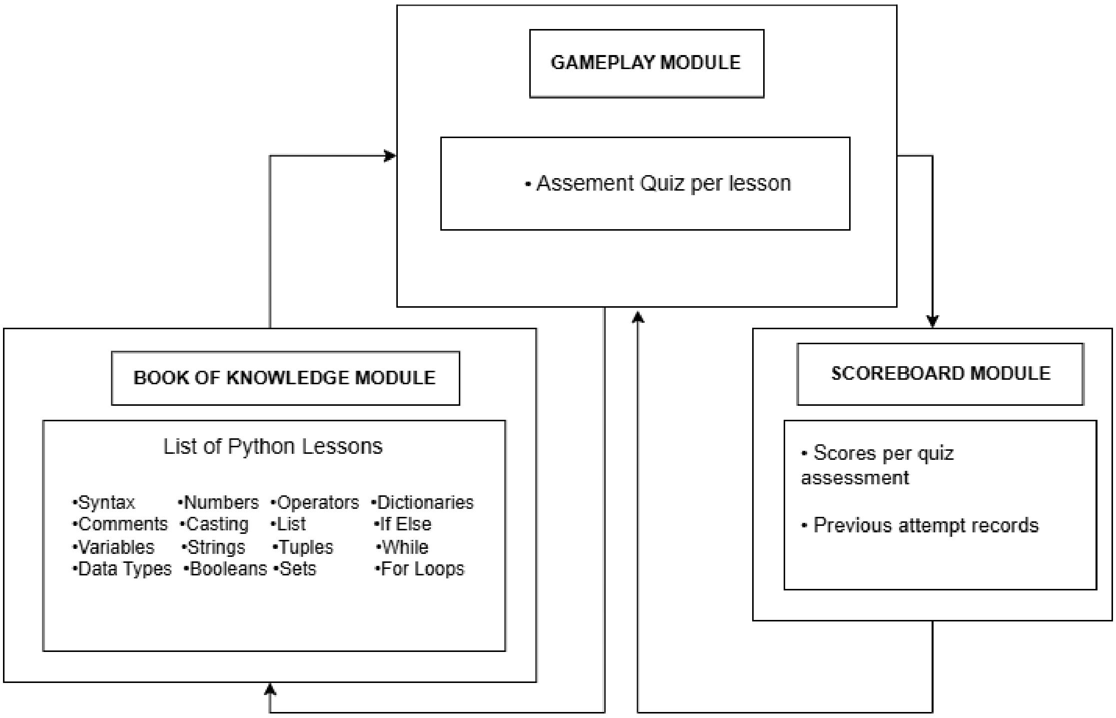



Figure 5. System architecture of Py-Quest: An Android-Based Game Application for Python Programming Fundamentals

The book of knowledge module provides a lesson for the given topics. It allows users to have more ideas about specific topics. This module activates when the player clicks a specific button, which will call a pop-up providing players with lessons corresponding to their selected topic that serves as guidance to enhance their problem-solving abilities and gameplay experience. The scoreboard module contains all the attempts of the user after answering a topic, after completing a topic, all the results will be stored in the scoreboard so that the user can monitor their progress. This module can be viewed upon clicking the scoreboard button on the homepage of the application; they will then need to press the completed topic to view the results.

The data flow diagram is presented in Figure 6. The flow of the game begins with Process 0.0, which is initiated by the user when they run the installed application. The game then displays the introduction page; this page contains some information and a brief introduction about the application.

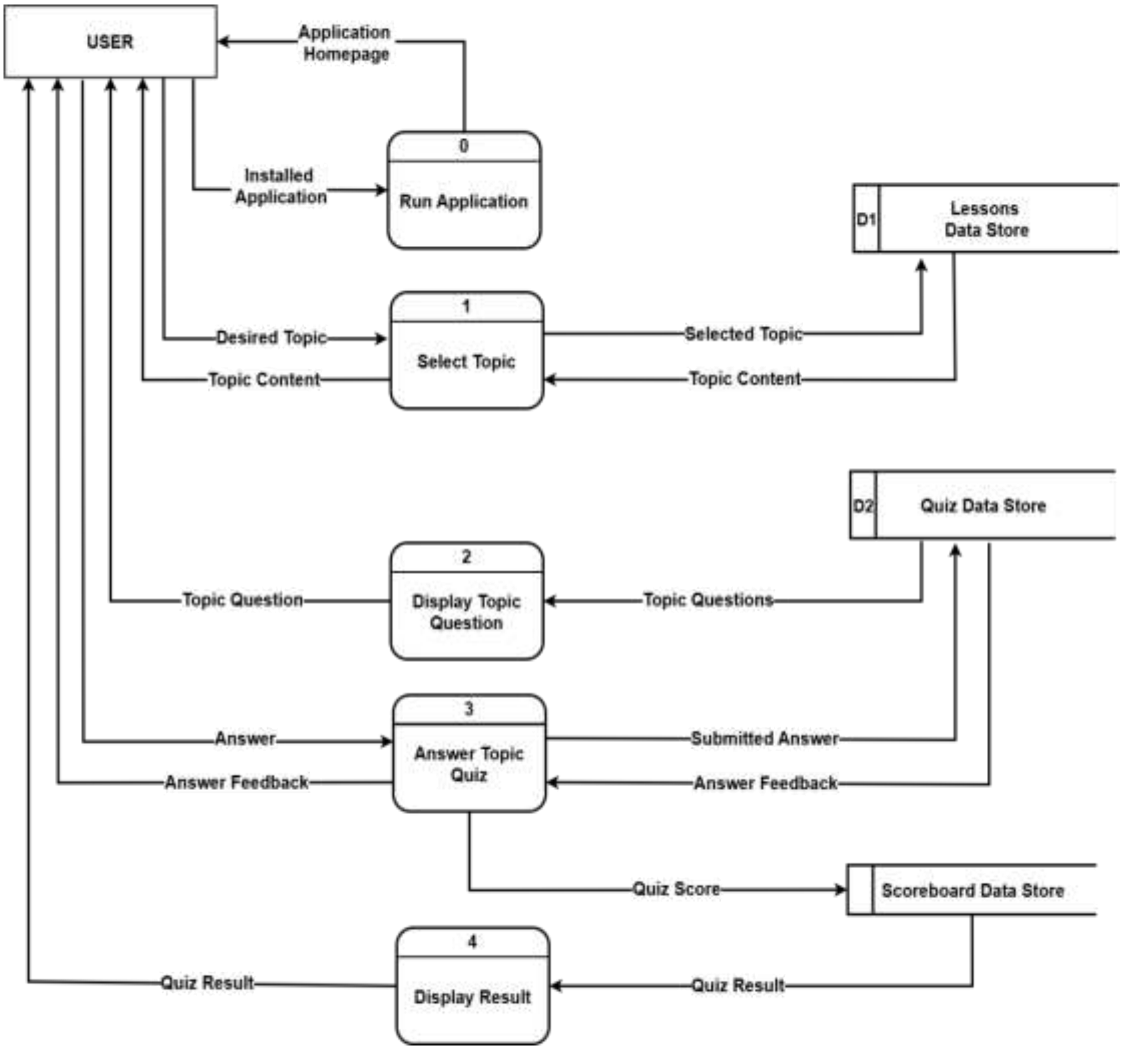



Figure 6. Data flow diagram of Py-Quest: An Android-Based Game Application for Python Programming Fundamentals

Once the user finished reading or skipped the introduction part, they will be directed to the home screen. The home screen is where the user can choose from the navigation buttons, which are, the start, scoreboard, avatar, info, and the settings button. In process 1.0, the user chooses the gameplay module. Once the user selects the start button, they will be directed to the topic selection page, where they can select any available topic. These topics include the fundamentals of Python programming language that range from Python introduction up to loops.

Once the user selects a topic another category which contains the topic contents will pop up for them, it contains the book of knowledge and the play button. If the user selects the book of knowledge module the lesson about the selected topic will be displayed to the user; the user must finish reading the book of knowledge to be able to start answering the quiz. After finishing the lesson, the user can now start answering the quiz, which leads to process 2.0, where the questions are displayed to the user. In process 3.0 the user then provides an answer based on the given questions. The submitted answer is then verified by the application. After the answer has been verified, it will proceed in process 4.0, where the result is displayed to the user.

Figure 7 illustrates the system's use case diagram, depicting one actor referred to as the "user". To play the quiz, the user needs to run the application. Once the application is launched, the user will be prompted to select one of the listed topics provided by the game system. Upon selecting a topic, a new category will be shown to the user, the user will need to select the “Book of Knowledge” that contains the lesson about the selected topic, once the user finishes the lesson, and they will now be able to select the play button. Upon pressing the play button, the application will display a series of questions for the user to choose from, these questions correspond to the topic chosen by the user. The user is required to provide answers to complete the quiz.

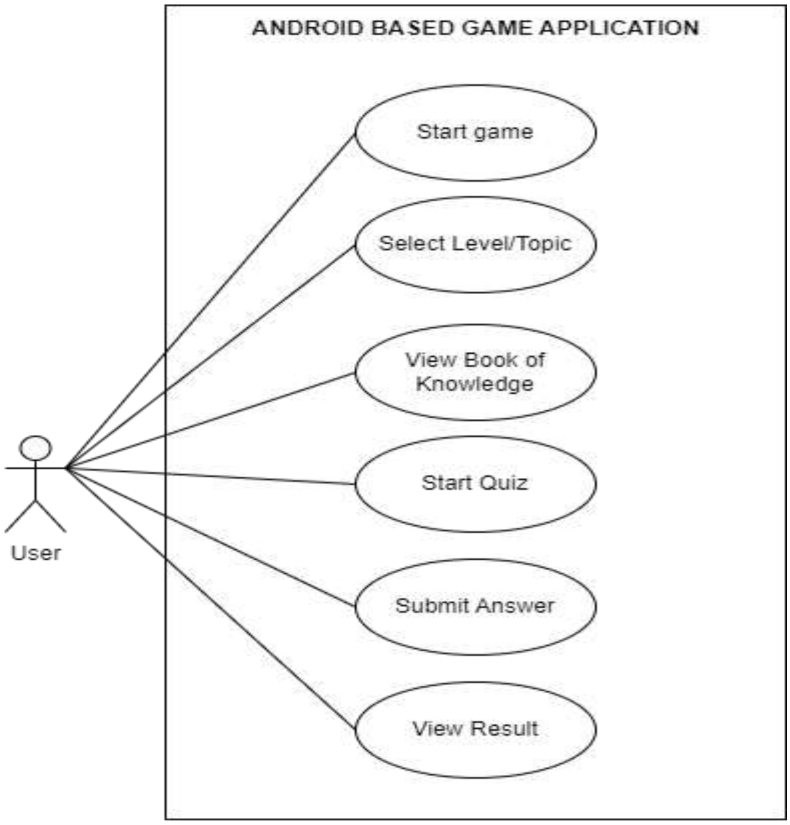



Figure 7. Use case diagram of Py-Quest: An Android-Based Game Application for Python Programming Fundamentals

Finally, after the user provides an answer and completes the quiz, the system then verifies the submitted answer and then displays the result for the user to view.

Figure 8 illustrates a sequence diagram that shows the system of the android-based game application that will be utilized to promote awareness to those who are unfamiliar with the Python programming language at Tanza National Trade School. This android-based game consists of a gameplay module, book of knowledge module, and scoreboard that can access by the user.

To access the Py-Quest, the user must run the game. The intro page will then appear, and a character in a dialogue box will appear in the center of the screen, presenting the introduction and the game mechanics. Following that, the user can now view the home page and choose whether to access the start, scoreboard, avatar, info, and settings page. When a user chooses the start button, the system will display the sixteen (16) topics which are: introduction, comments, variables, data types, numbers, casting, strings, Booleans, operators, lists, tuples, sets, dictionaries, if else while loops and for loops. When the user chooses to access the scoreboard, the user can view the collective scores of the finished topics and if not, the user can go back to the home page. When the user chooses to access the avatar, it will then lead to available characters or avatars awarded to the user; the back button is also available if the user chooses to go back to the home page. Users can also choose the information or info page wherein provides a comprehensive overview of the game, including the introduction and mechanics. Settings can also be accessed by the user when chooses to go back to the category. This will allow the user to manage their sound experience by turning on or off background music and sound effects and or going back to the home page.

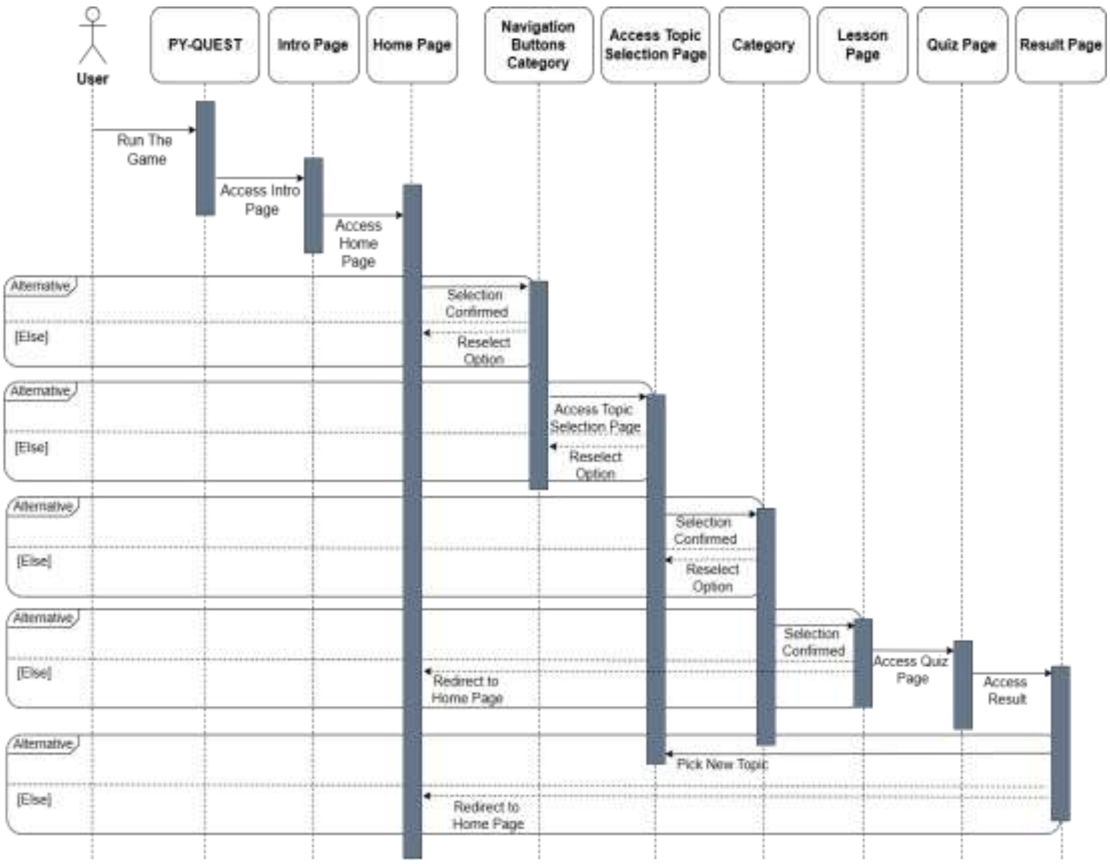



Figure 8. Sequence diagram of the Py-Quest: An Android-Based Game Application for Python Programming Fundamentals

When the user decides to access the available topic, the user can also choose to access the lesson, which will direct to the Book of Knowledge page where the user must finish reading the lesson to unlock the play button. The play button on the other hand directs the user to the ‘Game Page’ where they can start to answer the quiz. The back and home button will be available at both top corners of the screen.The following procedure is accessing the quiz game of the selected available topic. Once completed, a success message will appear; otherwise, the user can choose to go back to the navigation buttons.

**Software Design**

The system was designed in accordance with the software quality characteristics outlined in the ISO 25010 standard. These characteristics include functional suitability, reliability, performance efficiency, usability, security, maintainability, and portability. Each of these aspects plays a crucial role in ensuring the system achieves optimal quality and performance upon implementation. By following these guidelines, the system has meet high standards of excellence and efficiency.

Draw.io was employed for creating detailed wireframes and prototypes. This tool was instrumental in visualizing the system's architecture and user interface (UI) and user experience (UX) designs. The initial iteration of the software focused extensively on developing these architectures and UI/UX plans. Each design element and module were meticulously crafted and then transformed into prototypes.

These prototypes were then evaluated by selected participants, ensuring that all aspects of the system were thoroughly tested and refined. This iterative process of design, prototyping, and evaluation was crucial in honing the system’s functionality and usability, ensuring that it meets the needs and expectations of its users. By integrating feedback from these evaluations, the system was continuously improved, paving the way for a robust and user-friendly final product.

Figure 9 shows the home page of the application where all the navigation buttons are displayed, the following buttons will direct the user to the topic selection page, scoreboard page, avatar selection page, information page, and the settings page.


Figure 9. Prototype of Py-Quest home page

The Book of Knowledge page, illustrated in Figure 10, provides users with a detailed lesson on the topic they have selected. This page is designed to enhance the learning experience by allowing users to either read the lesson or listen to an audio narration of the content. To ensure thorough understanding, users are required to complete the entire lesson before they can unlock the play button. Once the lesson is finished, the play button will become accessible, enabling users to start the quiz associated with the topic.


Figure 10. Prototype of Py-Quest book of knowledge

Figure 11 shows the game page of the application; this page is where the user is given a set of questions about the selected topic. The user must then input an answer in the text field and then submit it. After answering all the questions, the result will then be displayed to the user. The egg serves as the user's lives; each wrong answer submitted by the user will deduct their lives. If the user run out of lives they will be directed to the next question.

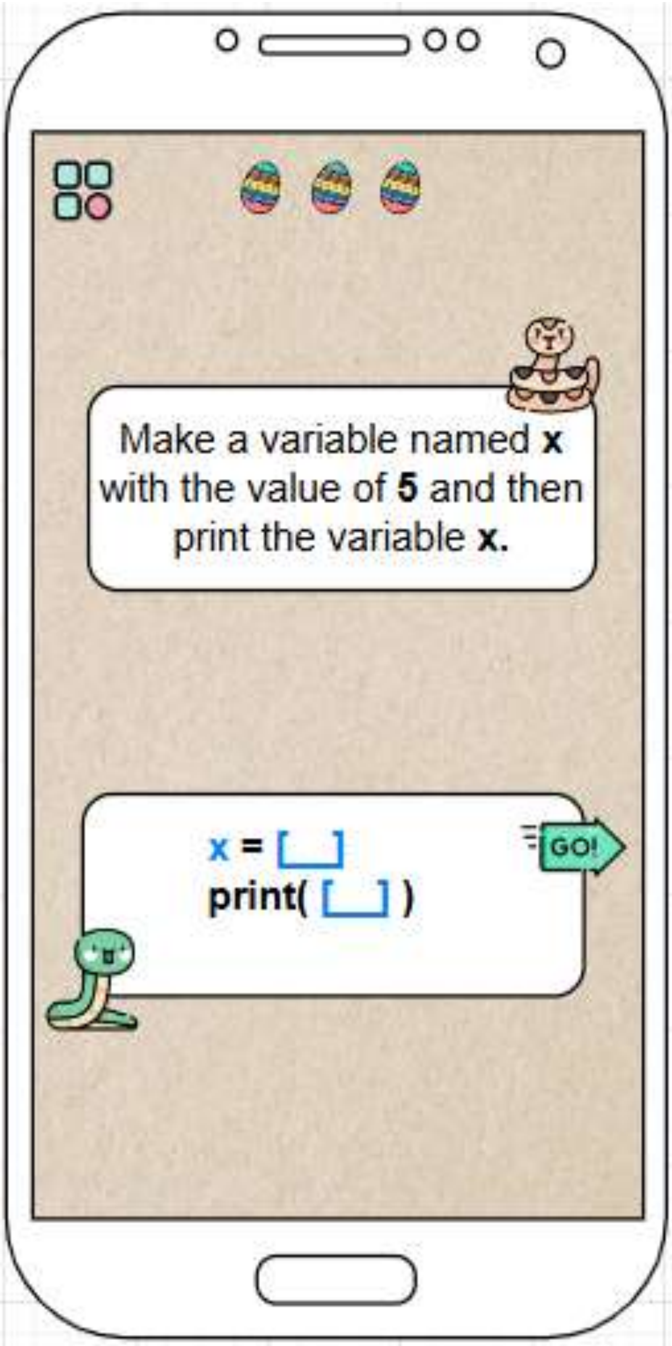



Figure 11. Prototype of Py-Quest game page

Figure 12 illustrates the scoreboard page of the application; this page is where the results of all the user's previous attempts are recorded. This page displays the number of attempts, the score, and the date when the user took the attempt.

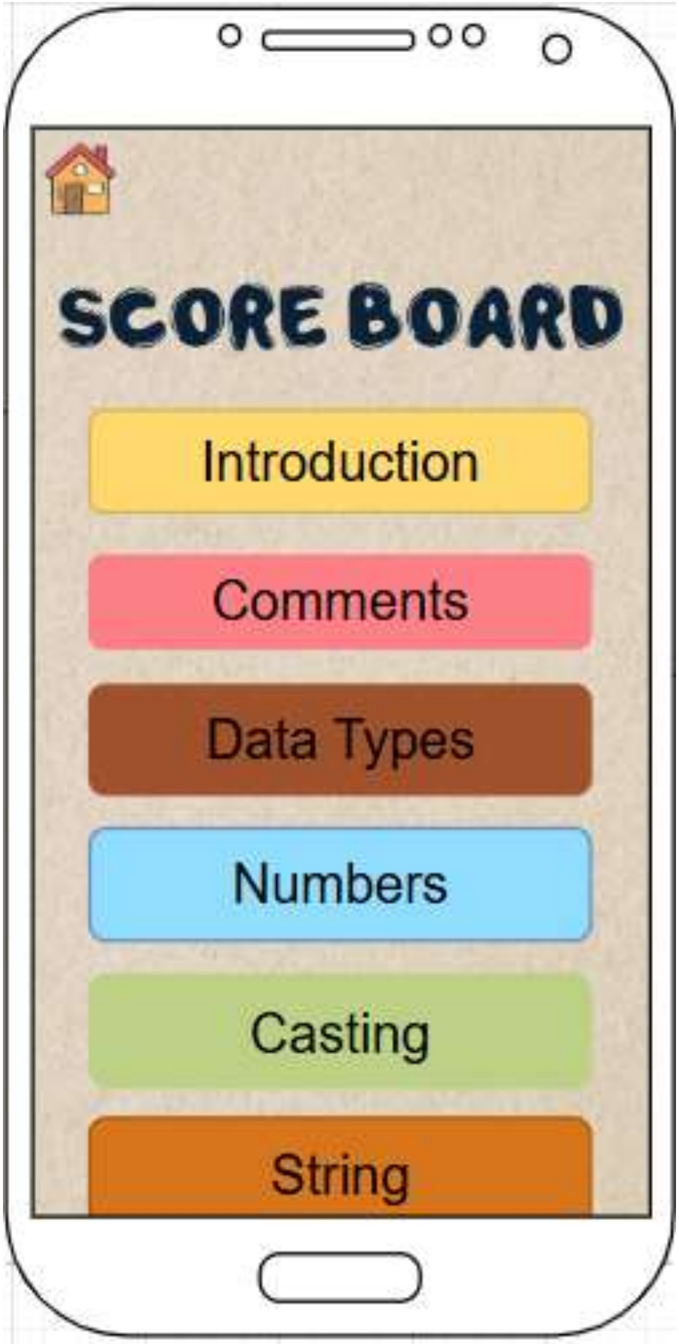



Figure 12. Prototype of Py-Quest scoreboard page

Figure 13 displays the avatar selection page where the user can select their desired avatar. The user is rewarded an avatar when they successfully reach the passing score needed for a topic, and the avatar rewards are placed randomly in selected topics.

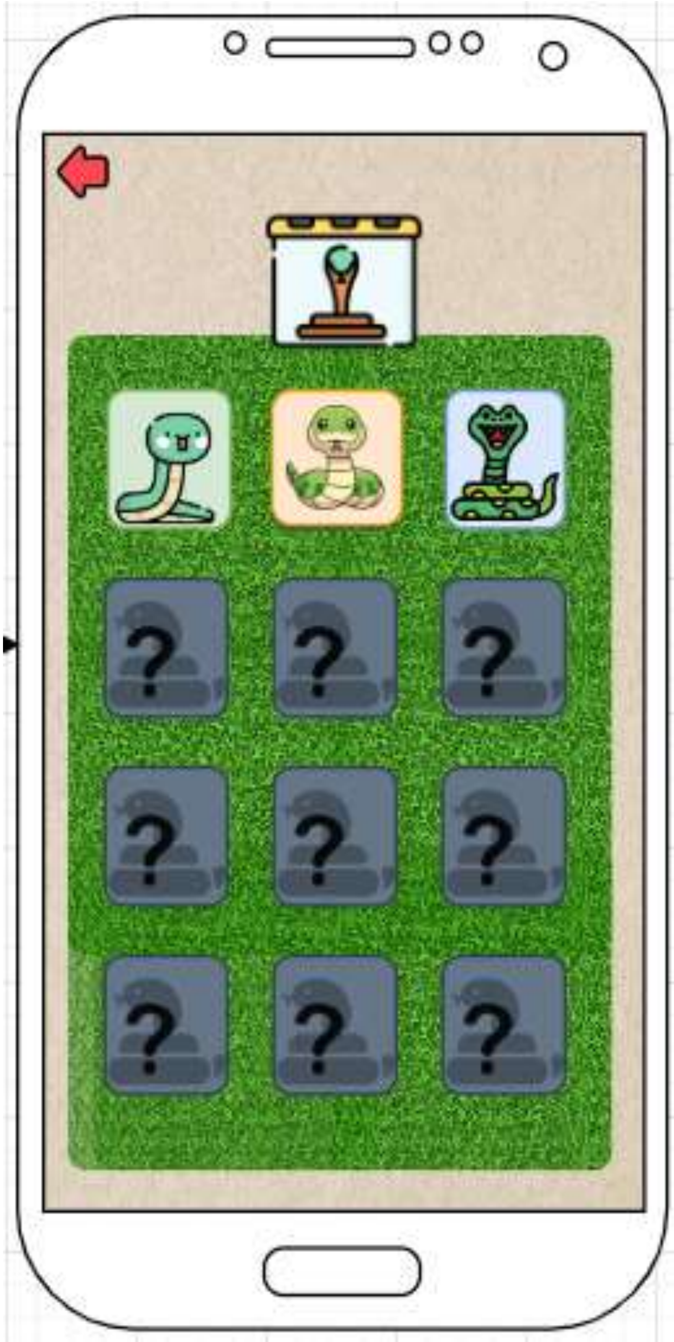


Figure 13. Prototype of Py-Quest avatar selection

Figure 14 illustrates the information page; the user can access this page by pressing the ‘Info’ button in the home page. This page provides a comprehensive overview of the game, including its introduction and mechanics.

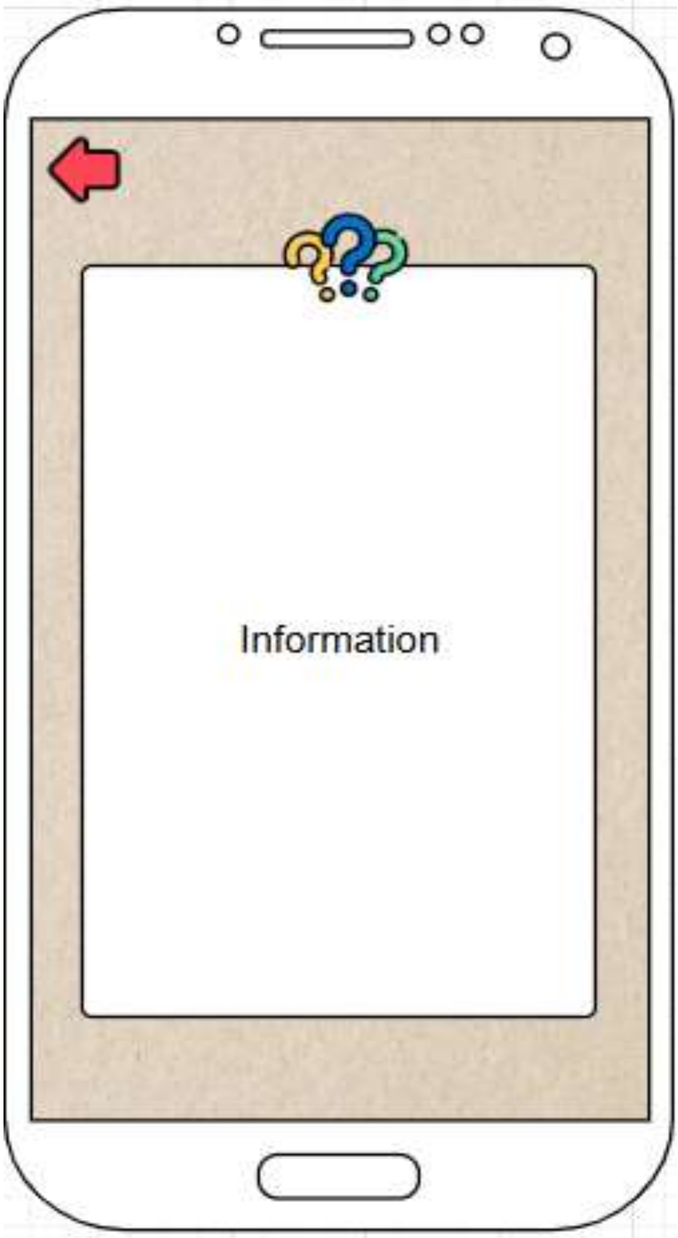



Figure 14. Prototype of Py-Quest information page

**System Development**

To establish the final design of the system, the researchers used a variety of technologies and tools. The development process incorporated programming languages such as PHP and Java, and utilized SQLite and XML for efficient database management. Android Studio was the designated Integrated Development Environment (IDE) used for coding and development.

The computer unit used in developing the application has the following hardware and software specifications: Microsoft Windows 10 Home Single Language

64-bit operating system, 4GB DDR3 RAM, Intel(R) Pentium(R) CPU N4200 @ 1.10GHz 6GB Intel HD Graphics 505 (Apollo Lake) x64-based processor. The software that will be utilized to be able to create the initial design of the system is draw.io, this software will also be used to create wireframes and diagrams that will be used in the development process of the application. The researchers will use the following software; Android Studio, SQLite, and XML for the development of the system. The system will be developed using PHP and Java as the programming languages.

**System Testing**

The application was tested for functionality, usability, and compatibility to ensure its effectiveness. Testing the functionality of the game means looking at every part of it to make sure everything works. This includes checking how users move around in the game, and play it from the main menu to the actual game activities. For usability testing, all navigation, such as icons and buttons should be accessible and consistent across all pages. Last is compatibility testing to ensure that the application functions and operates properly on android devices across various versions. This comprehensive testing procedure ensures that every aspect of the application functions correctly and without encountering any errors.

**Functionality Testing.** The following procedures were conducted to test the functionality of the application (See Appendix Table 1).

A. Test if the user can select a list of topics and quiz.
    1. Start the application.
    2. Select start button
    3. The application displays the list of topics for the user to choose from.

B. Test if the audio plays, and is accurate as what is written in the lesson.
    1. User selects a topic.
    2. Select book of knowledge
    3. Press audio button

4. A narration of the lesson will be played for the user.

C. Test if the scoreboard page will display accurate scoring after completing the quiz.
    1. User selects a topic.
    2. Select the play button.
    3. Complete the quiz.
    4. A display will pop up containing the user's result.

D. Verify if the user will be notified when it is the final chance to answer the given question.
    1. User submits two incorrect answers.
    2. A notice saying “Last Chance” will appear.

E. Test if all the buttons function is working.
    1. Users press all the buttons.
    2. Buttons function as intended.

F. Test if the lives get lessened with each mistake
    1. User submits an incorrect answer.
    2. User’s life will be deducted.

G. Test if the next topic will unlock after the user meet the passing score of the selected topic
    1. Users complete a topic.
    2. Next topic unlocks.

H. Verify if the user is able to enter text into the text field.
    1. User starts the quiz.
    2. User type answer in text field.

I. Test if the user is able to change avatars.
    1. User selects the avatar button.
    2. Select any available avatar.

J. Test if completing a random topic unlocks the avatar associated with it.

1. User complete random topic.
2. Avatar skin is rewarded.

There are 10 test cases conducted to test its functionality. The first test case is intended to test if the user can select a list of topics and Quiz. The second test aimed to verify whether the audio plays, and is accurate as what is written in the lesson. The third test case focused on verifying if the scoreboard page will display accurate scoring after completing the quiz. The fourth test case evaluates if the user will be notified when it is the final chance to answer the given question. The fifth test case focused on testing if all the buttons function is working. The sixth test case aimed to monitor if the lives get lessened with each mistake. The seventh test case was intended to test if the next topic will unlock after the user meets the passing score of the selected topic. The eighth test case was focused on verifying if the user is able to enter text into the text field. The test concluded that the user was able to insert text into every text field on the application. The ninth test case aimed to see if the user is able to change avatars, which is then verified as the avatar changed to exactly what the user selected. The tenth test case examined if completing a random topic unlocks the avatar associated with it. It was observed that completing random topics and reaching the passing score gives the user the reward avatar.

**Compatibility Testing.** The applications compatibility testing was conducted to determine which Android devices can operate and function properly across various Android versions (See Appendix Table 2).

A. Android 13
    1. Launch the application on the android 13
B. Android 12
    1. Launch the application on the android 12
C. Android 11
    1. Launch the application on the android 11

D. Android 10

1. Launch the application on the android 10

E. Android 9

1. Launch the application on the android 9

F. Android 8

1. Launch the application on the android 8

**Usability.** To assess the system's user-friendliness and ease of operation, usability testing was conducted as shown in Appendix Table 3. This testing specifically examined the usability and design of the system's interface. To test the usability of the application, the following procedures are conducted.

A. Test if buttons and icons are displayed based on their intended functions.

1. Check all the buttons and icons function.

B. To test if each lesson within a topic has an audio button.

1. Check if all the topics lesson has an audio button

C. Verify if all fonts are readable.

1. Check the text readability.

**System Evaluation**

To prepare for the system evaluation, the researchers submitted a letter requesting permission to conduct the evaluation to the dean of Cavite State University (see Appendix 9), the ICT professor of Tanza National Trade School. Upon receiving approval, the researchers utilized an evaluation form based on the ISO 25010 software standard to assess the system. The developed application were evaluated by 10 IT/CS instructors as technical evaluators, 100 ICT students (see appendix 13) from Tanza National Trade School. The application is demonstrated to both technical and non-technical evaluations of the system. Py - Quest was evaluated by the researchers, with ISO25010, the international standard for evaluating software.

Table 4 shows the Likert scale that can be used to generate the overall interpretation of the mean score of each item.

Table 4. Likert scale

| RANGE OF WEIGHTED MEAN | INTERPRETATION |
|---|---|
| 4.51 - 5.00 | Excellent |
| 3.51 - 4.50 | Very Good |
| 2.51 - 3.50 | Good |
| 1.51 – 2.50 | Fair |
| 1.50 and below | Poor |

**Implementation Plan**

After the development of the mobile game, it will be sent to the Department of Information Technology for the evaluation of the system. If the system passed the evaluation, it will be presented to the user of the system. If Tanza National Trade School decides to use the mobile game, the researchers will hand over the documentation and mobile game altogether. It will serve as guidance and to provide necessary ideas on how the mobile game works as well as how to update and take maintenance. A letter of agreement will state that the school will receive the application, releasing the researchers from any responsibility towards it. These approaches are detailed below.

Table 8 presents the implementation for the Development of Py - Quest. The following strategies and activities created by the researchers that would be followed as well as the persons involved in order to implement the system effectively.

The initial step involves submitting a system evaluation approval form to the representatives of Tanza National Trade School, a process anticipated to take three days. The next strategy entails demonstrations the system to research instructors and ICT students for their evaluation, which is expected to last five days, following this, the system will be evaluated by the ICT students, professor representatives, and panelists, using a provided evaluation form, a process taking one to three days. Upon completion of the demonstration and evaluation, the final step involves deploying the system application in Tanza National Trade School. The implementation will require a mobile phone for testing and will be carried out in this school. To successfully implement the application, it is essential to ensure that the device is running at least Android version 8, but not exceeding Android version 13. Additionally, the application requires a minimum of 97 MB of available storage space to function properly. This compatibility range and storage requirement are critical for the optimal performance and stability of the application.

Table 5. Implementation plan

| STRATEGY | ACTIVITIES | PERSONS INVOLVED | DURATION |
|---|---|---|---|
| Approval for evaluation | Requesting for approval of letter for the evaluation of the system | Researchers, research instructors, ICT professor | 1- 3 days |
| Testing | Presenting the system demonstration to the DIT, research panels, and instructors | Researchers, research instructors, ICT students | 1-3 days |
| System Evaluation | Evaluation of the system by providing the evaluation forms | Researchers, users, research panels | 1-5 days |
| System Deployment | The system is deployed after testing demonstration and will be deployed for the usability of the user | Researchers, user | 1-2 days |
| Document and system implementation agreements | The agreements and files about the system will be made. The files and the system will be handed over to the organization for maintenance and updating purposes. | Researcher and DIT | 1-3 days |

# RESULTS AND DISCUSSION

This chapter presents the presentation, analysis, interpretation, and discussion of the results of the study.

## System Design

The Py-Quest: An Android-Based Game Application for Python Programming Fundamentals was developed for the ICT Students of Tanza National Trade School, Cavite with the aim to assist them in learning and understanding the basics of Python. By using interactive gameplay, the project made learning accessible anytime and anywhere, and also engaging. It provides a fun way for young people to gain valuable coding skills, which could help them in their future education and career opportunities. The application is composed of three main modules: gameplay module, scoreboard module and book of knowledge module.

During the designing phase, the researchers took several steps to ensure the application's effectiveness and user-friendliness. They created drafts and prototypes to visualize the layout and functionality. The process involved analyzing requirements to understand the needs of users, developing initial sketches and wireframes for the layout, and building interactive prototypes to show how the system would work. The researchers used several technologies to develop the application, including Java and PHP for the interface, Android Studio for coding and building the system, and SQLite and XML for the database. These efforts aimed to create a well-structured, efficient, and user-friendly Android-based mobile application that meets the needs of the ICT student in Tanza National Trade School.

## System Development

The researchers created an Android-based game for ICT students. They used a combination of software and tools, including Android Studio, Java, PHP, and draw.io. By utilizing these tools, the researchers developed an effective educational resource for the students.

Figure 15 shows the welcome page, displaying the name and logo of the system, which is “Py-Quest”. This page will also serve as the loading screen. Upon completion of the loading process, users will automatically be redirected to the next page, which is the introduction page (refer to Figure 16).


Figure 15. Screenshot of Py-Quest Application’s welcome page

Figure 16 shows the introduction and mechanics page. A character with a dialogue box will appear in the center of the screen, presenting the introduction content. The dialogue box will have two buttons; clicking the left button will show the previous dialogue, while the right button will reveal the next one. A settings button will be located in the top left corner for accessing the settings page and on the top right corner the skip button will be available.

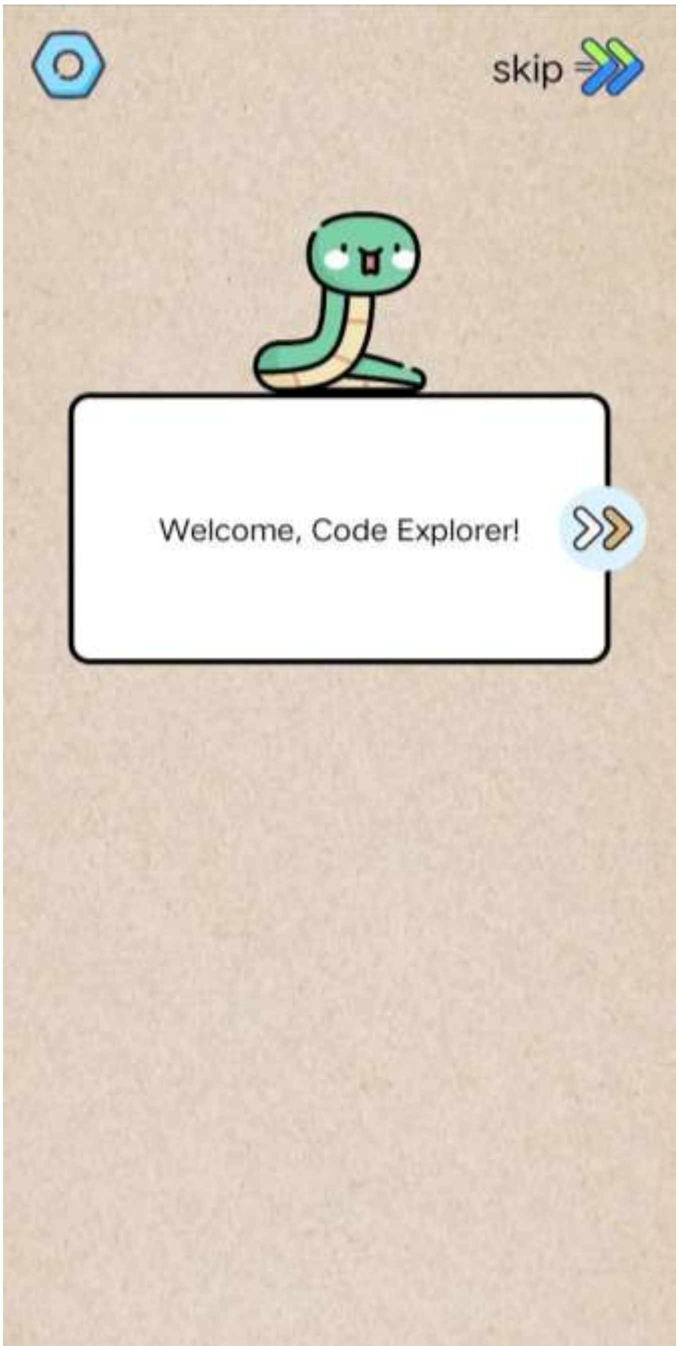



Figure 16. Screenshot of Py-Quest application's intro page

The homepage is presented in figure 17. It serves as the central hub of the application. Located at the top left corner is the Cavite State University Logo (CvSU), and on the center is the applications logo and the navigation buttons. The buttons are labeled as 'Start', 'Scoreboard', 'Avatar', 'Info', and 'Settings', these buttons allow users to navigate the application with ease by giving them direct access to the pages that correspond to them.


Figure 17. Screenshot of Py-Quest application’s home page

Additionally, figure 18 focuses on recognizing the importance of user guidance, the homepage includes a comprehensive tutorial designed to assist new users in understanding the use of each button.


Figure 18. Screenshot of Py-Quest application's sample tutorial

The topic selection page is presented in figure 19, it is where the levels are kept, the user can choose any available topics to access. Only The first topic will be initially available. the user must finish answering the first topic to unlock the next one.

Upon pressing any of the topics, the user will be directed to the topic contents page (see Figure 20). The home button will also be placed on the top-left corner of the screen, this button will direct the user to the home page.

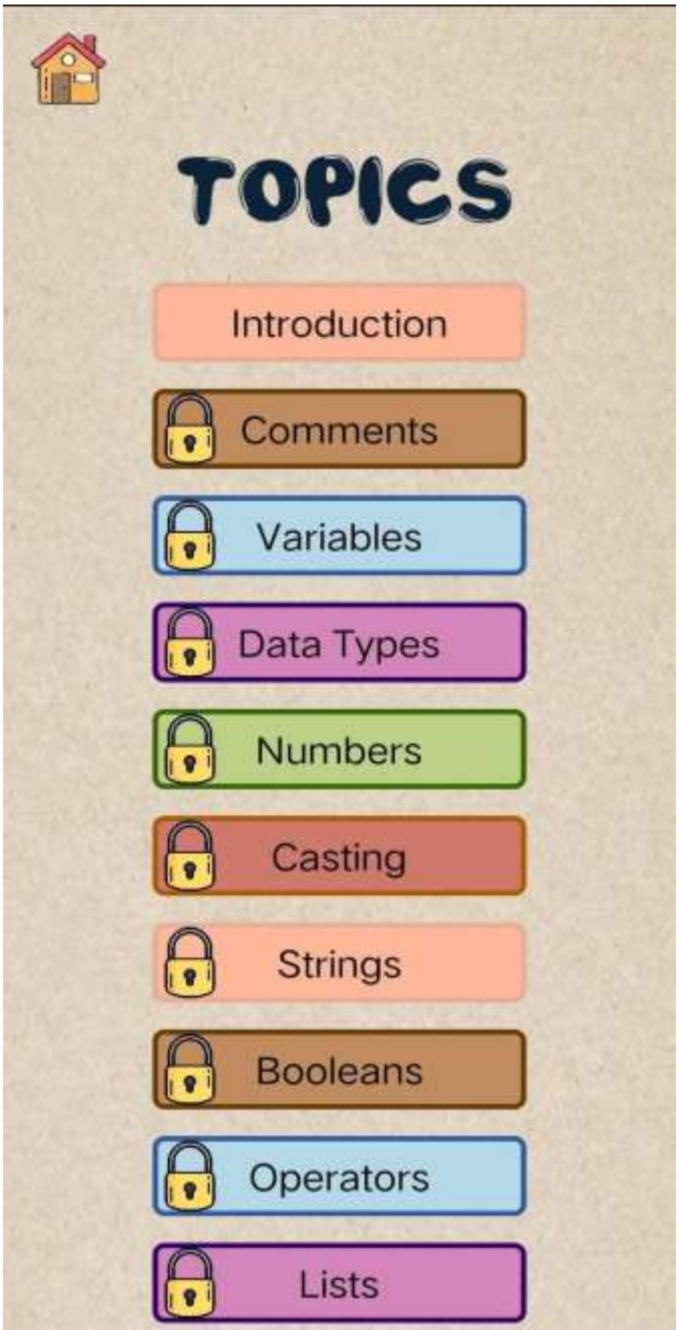



Figure 19. Screenshot of Py-Quest application's topic selection page

Figure 20 contains the book of knowledge and the play button. When the Book of Knowledge button is pressed, it will direct the user to the book of knowledge page where the user must finish reading the lesson to unlock the play button. The play button on the other hand directs the user to the game page where they can start to answer the quiz. The back and home button will be available at both top corners of the screen.

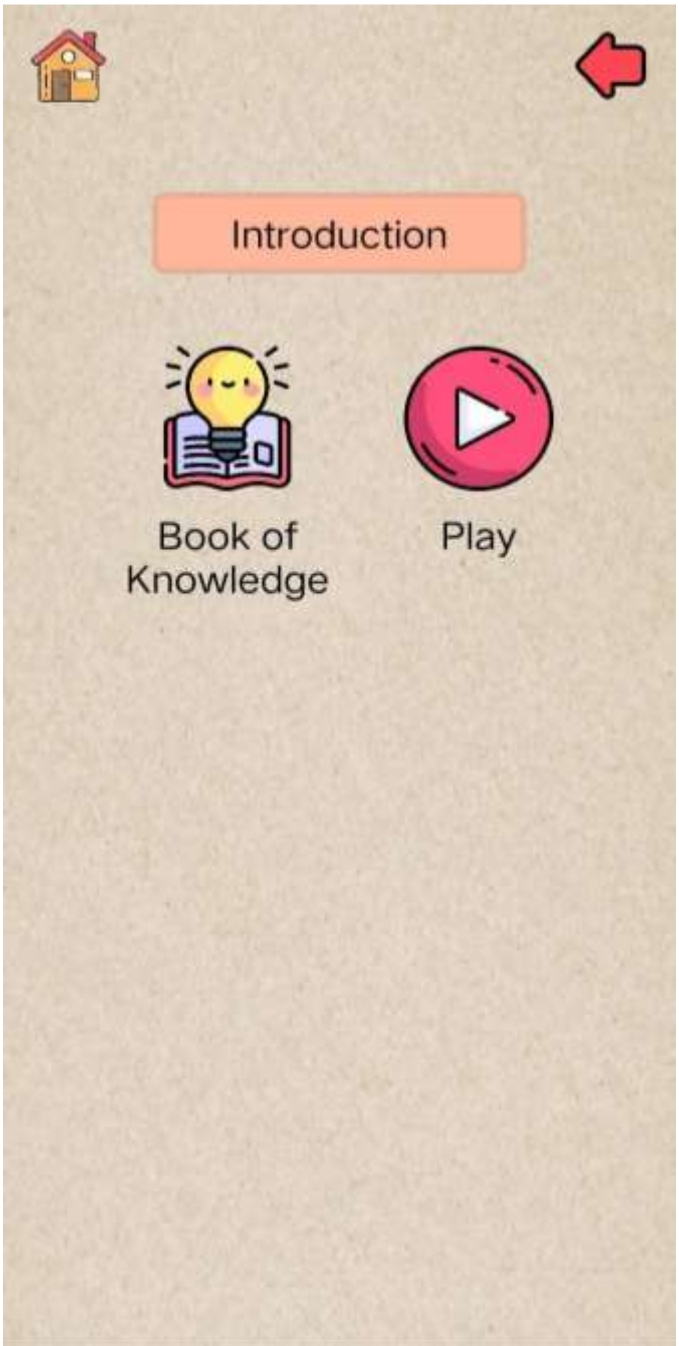



Figure 20. Screenshot of Py-Quest application's topic contents page

Figure 21 depicts the book of knowledge that serves as the lesson for the selected topic, the user must finish reading the lesson provided to unlock the play button. A back button will also be available on the top right corner of the screen, and on the left top corner is the sound button, that when pressed by the user, a narration sound about the lesson will play.


Figure 21. Screenshot of Py-Quest application's book of knowledge

Figure 22 shows the quiz details page of the application, where users are introduced to the topic they will be engaging with. At the top center of the page, there is an orange rectangular button labeled instructions, serving as the header for the page. Below this header, a bordered dialogue box occupies the central area of the screen. The data contained in the box includes the topic's title, number of questions, and the directions for answering the selected topic. A continue button is placed at the bottom of the page. When pressed, this button will guide the user to the next step or screen in the application.

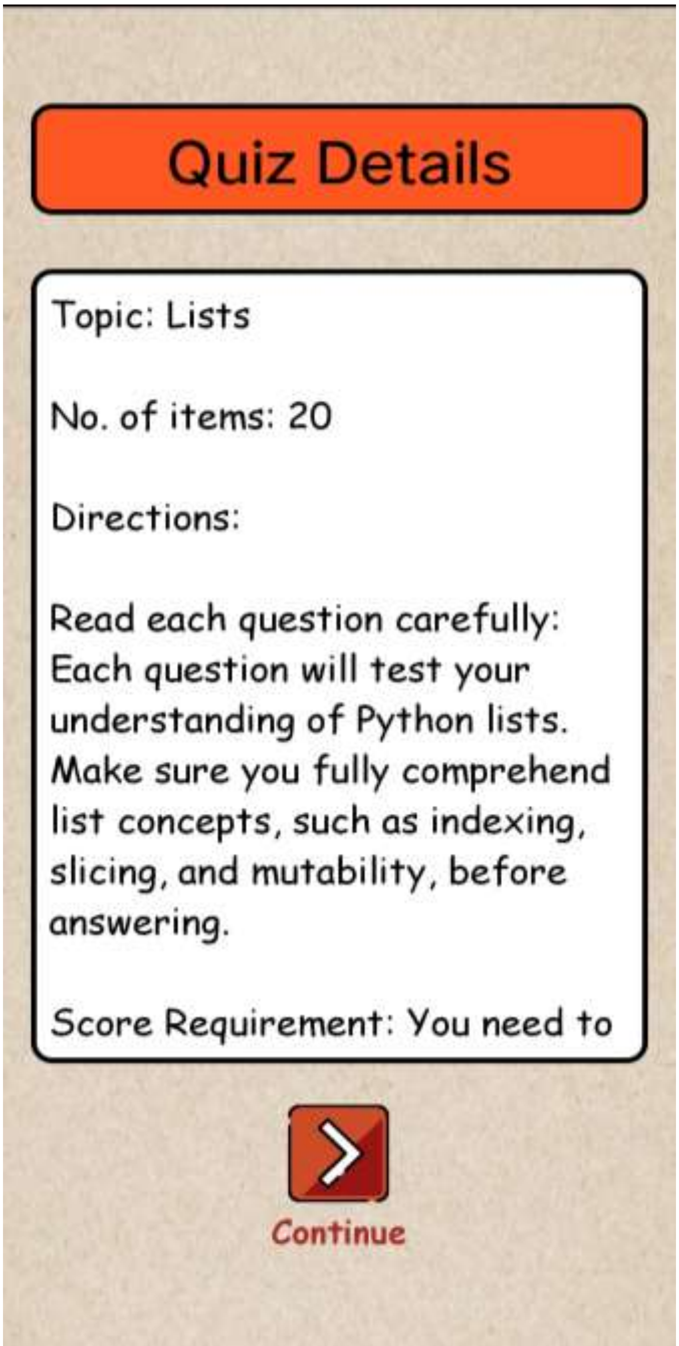



Figure 22. Screenshot of Py-Quest application's quiz details page

Figure 23 shows the game page of the application, in this page, two characters will be displayed on the screen, each equipped with a dialogue box. The character positioned in the upper section of the screen provides the questions, while the other represents the player. A submit button will be placed on the right side of the player's dialogue box, for the submission of answers. A menu button is provided at the top-left corner of the page that will help them navigate through pages (see Figure 28).

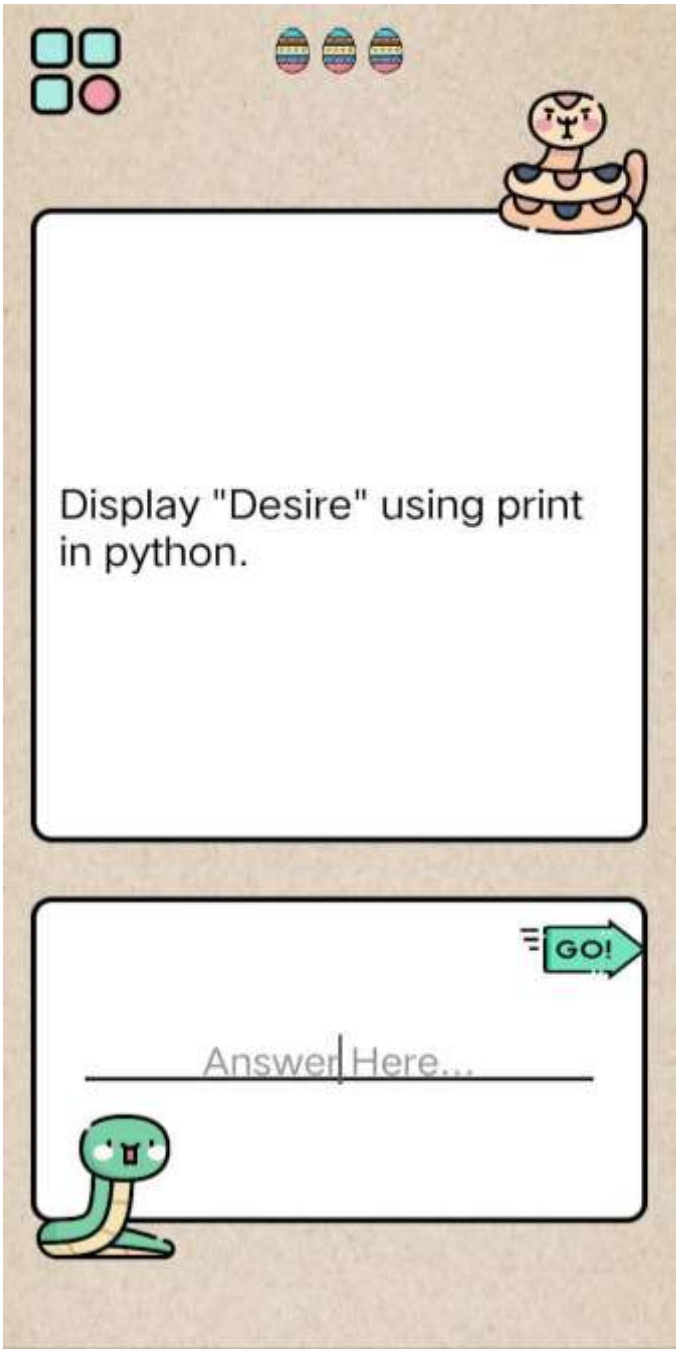



Figure 23. Screenshot of Py-Quest application's game page

Figure 24 is an illustration where, if the user gives the right answer their dialogue box turns green. The explanation page will then will be displayed to them (refer to Figure 25).

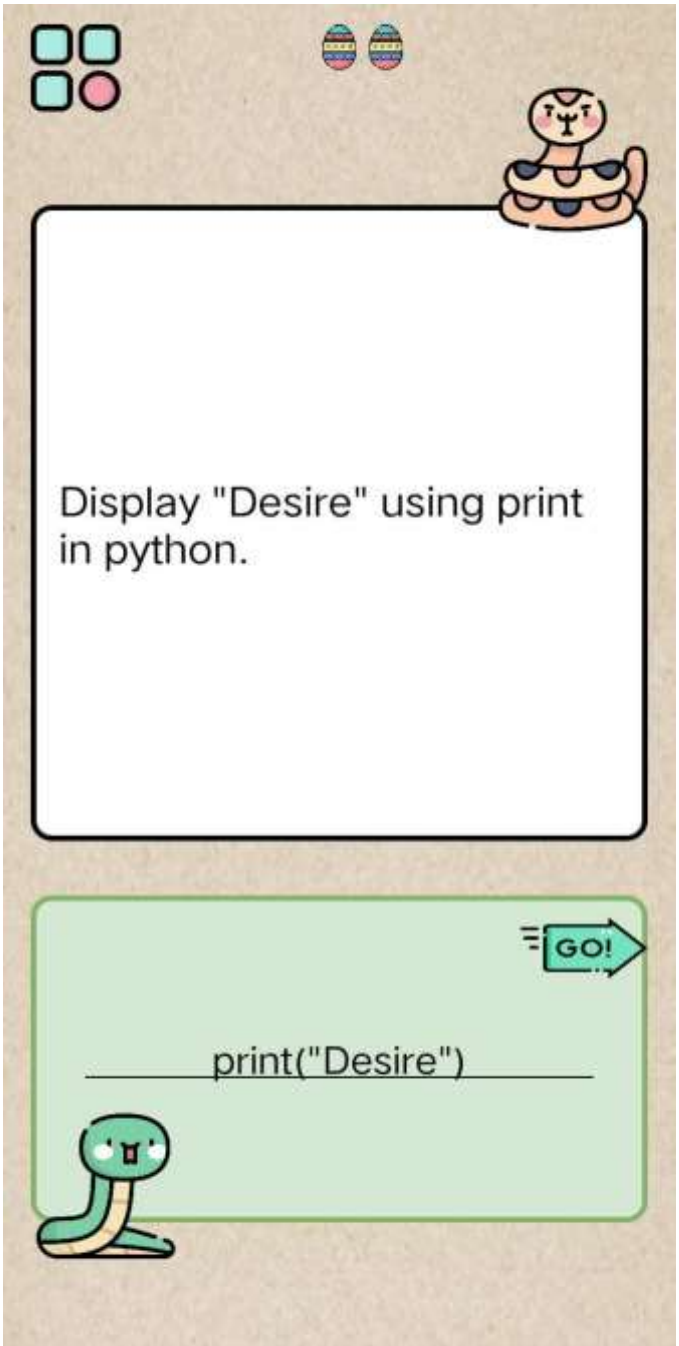



Figure 24. Screenshot of Py-Quest application’s game page - correct answer visualization

Figure 25 represents that if the user gives the wrong answer their dialogue box turns red. The explanation page will then will be displayed to them (refer to Figure 26). The first time the user input a wrong answer, the dialogue box will turn red, if the user then input incorrect answer again, the wrong part of their answer will also turn red.

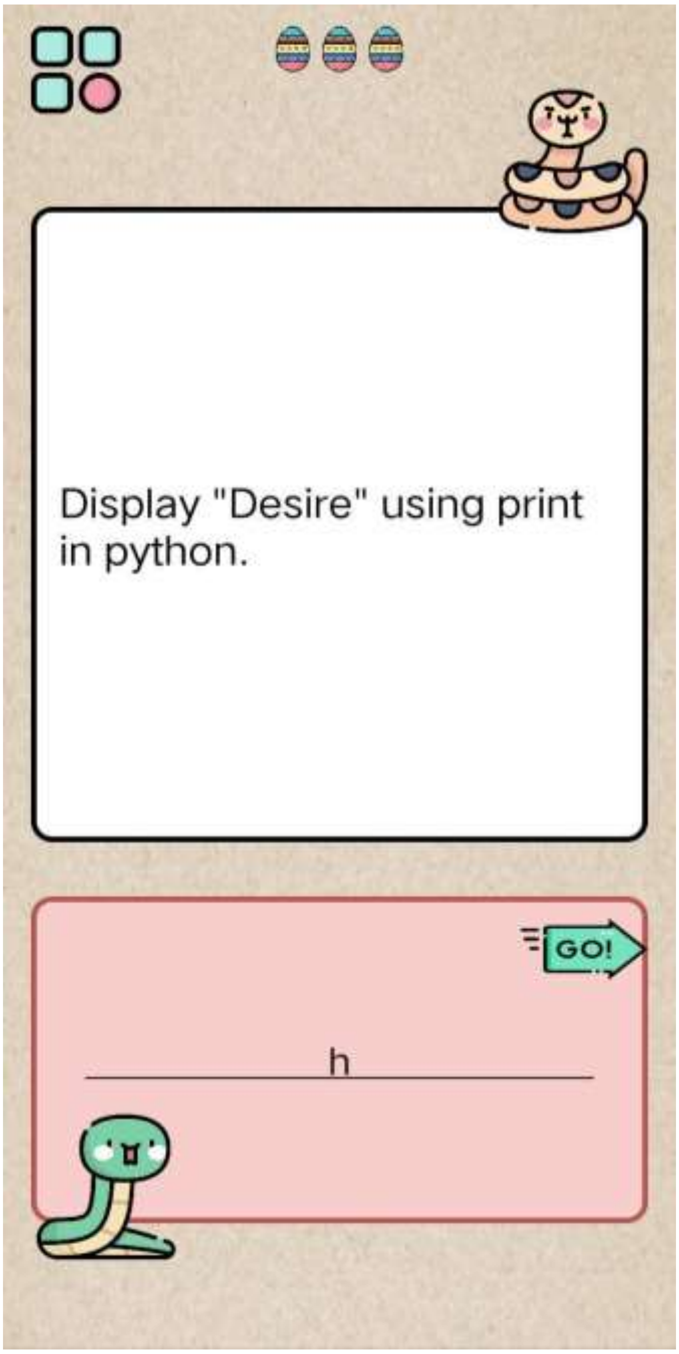



Figure 25. Screenshot of Py-Quest application's Game page incorrect answer visualization

Figure 26 depicts the key to remember page of the application, where the answer to each question is explained. Positioned at the top center part of the page, an icon serves as a visual indicator, written as key to remember, it will serve as hint for the user. Below is the dialogue box that contains the explanation to the question's answer. And on the bottom part is the next button. When pressed by the player, this button directs them to the next question. Below this icon is the dialogue box that contains the explanation to the question's answer. And on the bottom part is the "next" button. When pressed by the player, this button directs them to the next question.

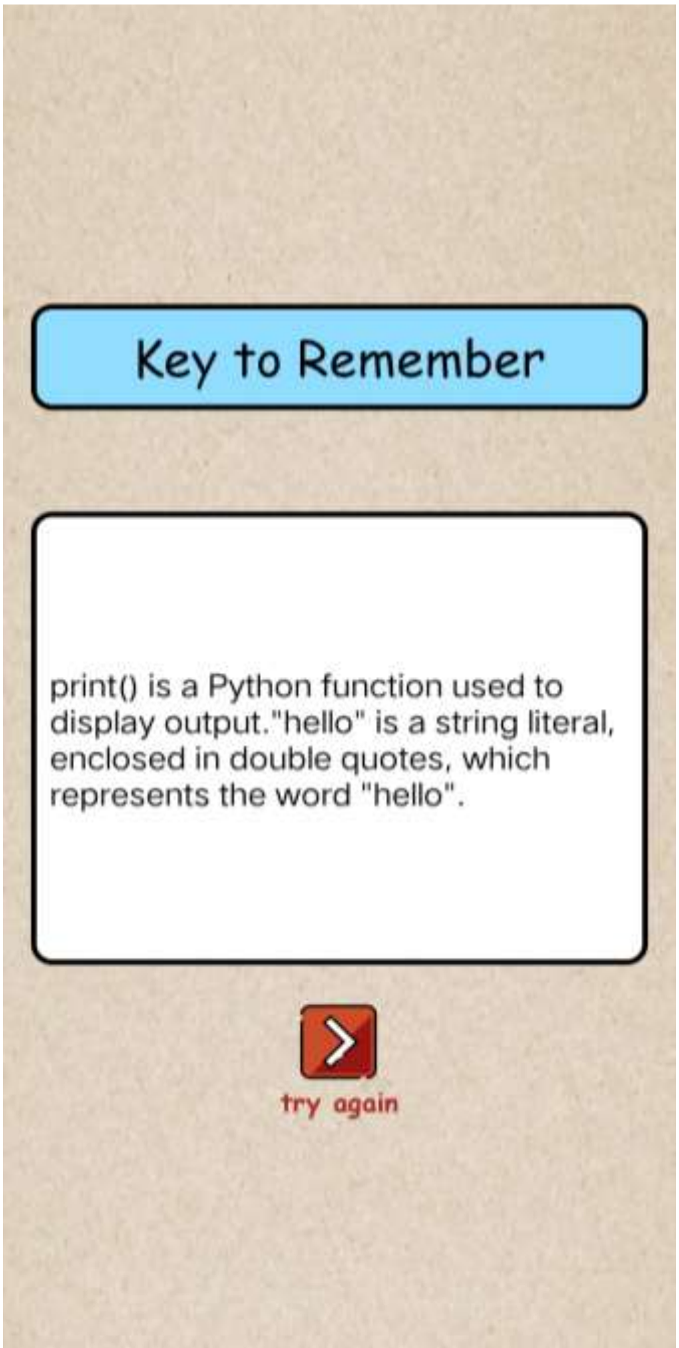



Figure 26. Screenshot of Py-Quest application's game page - explanation page

Figure 27 shows the result page of the application where the score of the user is displayed after answering all the questions in a topic. The number of attempts is recorded along with the score and the date they answered the topic. The user can view their previous results in the scoreboard page.


Figure 27. Screenshot of Py-Quest application's game page - result page

The primary function of the pop-up menu in Figure 28 is to provide players with a means of navigating between different pages within the game. The pop-up will feature four buttons: the back button will take the user back to the previous page; the home button will redirect the user to the homepage. Another one is the settings button that directs the user to the settings page, and lastly is the info button, where the user can view the information page.

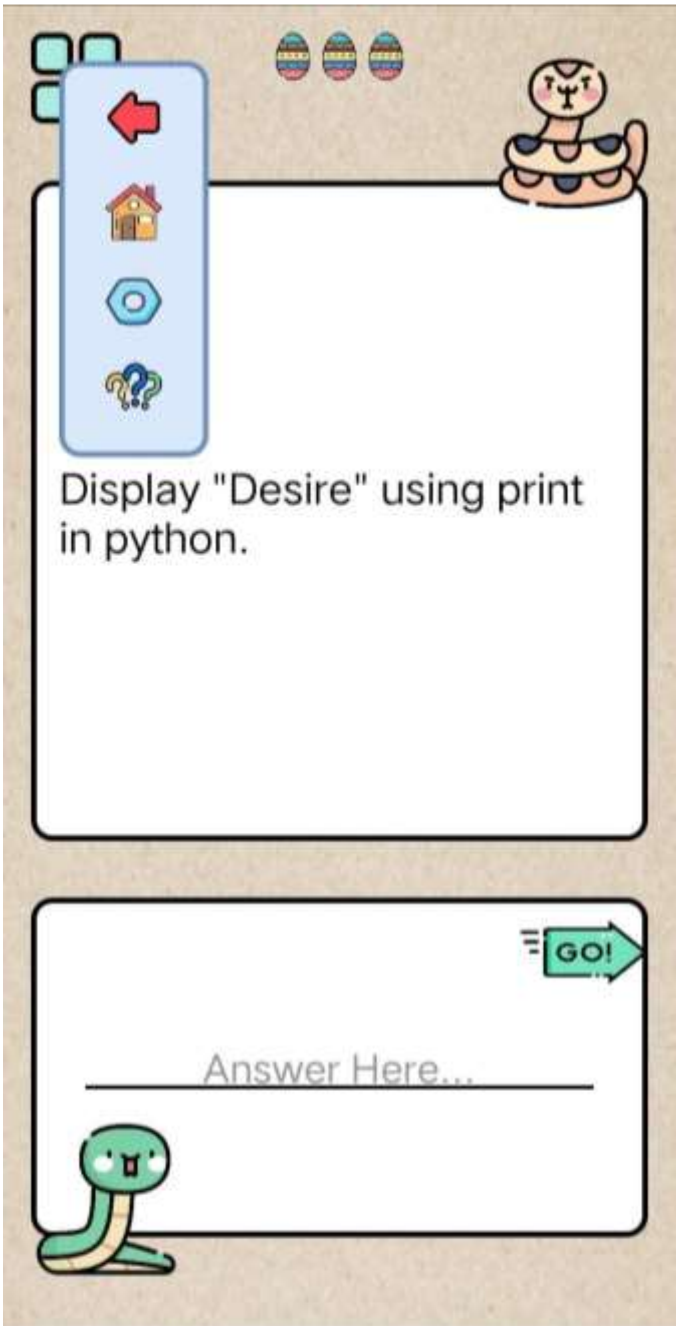



Figure 28. Screenshot of Py-Quest application's pop-up menu page

The scoreboard page in figure 29 can be accessed by pressing the scoreboard button in the homepage of the application. Recorded scores can be viewed by clicking a topic, the attempt number, score and date will be displayed for the user upon clicking a topic. A back button will be provided to navigate back to the previous page.

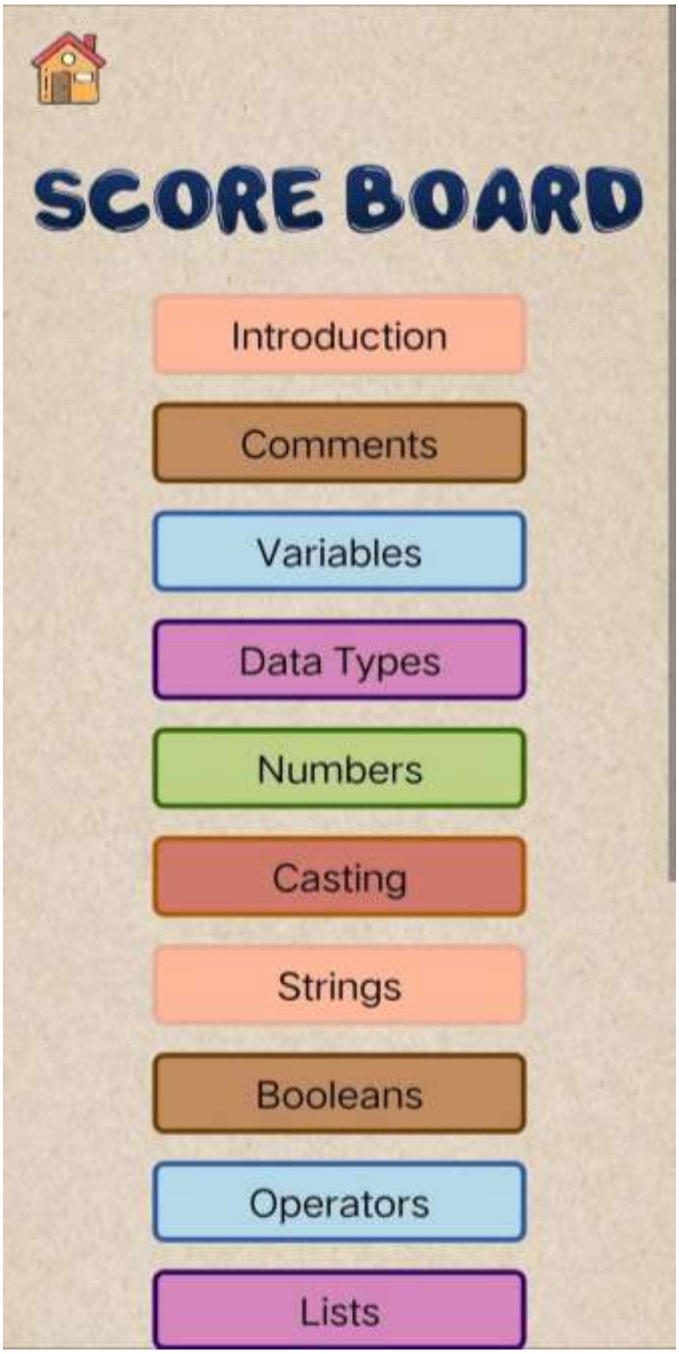



Figure 29. Screenshot of Py-Quest application's scoreboard page

Figure 30 shows the screenshot of the Avatar Page. It showcases all the available characters or avatars awarded to the user. Only the default avatar will be initially available, with other avatars remaining locked until the player achieves specific accomplishments, typically through successful completing levels/topics. Additionally, a back button will be placed in the top-left corner of the screen, upon pressing the button, the user will be directed to the home page.

Figure 30. Screenshot of Py-Quest application’s avatar skin page

Figure 31 illustrates the information page; the user can access this page by pressing the info button in the home page. This page provides a comprehensive overview of the game, including its introduction and mechanics. It is designed to serve as a one-stop shop for players who want to learn more about the game or refresh their memory on the key concepts. The back button will also be placed on the top-left corner of the screen, upon pressing the button the user will be directed to the home page

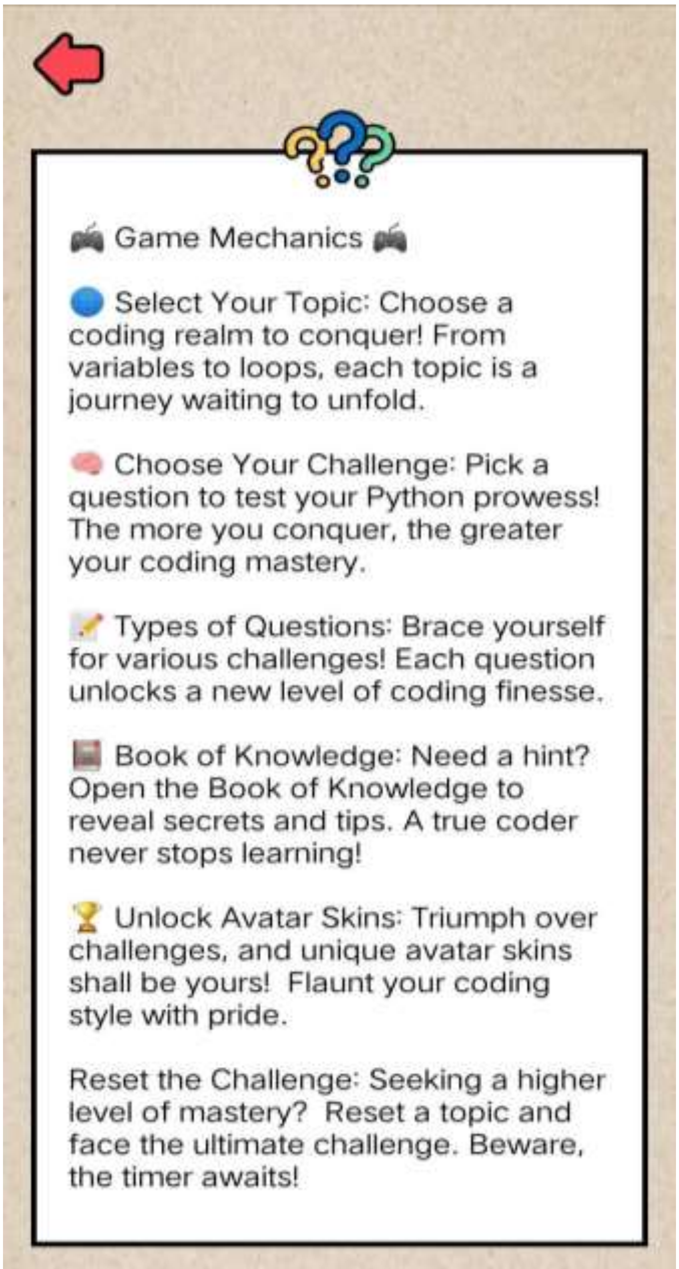



Figure 31. Screenshot of Py-Quest application's information page

Figure 32 shows the settings page. This page lets users manage their sound experience by turning on or off background music and sound effects. It's useful when users are in a quiet place and want to avoid bothering others, or when they need to focus on a task without any distractions.

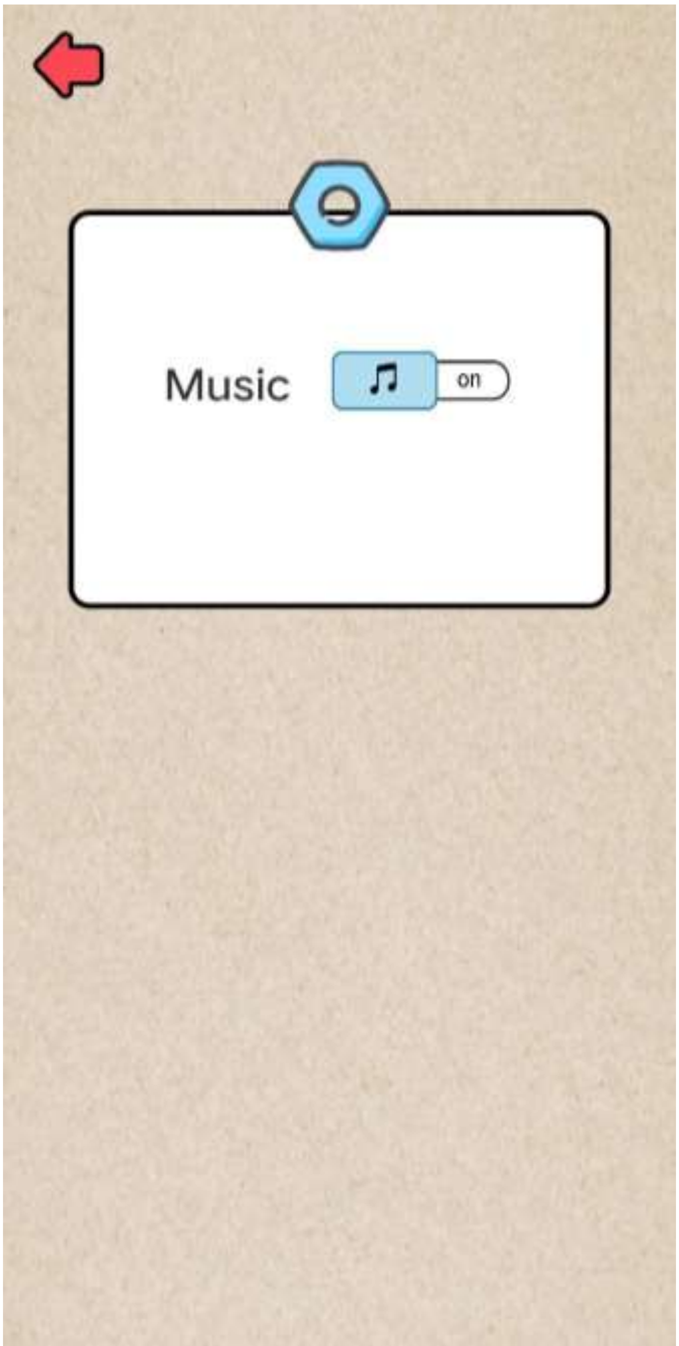



Figure 32. Screenshot of Py-Quest application’s settings page

Figure 33 shows the topic completed page notification. Upon the player's successful completion of all questions within a specific topic or level, they will be greeted with a dedicated page. Positioned at the center of the screen, the player's designated character will make an appearance, accompanied by a message encased within a ribbon, which reads topic completed.


Figure 33. Screenshot of Py-Quest application's topic completed page

Figure 34 shows the home page upon topic completion This page indicates that the user has completed all the questions within a particular topic. As a result, the next topic will be unlocked, allowing them to answer the new set of questions and further enhance their understanding of this new topic.

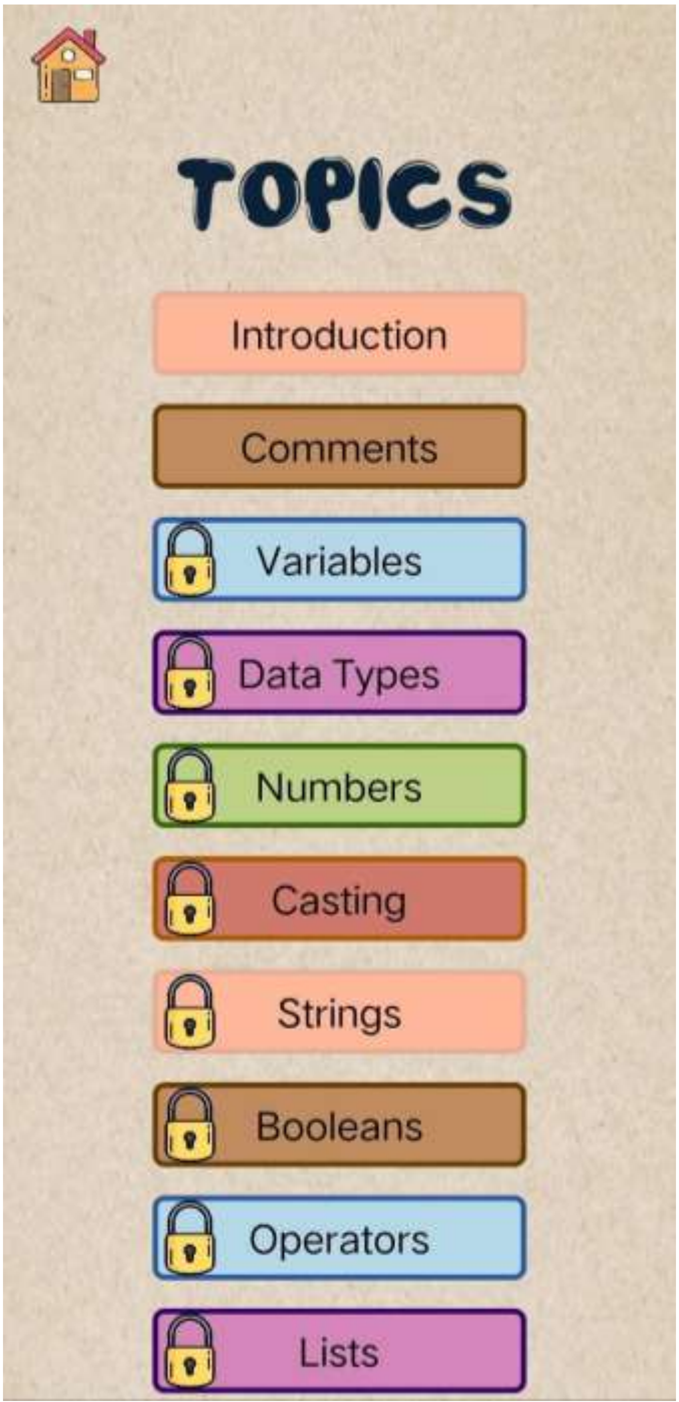



Figure 34. Screenshot of Py-Quest application’s home page (upon topic completion)

**System Testing**

The Android based game application called Py Quest was tested according to its functionality, usability and compatibility to make sure that the application is working properly.

**Functionality Testing.** The functionality testing of the application involved ten test cases to verify different features. The tests confirmed that users could select topics and quizzes, and that audio playback matched the written lessons accurately. The scoreboard reliably displayed quiz scores, and users were notified when it was their last chance to answer a question. All buttons functioned correctly, user lives decreased with each wrong answer, and the next topic only unlocked after passing the current quiz. Additionally, users could enter text into text fields and change avatars as intended, and completing random topics unlocked associated avatars. The application passed all test cases, demonstrating successful implementation of all specified usability requirements (see Appendix Table 4).

**Compatibility Testing.** Appendix Table 5 presents a summary of tests conducted to verify the application's functionality across different versions of Android. The objective was to ensure that all features of the application operated correctly. Verification tests were performed on various Android versions, including Android 13, 12, 11, 10, 9, and 8. The results indicated that the application functions perfectly on all the tested devices. All tests received a "Passed" remark, confirming that the application performed as expected without any issues across these Android versions.

**Usability Testing.** Usability testing involved three key areas. Firstly, it ensured that all buttons and icons in the application were correctly aligned with their intended functions, achieving a positive assessment. This ensures users can easily comprehend the purpose of each button. The second test focused on verifying if every lesson within each topic included an audio button, meeting specified requirements. This was confirmed successfully. The final test specifically assessed the readability of all fonts used in the application, with the application passing this evaluation. Overall, the results of the usability testing were positive, as all test cases were successfully completed. This indicates that the educational Android application maintains a high level of user-friendliness. The result of the usability testing is shown in Appendix Table 6.

**System Evaluation**

Table 6 displays the mean and standard deviation for each question regarding functional suitability. The overall average depicted in the table suggests that the system's performance in terms of functional suitability is Very Good with mean equivalent to 4.60 and standard deviation of 0.60. Functional completeness is rated as Excellent with 4.80 mean and 0.45 standard deviation, while functional correctness and functional appropriateness are both assessed as Very Good, with means of 4.50 (SD = 0.63) and 4.49 (SD = 0.70) respectively.

Table 6. Results of the evaluation of non-technical participants for the functional suitability of the system

| FUNCTIONAL SUITABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Functional completeness (Degree to which the set of functions covers all the specified tasks and user objectives.) | 4.80 | 0.45 | Excellent |
| Functional correctness (Degree to which a product or system provides the correct results with the needed degree of precision.) | 4.50 | 0.63 | Very Good |
| Functional appropriateness (Degree to which the functions facilitate the accomplishment of specified tasks and objectives.) | 4.49 | 0.70 | Very Good |
| **Average** | **4.60** | **0.60** | **Very Good** |

Table 7 provides a detailed breakdown of performance efficiency including both the mean and standard deviation for each indicator. The system's performance efficiency shows a very positive rating, achieving an average mean of 4.33 and standard deviation of 0.64, which interprets as “Very Good” on the evaluation scale.

Table 7. Results of the evaluation of non-technical participants for the performance efficiency of the system

| PERFORMANCE EFFICIENCY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Time behavior (Degree to which the response and processing times and throughput rates of a product or system, when performing its functions, meet requirements.) | 4.42 | 0.62 | Very Good |
| Resource utilization (Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.) | 4.31 | 0.60 | Very Good |
| Capacity (Degree to which the maximum limits of a product or system parameter meets requirements.) | 4.25 | 0.71 | Very Good |
| **Average** | **4.33** | **0.64** | **Very Good** |

Table 8 shows the mean and standard deviation for each indicator under compatibility. The system's compatibility received an average mean of 4.33 with a “Very Good” interpretation and a standard deviation of 0.72, which means users have a consistent experience in terms of compatibility with the software.

Table 8. Results of the evaluation of non-technical participants for the compatibility of the system

| COMPATIBILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Co-existence (Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product. ) | 4.35 | 0.70 | Very Good |

Table 8. Continued

| COMPATIBILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Interoperability (Degree to which two or more systems, products or components can exchange information that has been exchanged.) | 4.30 | 0.73 | Very Good |
| **Average** | **4.33** | **0.72** | **Very Good** |

Table 9 shows the mean and standard deviation for each indicator under usability. The system's usability received an average mean of 4.44 with a Very Good interpretation and a standard deviation of 0.64, which means users can easily understand and access the software's design.

Table 9. Results of the evaluation of non-technical participants for the usability of the system

| USABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Appropriateness recognizability (Degree to which users can recognize whether a product or system is appropriate for their needs.) | 4.51 | 0.62 | Excellent |
| Learnability (Degree to which a system can be used by specified users to achieve specified goals of learning to use the system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.) | 4.48 | 0.54 | Very Good |
| Operability (Degree to which a product or system has attributes that make it easy to operate and control.) | 4.28 | 0.75 | Very Good |

Table 9. Continued

| USABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| User error protection (Degree to which a system protects users against making errors.) | 4.39 | 0.63 | Very Good |
| User interface aesthetics (Degree to which a user interface enables pleasing and satisfying interaction for the user.) | 4.56 | 0.62 | Excellent |
| Accessibility (Degree to which a product or system can be used by people with the widest range of capabilities to achieve a specified goal in a specified context of use.) | 4.41 | 0.69 | Very Good |
| **Average** | **4.44** | **0.64** | **Very Good** |

Table 10 presents the mean and standard deviation for each indicator under portability. The system's portability received an average mean of 4.30 with a "Very Good" interpretation and a standard deviation of 0.68. This suggests that the application can be used across different android versions.

Table 10. Results of the evaluation of non-technical participants for the portability of the system

| PORTABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Adaptability (Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.) | 4.46 | 0.62 | Very Good |
| Installability (Degree of effectiveness and efficiency with which a product | 4.40 | 0.62 | Very Good |

Table 10. Continued

| PORTABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| or system can be successfully installed and/or uninstalled in a specified environment.) | | | |
| Replaceability (Degree to which a product can replace another specified software product for the same purpose in the same environment.) | 4.03 | 0.81 | Very Good |
| **Average** | **4.30** | **0.68** | **Very Good** |

Table 11 provides a detailed overview of functional suitability metrics, detailing mean and standard deviation values for each evaluated question. Overall, the system demonstrates excellent performance in functional suitability, achieving a mean score of 4.77 with a standard deviation of 0.42. Specifically, functional completeness, correctness, and appropriateness are all rated as excellent, each with mean scores of 4.80 and standard deviations of 0.40.

Table 11. Results of the evaluation of technical participants for the functional suitability of the system

| FUNCTIONAL SUITABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Functional completeness (Degree to which the set of functions covers all the specified tasks and user objectives.) | 4.7 | 0.46 | Excellent |
| Functional correctness (Degree to which a product or system provides the correct results with the needed degree of precision.) | 4.8 | 0.4 | Excellent |

Table 11. Continued

| FUNCTIONAL SUITABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Functional appropriateness (Degree to which the functions facilitate the accomplishment of specified tasks and objectives.) | 4.8 | 0.4 | Excellent |
| **Average** | **4.77** | **0.42** | **Excellent** |

Table 12 shows the system's strong performance efficiency, with an overall "Very Good" rating and an average score of 4.37. Key metrics like time behavior, resource utilization, and capacity are also rated "Very Good," with mean scores of 4.3, 4.5, and 4.3, respectively. The standard deviations, ranging from 0.64 to 0.78, indicate consistent performance across these areas.

Table 12. Results of the evaluation of technical participants for the performance efficiency of the system

| PERFORMANCE EFFICIENCY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Time behavior (Degree to which the response and processing times and throughput rates of a product or system, when performing its functions, meet requirements.) | 4.3 | 0.64 | Very Good |
| Resource utilization (Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.) | 4.5 | 0.67 | Very Good |
| Capacity (Degree to which the maximum limits of a product or system parameter meets requirements.) | 4.3 | 0.78 | Very Good |
| **Average** | **4.37** | **0.70** | **Very Good** |

Table 13 details the mean and standard deviation values for each aspect related to compatibility. Overall, the system's compatibility is evaluated as excellent, achieving a mean score of 4.65 with a standard deviation of 0.56. Specifically, co-existence and interoperability are rated excellently, with mean scores of 4.60 (SD = 0.66) and 4.70 (SD = 0.46) respectively. These metrics underscore the system's compatibility, indicating effective integration and interaction capabilities across different environments.

Table 13. Results of the evaluation of technical participants for the compatibility of the system

| COMPATIBILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Co-existence (Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product. ) | 4.6 | 0.66 | Excellent |
| Interoperability (Degree to which two or more systems, products or components can exchange information that has been exchanged.) | 4.7 | 0.46 | Excellent |
| **Average** | **4.65** | **0.56** | **Excellent** |

Table 14 summarizes the usability assessment of the system. Overall, the system's usability is highly rated as excellent, with an average score of 4.72 and a standard deviation of 0.48, indicating consistent positive feedback across different usability measures. Specific aspects such as appropriateness recognizability (excellent, mean = 4.90, SD = 0.30), learnability (excellent, mean = 4.90, SD = 0.30), operability (excellent, mean = 4.80, SD = 0.40), user error protection (very good, mean

= 4.30, SD = 1.01), user interface aesthetics (excellent, mean = 4.80, SD = 0.40), and accessibility (excellent, mean = 4.60, SD = 0.49) are individually evaluated.

Table 14. Results of the evaluation of technical participants for the functional suitability of the system

| USABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Appropriateness recognizability (Degree to which users can recognize whether a product or system is appropriate for their needs.) | 4.9 | 0.3 | Excellent |
| Learnability (Degree to which a system can be used by specified users to achieve specific goals of learning to use the system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.) | 4.9 | 0.3 | Excellent |
| Operability (Degree to which a product or system has attributes that make it easy to operate and control.) | 4.8 | 0.4 | Excellent |
| User error protection (Degree to which a system protects users against making errors.) | 4.3 | 1.01 | Very Good |
| User interface aesthetics (Degree to which a user interface enables pleasing and satisfying interaction for the user.) | 4.8 | 0.4 | Excellent |
| Accessibility (Degree to which a product or system can be used by people with the widest range of capabilities to achieve a specified goal in a specified context of use.) | 4.6 | 0.49 | Excellent |
| **Average** | **4.72** | **0.48** | **Excellent** |

Table 15 presents the mean and standard deviation results for the system's reliability. Overall, the system demonstrates excellent reliability, with a mean score of 4.48 and a standard deviation of 0.59. Specific aspects like maturity and availability are graded as excellent, achieving means of 4.70 and 4.60 respectively, with moderate standard deviations. Fault tolerance is rated very good at 4.20 with a standard deviation of 0.75, while recoverability also scores very good at 4.40 with a standard deviation of 0.49.

Table 15. Results of the evaluation of technical participants for the reliability of the system

| RELIABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Maturity (Degree to which a system, product or component meets needs for reliability under normal operation.) | 4.7 | 0.64 | Excellent |
| Availability (Degree to which a system, product or component is operational and accessible when required for use.) | 4.6 | 0.49 | Excellent |
| Fault tolerance (Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.) | 4.2 | 0.75 | Very Good |
| Recoverability (Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.) | 4.4 | 0.49 | Very good |
| **Average** | **4.48** | **0.59** | **Very Good** |

The summary of results from Table 16 indicates that the system's maintainability is highly rated overall, achieving a mean score of 4.48 with a standard deviation of 0.59. Specific aspects of maintainability such as maturity and availability are rated excellent, with mean scores of 4.70 and 4.60 respectively, and modest standard deviations. Fault tolerance and recoverability are also strong, rated very good with mean scores of 4.20 and 4.40, each having varying but acceptable standard deviations.

Table 16. Results of the evaluation of technical participants for the maintainability of the system

| MAINTAINABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Modularity (Degree to which a system is composed of discrete components.) | 4.5 | 0.67 | Very Good |
| Reusability (Degree to which an asset can be used in more than one system, or building other assets.) | 4.6 | 0.49 | Excellent |
| Analyzability (Degree of effectiveness with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts.) | 4.5 | 0.50 | Very Good |
| Modifiability (Degree to which a system can be effectively and efficiently modified without degrading existing product quality.) | 4.2 | 0.66 | Very Good |
| Testability (Degree of effectiveness with which test criteria can be established for a system.) | 4.7 | 0.46 | Excellent |
| **Average** | **4.5** | **0.55** | **Very Good** |

The mean and standard deviation for portability are summarized in Table 17. The system received an excellent rating for its portability, with a mean score of 4.73 and a standard deviation of 0.43 according to the overall results. Specifically, adaptability is marked as excellent, with a mean of 4.60 and a standard deviation of 0.49. Installability is also graded as excellent, with a mean of 4.80 and a standard deviation of 0.40. Furthermore, the replaceability has been ranked as exceptional, with a mean score of 4.80 and a standard deviation of 0.40.

Table 17. Results of the evaluation of technical participants for the portability of the system

| PORTABILITY | MEAN | STANDARD DEVIATION | INTERPRETATION |
|---|---|---|---|
| Adaptability (Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.) | 4.6 | 0.49 | Excellent |
| Installability (Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.) | 4.8 | 0.4 | Excellent |
| Replaceability (Degree to which a product can replace another specified software product for the same purpose in the same environment.) | 4.8 | 0.4 | Excellent |
| **Average** | **4.73** | **0.43** | **Excellent** |

# SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This section provides a comprehensive overview of the key findings from the study, draws conclusions based on the results, and offers recommendations for future improvements and applications of the PY-Quest game application

## Summary

The capstone project, titled "Py-Quest: An Android-Based Game Application for Python Programming Fundamentals," was executed at Cavite State University from March 2022 to June 2024. The project's primary aim was to develop a mobile game designed to teach Python programming fundamentals to ICT students. This initiative addressed challenges related to the complexity of learning Python and the availability of offline resources, as identified through data collection and surveys. Despite many students being unfamiliar with Python basics, their strong interest in learning the language was evident. Motivated by this enthusiasm, the researchers embarked on creating Py-Quest, an accessible and engaging game built using Java and Android Studio. The game features two main modules: the book of knowledge, which offers lessons on Python fundamentals, and the gameplay module, which includes assessment features.

The methodology included requirement analysis through surveys approved by local leaders and face-to-face surveys to identify user difficulties. The system featured Python basics, various question types, a book of knowledge, audio narration, customizable settings, avatar rewards, scoreboard tracking for attempt management. Development includes using draw.io for prototypes and PHP, Java, SQLite, XML, and Android Studio for implementation, and followed ISO 25010 for quality assurance testing ensured functionality across Android versions 8 to 13, usability, and thorough evaluation by technical evaluators and users. The implementation plan involved demonstrations, evaluations, and deployment strategies with local stakeholders for effective integration and user acceptance.

After a series of tests, including functionality, usability, and compatibility assessments, the application was thoroughly evaluated. These tests covered various aspects such as user interaction, audio accuracy, scoreboard functionality, error notifications, and button operations. Usability testing focused on interface navigation and text readability, while compatibility testing ensured proper operation across Android versions 8 through 13. All tests were successfully passed, confirming that the application achieved the desired level of user-friendliness.

The system was evaluated using the ISO 25010 standard by technical instructors and out-of-school youths, achieving high quality ratings. The non-technical assessment showed impressive results in functional compatibility (4.60), performance efficiency (4.33), compatibility (4.33), usability (4.44), and portability (4.30). Technical evaluation highlighted excellent scores in functional suitability (4.64), performance efficiency (4.37), compatibility (4.65), usability (4.72), reliability (4.48), maintainability (4.50), and portability (4.73). Overall, the system demonstrated robust performance and quality, with most components rated excellent or very good.

**Conclusion**

The researchers determined that all goals were achieved as the evaluation results indicated that the game-based application successfully met the necessary requirements and accomplished the study's objectives. The statistical analysis employed mean and standard deviation to assess various metrics. Participants rated the developed platform as “Very Good” in terms of functional suitability, efficiency, compatibility, usability, reliability, security, maintainability, and portability. These high ratings underscore the application's robust design and user-centric approach, ensuring a reliable and efficient learning tool for Python programming fundamentals.

Challenges identified during the research included the difficulties faced by the students in understanding lessons, memorizing syntax, and dealing with distractions which hinder their grasp of programming concepts. Traditional teaching methods and a lack of engaging practice exercises contribute to this problem, leading to poor understanding and retention of essential programming knowledge. Several challenges students face in learning programming.

The creation of the Android-based game application for Python programming fundamentals effectively resolved the previously mentioned issues. The application includes a range of features designed to enhance learning and engagement: topics, various types of questions, a book of knowledge, settings, avatar skins, a scoreboard, lesson audio, and the ability to be accessed anytime and anywhere. These features collectively provide a comprehensive learning environment tailored to the needs of the users.

The application was built using several tools, including Java, PHP, MySQL, Android Studio, XML, and HTML. This technical foundation ensured a strong and flexible solution that could meet the various needs of its users. The development process involved various testing and evaluation, with the system passing all technical and non-technical assessments. These evaluations confirmed the application's reliability and effectiveness in delivering educational content.

The researchers concluded that the identified problems from the data gathering have been successfully addressed by the game-based application. The features incorporated into the application directly responded to the needs of the target users, providing an effective tool for learning Python programming fundamentals. The success of the application in these tests demonstrates its potential as a valuable educational resource, particularly for those with limited access to traditional learning environments.

With its proven capabilities, the Android-based game application shows promise as a valuable educational tool for anyone dedicated to learning Python fundamentals. It offers a flexible, accessible, and engaging learning platform that can be effectively implemented in various academic settings. This innovative approach to education not only bridges the gap for ICT students but also provides a scalable solution that can be adapted to future educational initiatives, ensuring broader access to quality learning resources.

**Recommendations**

From the end result, the previously mentioned conclusion, and the suggestions collected through the software assessment, the following recommendations are created:

1. Cross-platform Compatibility. The mobile app should have the capability to run on various operating systems and devices such as Android smartphones, tablets, and possibly even iOS devices or desktop computers. This feature allows users to access and enjoy educational content regardless of the device they use, thereby increasing accessibility and user engagement.

2. Enhanced Lesson Content and Strategy. The mobile app should improve lesson content and strategies to enhance the learning experience. This includes interactive lessons and practice sessions designed to make learning more engaging and effective.

3. Offline Social Interaction. The game should include a feature that allows users to interact with each other even when they are not connected to the internet.

# REFERENCES


Abou-Samra, O. (2020). *Mobile Technology*. PrepareCenter. Retrieved from https://preparecenter.org/topic/mobile-technology/

Bagunu, R., Buan, K. (2017). *Silicon Valley: 2D Android Based Mobile Adventure Game for Computer Fundamentals*. BS Information Technology, Cavite State University, Indang Cavite.

BasuMallick, C. (2024, March 19). Android OS: History, Features, Versions, and Benefits | Spiceworks. Spiceworks Inc. https://www.spiceworks.com/tech/tech-general/articles/android-os/#:~:text=The%20key%20advantages%20of%20Android,flexibility%20within%20the%20Android%20ecosystem.

Bonza, J., Lopio, M. (2019). *Binary Wars: A mobile Game Development About Binary and Decimal Conversion.* BS Information Technology, Cavite State University, Indang Cavite.

Cambridge.org. (2023). Quest. Retrieved from https://dictionary.cambridge.org/us/dictionary/english/ques

Cheung, S.Y., & Yin Ng, Kai (2021). *Application of the Educational Game to Enhance Student Learning.Frontiers in Education*. Retrieved from https://www.frontiersin.org/articles/10.3389/feduc.2021.623793/full

Chimpz, R. (2019). *The importance of Educational Games. Medium*. Retrieved from https://medium.com/@redchimpzindia/the-importance-of educationalgames-c6e948fcd6b6

Corbo, A. (2023). What is Android Development? Built In. Retrieved from https://builtin.com/software-engineering-perspectives/android-development?fbclid=IwAR2gXnhznP3NVtjA2a12_nxjZ4b_snd3mxX5Qrchl0A0Q3WUx7ZcOgoP37A

Dadheech, A. (2021*). The importance of game-based learning in modern education. The Knowledge Review*. Retrieved from https://theknowledgereview.com/ importance-game-based-learning-modern- education/

Dauz, D., Dela Cruz, C., Pacaña, J. (2019). *Progdungeon: An Android-Based Python Coding Game*. BS Computer Science, Cavite State University, Indang Cavite.

GeeksforGeeks. (2024, January 31). What is Android Studio? Retrieved from https://www.geeksforgeeks.org/overview-of-android-studio/

GeeksforGeeks. (2024, June 6). What is Software Testing? Retrieved from https://www.geeksforgeeks.org/software-testing-basics/

GeeksforGeeks. (2024, August 23). Introduction to Java Retrieved from https://www.geeksforgeeks.org/introduction-to-java/

IGI Global. (2018). What is Mobile Games. Retrieved from https://www.igi-global.com/dictionary/mobile-game-based-learning/47743

Jennifer Smith (2022). What is Photoshop? Retrieved from https://www.agitraining.com/adobe/photoshop/classes/what-is-photoshop

Koupritzioti, D., Xinogalos, S. (2019). PyDiophantus maze game*: Play it to learn mathematics or implement it to learn game programming in Python*. Retrieved from https://link.springer.com/article/10.1007/s10639019100871?fbclid=IwAR287lu4JPs0mRpxrbal0SfXN0MXF_MWg2wtIGxipt00qhcKHhXjHz75xU#citeas

Laura Paraschiv (2023). *Draw.io online – a step-by-step guide for users*. Retrieved from https://fotc.com/blog/drawioonlineguide/?fbclid=IwAR23ATzjq8gj84TorNBrMuoTg7M3smGdT-5g-eLBQl2AXjjbNezdawEL0EM

Margaret Rouse (2017). *Adobe Photoshop*. Retrieved from https://www.techopedia.com/definition/32364/adobephotoshop?fbclid=IwAR1wvZ9YBanh5zUpFOBthH_Mp-oGNjBGzn7hZYOS0Mt9D-JzJgm4ji-Hgxw

Mendiratta, P. (2023). What is an Android Based Applications? 10 Best Types of Android Applications. Innovative Web And Mobile App Development Technology Insights. Retrieved from https://pairroxz.com/blog/android-based-applications/#:~:text=An%20Android%20based%20application%20designed,and%20create%20Android%20based%20applications.

Merrill, M. (2023). *What Is Rapid Prototyping In UX? Step-by-Step Guide & Wise Tips. Devsquad.* https://devsquad.com/blog/what-is-rapid-prototyping-and-why-is-it-used-in-development?

Newzoo. (2023). Global Games Market Report. Retrieved from https://newzoo.com/games-market-reports-forecasts

Programming language. (2021). Retrieved fromhttps://www.computerhope.com/jargon/p/programming-language.htm

Rehman, J. (2021). Advantages and disadvantages of android operating system - IT Release. IT Release. Retrieved from https://www.itrelease.com/2019/09/advantages-and-disadvantages-of-android-operating-system/

Statology. (2023). Slovin's formula: Definition + examples. Retrieved September 3, 2024. Retrived from https://www.statology.org/slovins-formula/

Study.com. (2023). Educational games: Definition, history & benefits. Retrieved from https://study.com/academy/lesson/educational-games-definition-history-benefits.html

Tamosevicius, R. (2022). Why is Game-Based learning important? eLearning Industry. Retrieved from https://www.google.com/amp/s/elearningindustry.com/why-is-game-based-learning-important/amp

Vijay Kanade (2023). What Is XML (Extensible Markup Language)? Meaning, Elements, and Benefits. Retrieved from https://www.spiceworks.com/tech/tech-general/articles/what-is-xml/amp/?fbclid=IwAR3oHVBMK5FSGfR2pgvz7C9UuK00uEQWyLwUxvfWYX88WwZFbL81dcNU9E

Web Design Inspiration. (2021). Flaticon by Freepik review. Web Design Inspiration. https://www.webdesign-inspiration.com/blog/flaticon-by-freepik-review

# APPENDICES

# APPENDIX TABLES

Appendix Table 1. Functionality testing of the Py-Quest

| Test Case Summary | Test Data | Expected Result |
|---|---|---|
| Test if the user can select a list of topics and Quiz. | Selection List | The user is able to select a list for lessons and quiz. |
| Test if the audio play, and is accurate as what is written in the lesson | Lesson Audio | If the user clicks audio button, the audio will play and provide an accurate wording |
| Test if the scoreboard page will display accurate scoring after completing the quiz. | Score | The result of the quiz is reliable and accurate in the score displayed in the Scoreboard. |
| Verify if the user will be notified when it is the final chance to answer the given question. | Notification | User was notified if it was the last chance to answer the quiz correctly. |
| Test if all the buttons' function is working. | Buttons | The user was able to use the buttons of its function. |
| Test if the lives get lessened with each mistake. | Attempt | Lives of the user will lessen every wrong answer they made. |
| Test if the next topic will unlock after the user meet the passing score of the selected topic | Unlock topics | Users can only proceed to the next topic if they passed the quiz in the current topics. |
| Verify if the user is able to enter text into the text field. | Text Field | User is able to enter a text into a text field. |
| Test if the user is able to change avatars. | Change Avatar | User is able to pick any of the unlocked avatar. |
| Test if completing a random topic unlocks the avatar associated with it. | Unlock Avatars | User is given an avatar by completing random topics |

Appendix Table 2. Compatibility testing of the Py-Quest

| Version | Expected Result |
|---|---|
| Android 13 | All functions of the application are working properly. |
| Android 12 | All functions of the application are working properly. |
| Android 11 | All functions of the application are working properly. |
| Android 10 | All functions of the application are working properly. |
| Android 9 | All functions of the application are working properly. |
| Android 8 | All functions of the application are working properly. |

Appendix Table 3. Usability testing of the Py-Quest

| Test Case Summary | Test Data | Expected Result |
|---|---|---|
| Test if buttons and icons are displayed based on their intended functions. | Buttons and Icons | All buttons and icons are displayed based on their intended functions. |
| To test if each lesson within a topic has an audio button. | Audio Button | Each lesson within a topic has an audio button. |
| Verify if all fonts are readable. | Font | All fonts are readable. |

Appendix Table 4. Results of Functionality testing of the Py-Quest

| Test Case Summary | Test Data | Expected Result | Actual Result | Remarks |
|---|---|---|---|---|
| Test if the user can select a list of topics and Quiz. | Selection List | The user is able to select a list for lessons and quiz. | The user was able to select a lesson and start a quiz. | Passed |
| Test if the audio play, and is accurate as what is written in the lesson | Lesson Audio | If the user clicks audio button, the audio will play and provide an accurate wording | The lesson audio plays, and is accurate as to what is written in the lesson. | Passed |
| Test if the scoreboard page will display accurate scoring after completing the quiz. | Score | The result of the quiz is reliable and accurate in the score displayed in the Scoreboard. | The scoreboard displays accurate scores and attempted data of the user. | Passed |
| Verify if the user will be notified when it is the final chance to answer the given question. | Notification | User was notified if it was the last chance to answer the quiz correctly. | The user is notified after giving two incorrect responses. | Passed |
| Test if all the buttons' function is working. | Buttons | The user was able to use the buttons of its function. | All intended function of the buttons is perfectly working | Passed |
| Test if the lives get lessened with each mistake. | Attempt | Lives of the user will lessen every wrong answer they made. | Lives get lessened each time the user answers incorrectly. | Passed |
| Test if the next topic will unlock after the user meet | Unlock topics | Users can only proceed to the next topic if they | The next topic is unlocked after the user finished | Passed |

| | | | | |
|---|---|---|---|---|
| the passing score of the selected topic | | passed the quiz in the current topics. | answering the quiz and meeting the passing score. | |
| Verify if the user is able to enter text into the text field. | Text Field | User is able to enter a text into a text field. | User was able to enter text to all the text fields in each question. | Passed |
| Test if the user is able to change avatars. | Change Avatar | User is able to pick any of the unlocked avatar. | The avatar changed to exactly what the user selected. | Passed |
| Test if completing a random topic unlocks the avatar associated with it. | Unlock Avatars | User is given an avatar by completing random topics | The user received an avatar after answering random topics. | Passed |

Appendix Table 5. Results of Compatibility testing of the Py-Quest

| Version | Expected Result | Actual Result | Remarks |
|---|---|---|---|
| Android 13 | All functions of the application are working properly. | All functions of the application are working properly. | Passed |
| Android 12 | All functions of the application are working properly. | All functions of the application are working properly. | Passed |
| Android 11 | All functions of the application are working properly. | All functions of the application are working properly. | Passed |
| Android 10 | All functions of the application are working properly. | All functions of the application are working properly. | Passed |
| Android 9 | All functions of the application are working properly. | All functions of the application are working properly. | Passed |

| | | | |
|---|---|---|---|
| Android 8 | All functions of the application are working properly. | All functions of the application are working properly. | Passed |

Appendix Table 6. Results of Usability testing of the Py-Quest

| Test Case Summary | Test Data | Expected Result | Actual Result | Remarks |
|---|---|---|---|---|
| Test if buttons and icons are displayed based on their intended functions. | Buttons and Icons | All buttons and icons are displayed based on their intended functions. | All buttons and icons in the application are displayed based on their intended functions. | Passed |
| To test if each lesson within a topic has an audio button. | Audio Button | Each lesson within a topic has an audio button. | An audio button is available in every lesson. | Passed |
| Verify if all fonts are readable. | Font | All fonts are readable. | All fonts in the application are readable. | Passed |

Appendix Table 7. Profile of the Technical Respondents

| NO. | RESPONDENT'S NAME | POSITION |
|---|---|---|
| 1. | Prof. Charles Benedick Sotto | Instructor |
| 2. | Prof. Jc De Leon | Instructor |
| 3. | Prof. Hernz Ramonez | Instructor |
| 4. | Prof. Etai Ramos | Instructor |
| 5. | Prof. Marc Elvin Cerezo | Instructor |
| 6. | Prof. Christian Jed Quejado | Instructor |
| 7. | Prof. Julie Lontoc | Instructor |
| 8. | Prof. Stephen Rocillo | Instructor |
| 9. | Prof. Jose Andrew A. Bunan | Instructor |
| 10. | Prof. Carlo Astilla | Instructor |

Appendix Table 8. Overall Frequency Distribution Table for Non-Technical Evaluation based on Functional Suitability

| **INDICATORS** | **5** | | **4** | | **3** | | **2** | | **1** | | **TOTAL** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** |
| Degree to which the set of functions covers all the specified tasks and user objectives. | 82 | 82 | 16 | 16 | 6 | 6 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which a product or system provides the correct results with the needed degree of precision. | 57 | 57 | 36 | 36 | 21 | 21 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which the functions facilitate the accomplishment of specified tasks and objectives. | 61 | 61 | 27 | 27 | 12 | 12 | 0 | 0 | 0 | 0 | 100 | 100 |

Appendix Table 9. Overall Frequency Distribution Table for Non-Technical Evaluation based on Performance Efficiency

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** |
| Degree to which the response and processing times and throughput rates of a product or system, when performing its functions, meet requirements. | 49 | 49 | 44 | 44 | 7 | 7 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements. | 38 | 38 | 55 | 55 | 7 | 7 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which the maximum limits of a product or system parameter meet requirements. | 41 | 41 | 43 | 43 | 16 | 16 | 0 | 0 | 0 | 0 | 100 | 100 |

Appendix Table 10. Overall Frequency Distribution Table for Non-Technical Evaluation based on Compatibility

| **INDICATORS** | **5** | | **4** | | **3** | | **2** | | **1** | | **TOTAL** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** |
| Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product. | 49 | 49 | 37 | 37 | 14 | 14 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which two or more systems, products or components can exchange information that has been exchanged. | 46 | 46 | 38 | 38 | 16 | 16 | 0 | 0 | 0 | 0 | 100 | 100 |

Appendix Table 11. Overall Frequency Distribution Table for Non-Technical Evaluation based on Usability

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F | % | F | % | F | % | F | % | F | % | F | % |
| Degree to which users can recognize whether a product or system is appropriate for their needs. | 59 | 59 | 33 | 33 | 8 | 8 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which a system can be used by specified users to achieve specified goals of learning to use the system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use. | 51 | 51 | 46 | 46 | 3 | 3 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which a product or system has attributes that make it easy to operate and control. | 46 | 46 | 36 | 36 | 18 | 18 | 0 | 0 | 0 | 0 | 100 | 100 |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Degree to which a system protects users against making errors. | 48 | 48 | 43 | 43 | 9 | 9 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which a user interface enables pleasing and satisfying interaction for the user. | 60 | 60 | 36 | 36 | 4 | 4 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which a product or system can be used by people with the widest range of capabilities to achieve a specific goal in a specified context of use. | 53 | 53 | 35 | 35 | 12 | 12 | 0 | 0 | 0 | 0 | 100 | 100 |

Appendix Table 12. Overall Frequency Distribution Table for Non-Technical Evaluation based on Portability

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F | % | F | % | F | % | F | % | F | % | F | % |
| Degree to which a product or system can effectively and efficiently be adapted for different or evolvinghardware, software or other operational or usage environments. | 53 | 53 | 40 | 40 | 7 | 7 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment. | 47 | 47 | 46 | 46 | 7 | 7 | 0 | 0 | 0 | 0 | 100 | 100 |
| Degree to which a product can replace another specified software product for the same purpose in the same environment. | 34 | 34 | 35 | 35 | 31 | 31 | 0 | 0 | 0 | 0 | 100 | 100 |

Appendix Table 13. Overall Frequency Distribution Table for Technical Evaluation based on Functional Suitability

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F | % | F | % | F | % | F | % | F | % | F | % |
| Degree to which the set of functions covers all the specified tasks and user objectives. | 7 | 70 | 3 | 30 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a product or system provides the correct results with the needed degree of precision. | 8 | 80 | 2 | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which the functions facilitate the accomplishment of specified tasks and objectives. | 8 | 80 | 2 | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |

Appendix Table 14. Overall Frequency Distribution Table for Technical Evaluation based on Performance Efficiency

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F | % | F | % | F | % | F | % | F | % | F | % |
| Degree to which the response and processing times and throughput rates of a product or system, when performing its functions, meet requirements. | 4 | 40 | 5 | 50 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements. | 6 | 60 | 3 | 30 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which the maximum limits of a product or system parameter meet requirements. | 5 | 50 | 3 | 30 | 2 | 20 | 0 | 0 | 0 | 0 | 10 | 100 |

Appendix Table 15. Overall Frequency Distribution Table for Technical Evaluation based on Compatibility

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F | % | F | % | F | % | F | % | F | % | F | % |
| Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product. | 7 | 70 | 2 | 20 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which two or more systems, products or components can exchange information that has been exchanged. | 7 | 70 | 3 | 30 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 100 |

Appendix Table 16. Overall Frequency Distribution Table for Technical Evaluation based on Usability

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F | % | F | % | F | % | F | % | F | % | F | % |
| Degree to which users can recognize whether a product or system is appropriate for their needs. | 9 | 90 | 1 | 10 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a system can be used by specified users to achieve specified goals of learning to use the system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use. | 9 | 90 | 1 | 10 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a product or system has attributes that make it easy to operate and control. | 8 | 80 | 2 | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a system protects users against making errors. | 6 | 60 | 2 | 20 | 1 | 10 | 1 | 10 | 0 | 0 | 10 | 100 |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Degree to which a user interface enables pleasing and satisfying interaction for the user. | 8 | 80 | 2 | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a product or system can be used by people with the widest range of capabilities to achieve a specified goal in a specified context of use. | 6 | 60 | 4 | 40 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |

Appendix Table 17. Overall Frequency Distribution Table for Technical Evaluation based on Reliability

| **INDICATORS** | **5** | | **4** | | **3** | | **2** | | **1** | | **TOTAL** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** |
| Degree to which a system, product or component meets needs for reliability under normal operation. | 8 | 80 | 1 | 10 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a system, product or component is operational and accessible when required for use. | 6 | 60 | 4 | 40 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Degree to which a system, product or component operates as intended despite the presence of hardware or software faults. | 4 | 40 | 4 | 40 | 2 | 20 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system. | 4 | 40 | 6 | 60 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |

Appendix Table 18. Overall Frequency Distribution Table for Technical Evaluation based on Maintainability

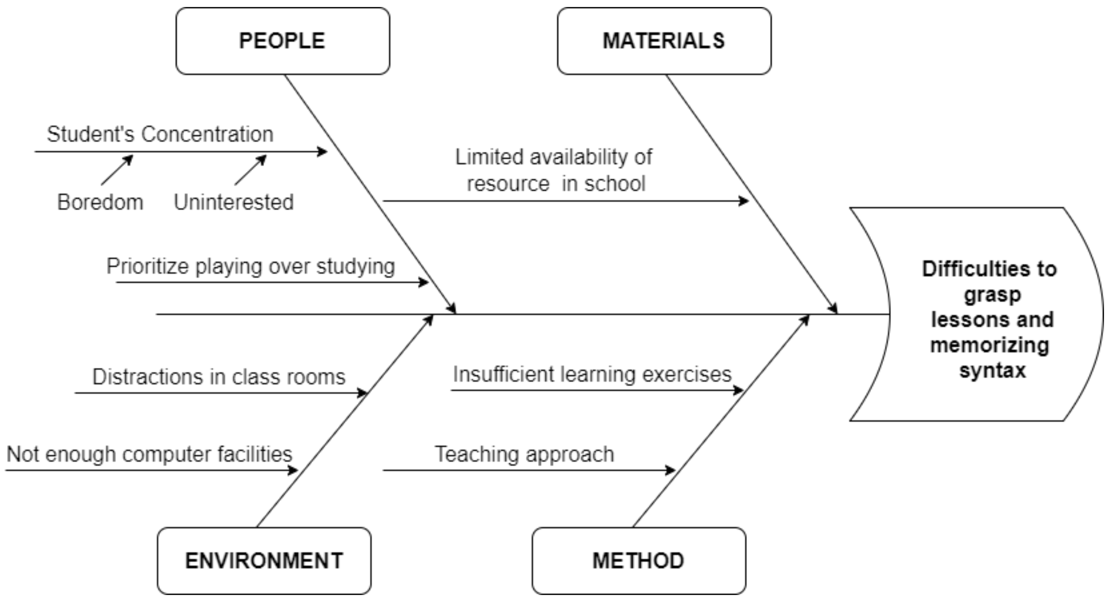
| **INDICATORS** | **5** | | **4** | | **3** | | **2** | | **1** | | **TOTAL** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** | **F** | **%** |
| Degree to which a system is composed of discrete components. | 6 | 60 | 3 | 30 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which an asset can be used in more than one system, or building other assets. | 6 | 60 | 4 | 40 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Degree of effectiveness with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts. | 5 | 50 | 5 | 50 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a system can be effectively and efficiently modified without degrading existing product quality. | 7 | 70 | 2 | 20 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree of effectiveness with which test criteria can be established for a system. | 7 | 70 | 3 | 30 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |

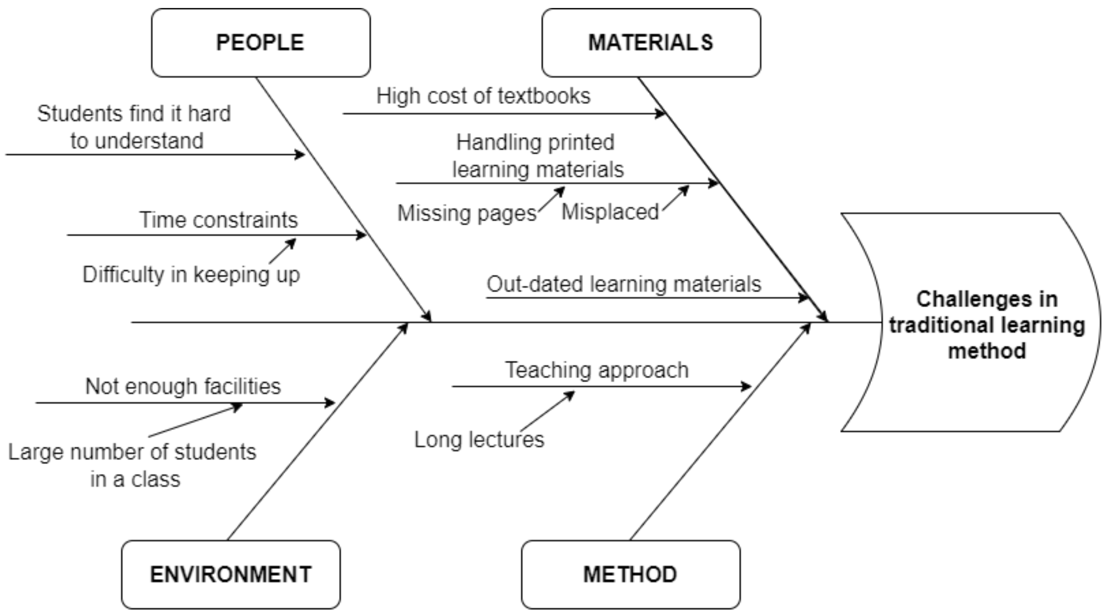
Appendix Table 19. Overall Frequency Distribution Table for Technical Evaluation based on Portability

| INDICATORS | 5 | | 4 | | 3 | | 2 | | 1 | | TOTAL | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F | % | F | % | F | % | F | % | F | % | F | % |
| Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments. | 6 | 60 | 4 | 40 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment. | 8 | 80 | 2 | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |
| Degree to which a product can replace another specified software product for the same purpose in the same environment. | 8 | 80 | 2 | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 100 |

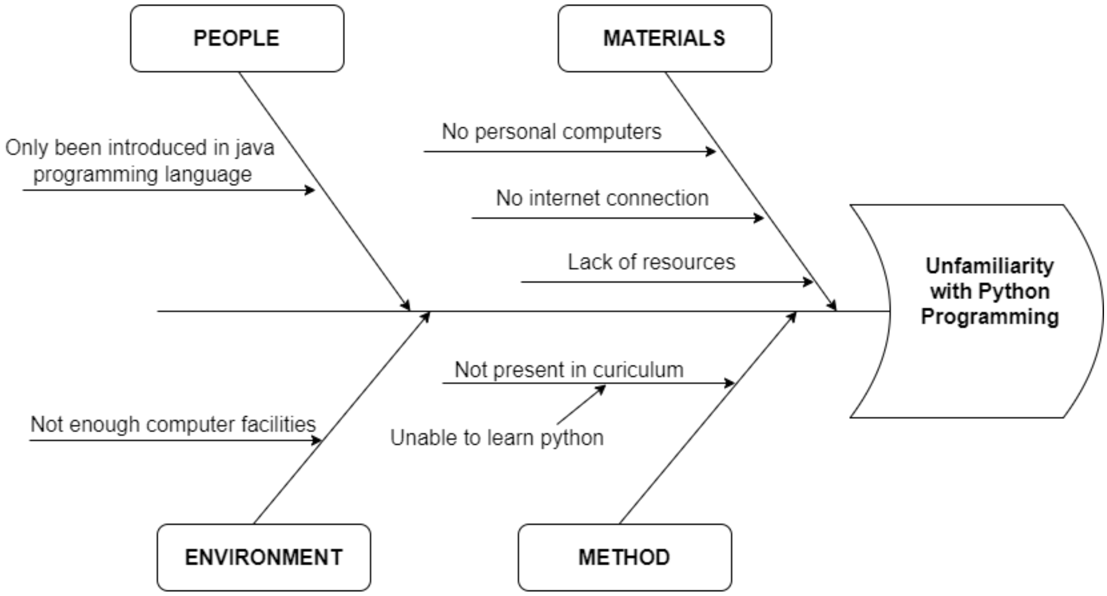
# APPENDIX FIGURE




Appendix Figure 1. Fishbone diagram of difficulties to grasp lessons and memorizing syntax

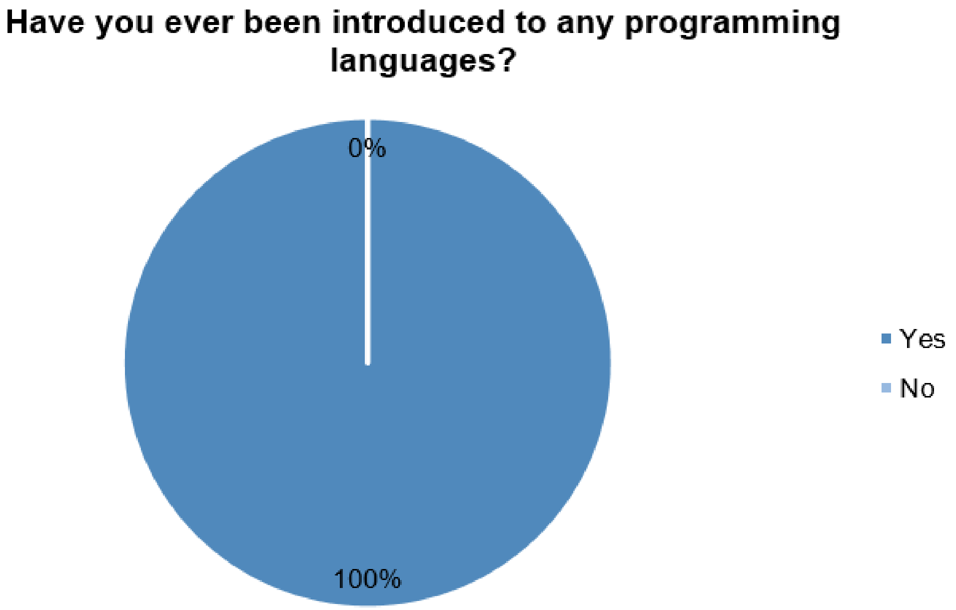



Appendix Figure 2. Fishbone diagram about challenges in traditional learning method

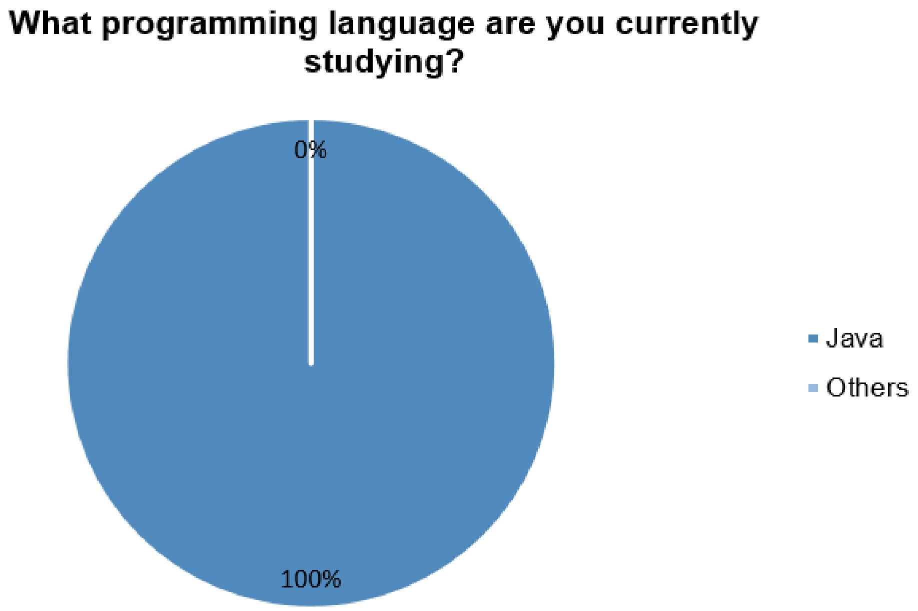



Appendix Figure 3. Fishbone diagram of unfamiliarity with Python programming

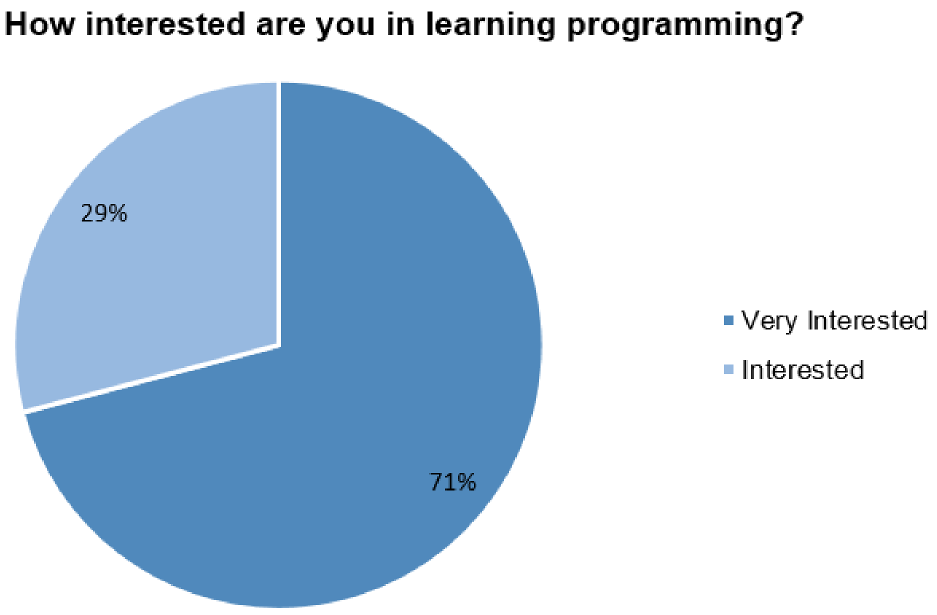

Appendix Figure 4. Timetable of the development of the study



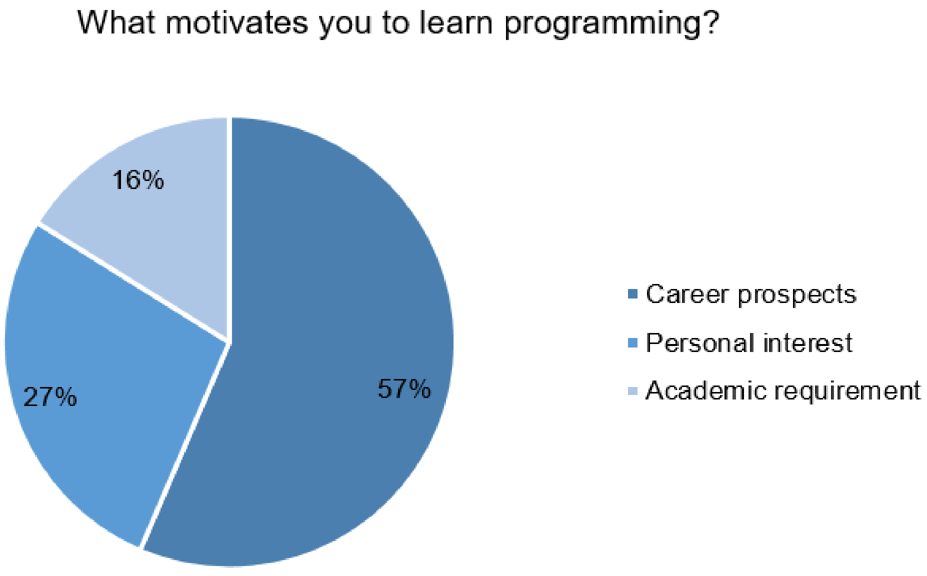

Appendix Figure 5. Survey result of respondents that has been introduce to any programming language



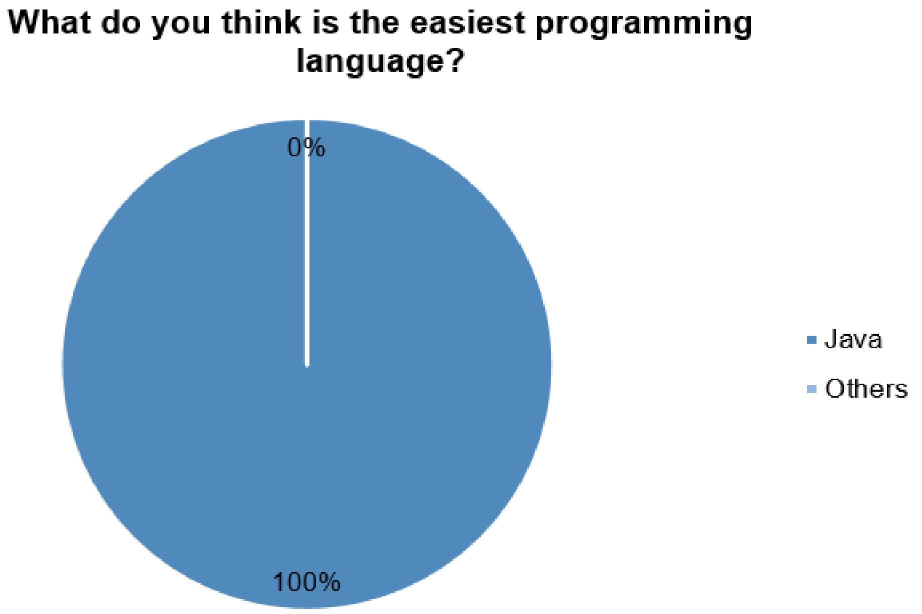

Appendix Figure 6. Survey result about the programming language that the respondents are presently learning



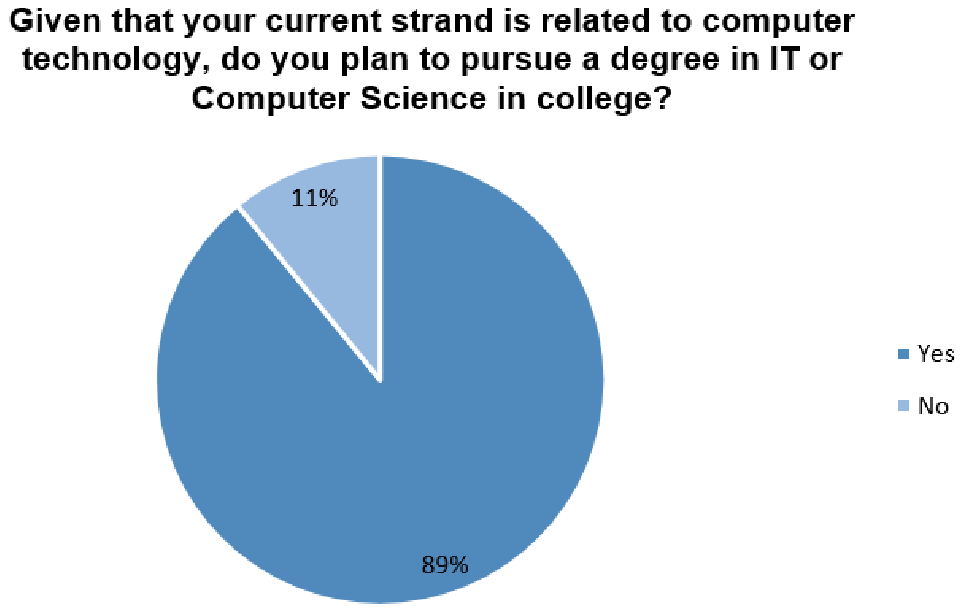

Appendix Figure 7. Survey result of respondents' interest in learning programming language



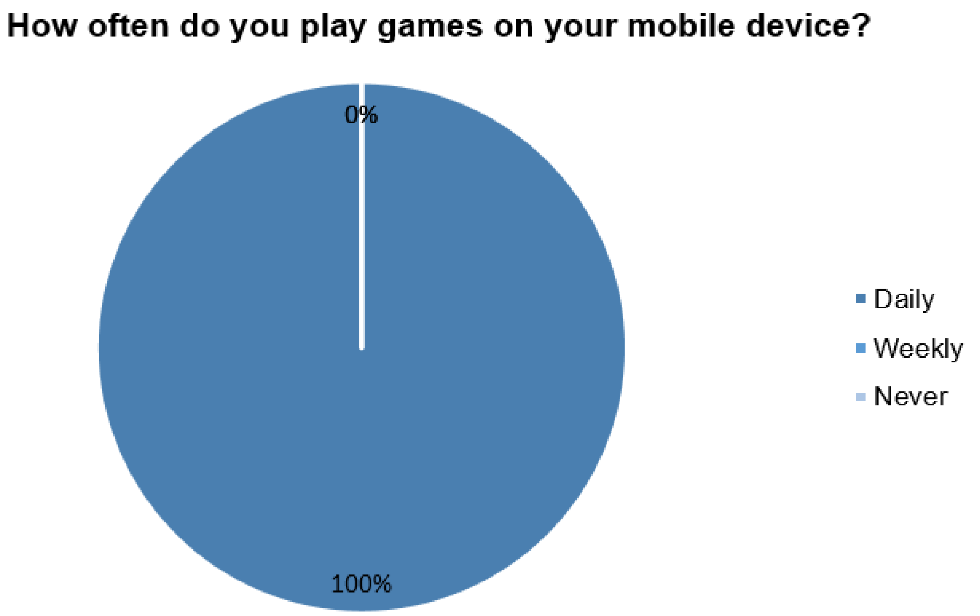

Appendix Figure 8. Survey result for respondents' motivation in learning programming



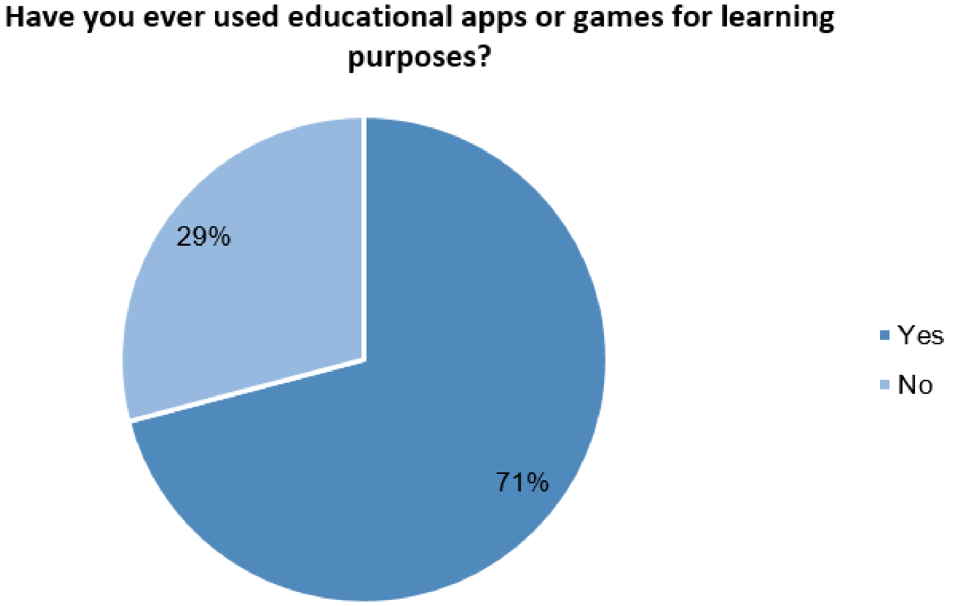

Appendix Figure 9. Survey result of what the respondents think which programming language is the easiest



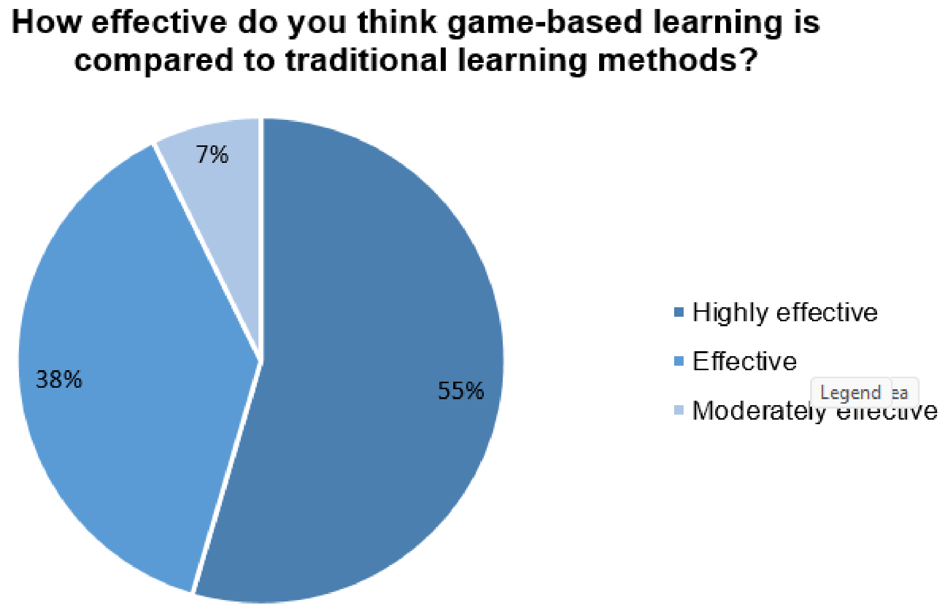

Appendix Figure 10. Survey result for respondents who considered to pursue IT or Computer Science in college



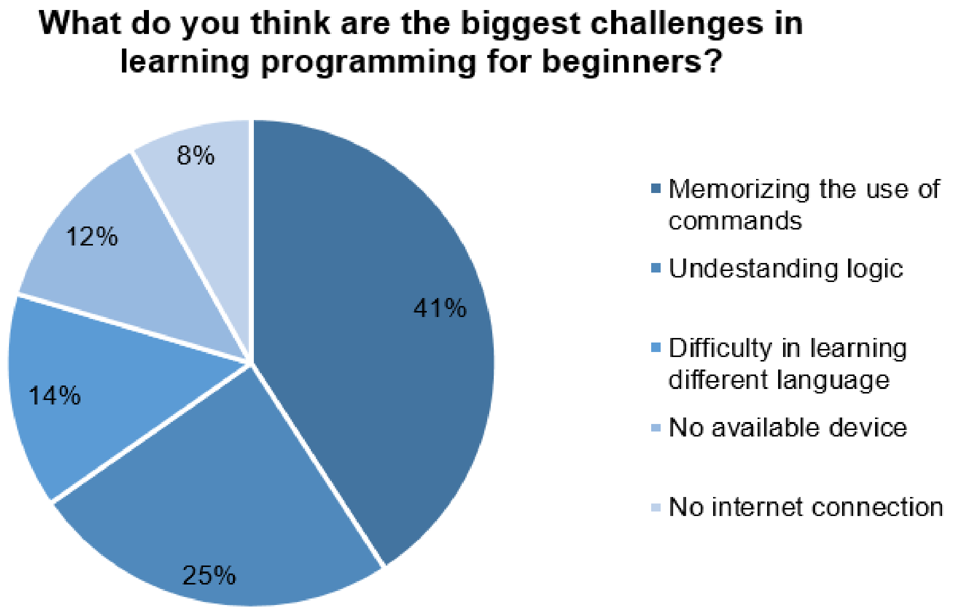

Appendix Figure 11. Survey result of how often the respondents play games on their mobile devices



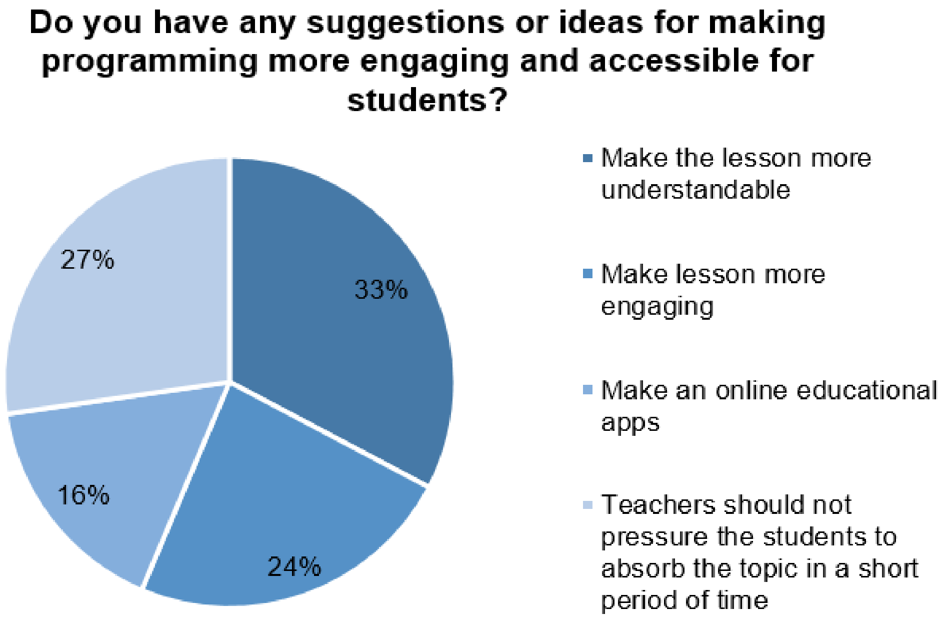

Appendix Figure 12. Survey result for number of respondents who used educational apps or games for learning purposes



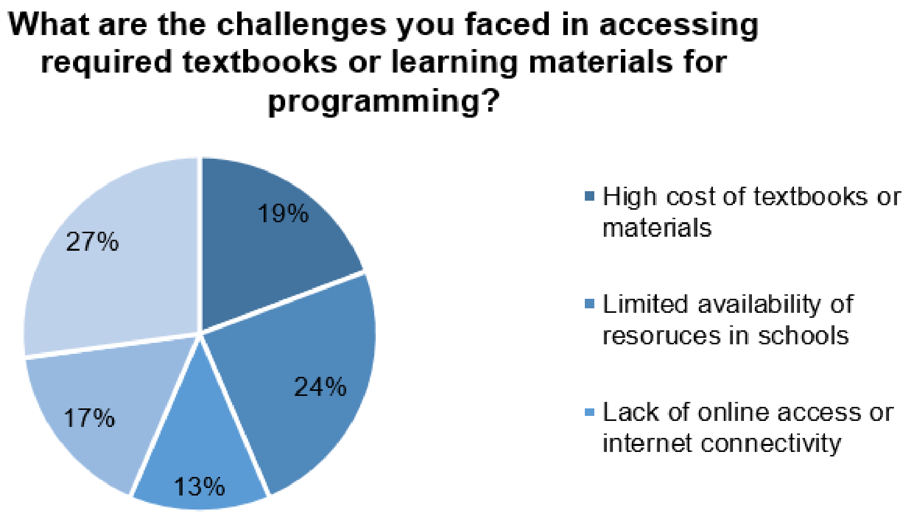

Appendix Figure 13. Survey result of respondents' opinion on how effective is game-based learning compared to traditional learning methods



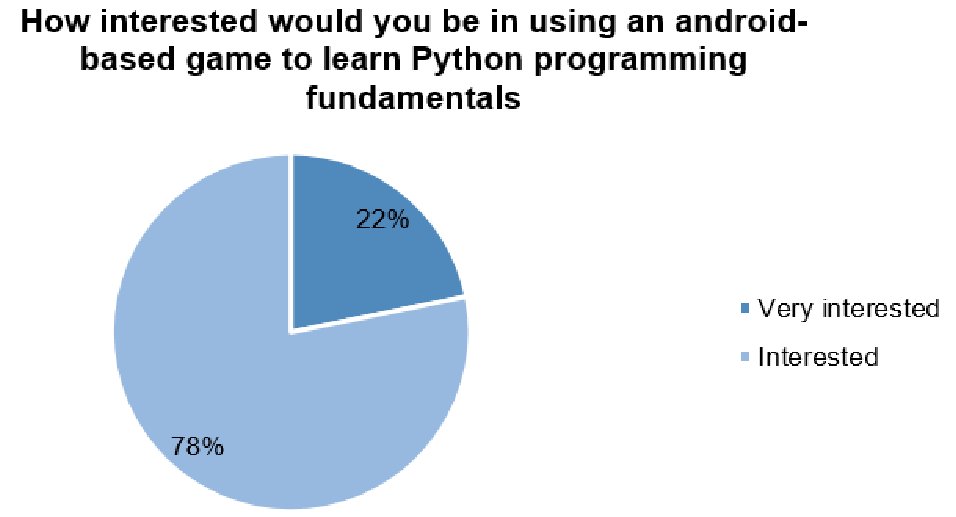

Appendix Figure 14. Survey result of what the respondents think is the biggest challenge in learning programming for beginners



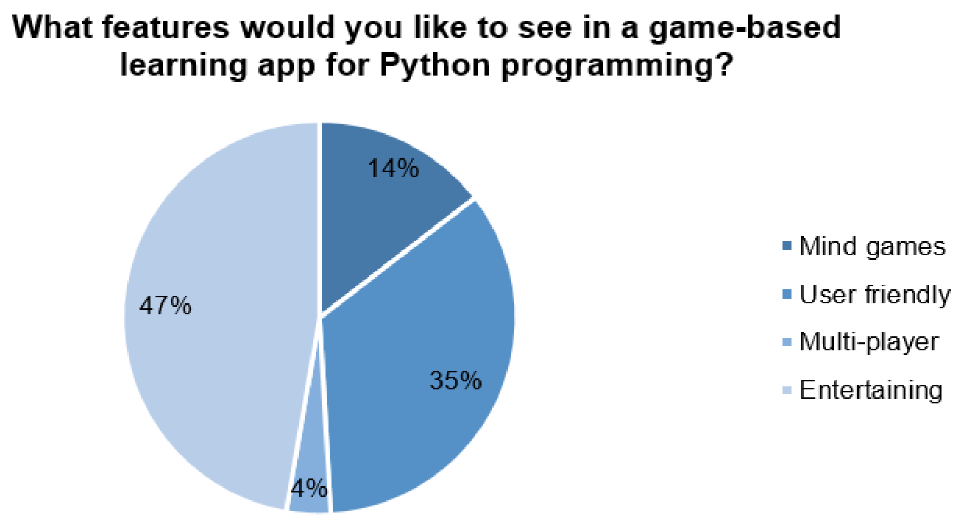

Appendix Figure 15. Survey result of respondents' suggestion on making programming more engaging and accessible for students




Appendix Figure 16. Survey result of respondents' challenges faced in accessing required textbooks or learning materials for programming




Appendix Figure 17. Survey result for respondents' interest in using an android-based game application to learn Python programming fundamentals




Appendix Figure 18. Survey result regarding the features that the respondents would want to see in a game-based learning application for Python programming

Appendix Figure 19. Pictures taken during data gathering to ICT professor

Appendix Figure 20. Pictures taken during data gathering to ICT students

Appendix Figure 21. Pictures taken during game demonstration

Appendix Figure 22. Pictures taken during  Non-Technical Evaluation

Appendix Figure 23. Pictures taken during the first day of Non-Technical Evaluation

Appendix Figure 24. Pictures taken during the first day of Non-Technical Evaluation of students

Appendix Figure 25. Pictures taken during Technical Evaluation

Appendix Figure 26. Pictures taken during Technical Evaluation of DIT professors

Appendix Figure 27. Pictures taken during the second day of Technical Evaluation


Appendix Figure 28. Pictures taken during the second day of Technical Evaluation of DIT professors


Appendix Figure 29. Pictures taken during the second day of Technical Evaluation of professors

Appendix Figure 30. Pictures taken during the presentation of the game to Tanza National Trade School ICT Professor

# OTHER APPENDICES

Appendix 1. Interview Report with the ICT professor of Tanza National Trade School

Republic of the Philippines
**CAVITE STATE UNIVERSITY**
**Don Severino delas Alas Campus**
Indang, Cavite

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**
**Department of Information Technology**

Title of Study: PY - QUEST: An Android-Based Game Application for Python Programming Fundamentals.

Researchers: Miriam Faith D. Medina
Mayvel T. Landicho
Michael D. Blanco

Interviewee: Eliza Furanil
Designation: ICT Teacher SHS - TNTS

**INTERVIEW QUESTIONAIRE**

1. How long have you been teaching programming languages?
**Answer:**
8 years

2. In your opinion, what are the main challenges students face when learning programming?
**Answer:**
equipment - computer facilities
class size

3. What is the most effective way for beginners to start learning programming?
**Answer:**
Start teaching resources
learn the basics
learn the introduction
have ~~[illegible]~~ a computer, laptop or even smart phone

4. Which aspects of programming tend to confuse students the most, and how can these be addressed?
**Answer:**
loopings →
Arrays →
List →
through repetition of practice
support with many examples
breakdown the code
to step by step.

5. In your opinion, what is the impact of using a programming language that may not align with a student's level of understanding?

**Answer:**

There should be student's interest with the course, without interest and enthusiasm worthless is the lesson and course

6. What is your view on the importance of having a balanced knowledge of different programming languages before entering college?

**Answer:**

This is an advantage for students before entering IT of CS or computer programming course

7. What techniques do you use to make programming concepts easier for students to understand?

**Answer:**

step by step teaching

explaining each step, syntax, code on how to use it

Providing more and more examples and practice activity

8. In your opinion, how can programming education be improved to be more effectively learned by students?

**Answer:**

* motivate the students
* make the topics interesting to the students
* Conditioning the students

9. Which programming languages do you usually recommend to beginners, and why?

**Answer:**

C++ → related to Java → same concept

HTML for beginners of web development

10. What is your perspective on using Python as an introductory programming language for beginners?

**Answer:**

This is a good start for programming course but introduction must explore also some programming language before python to know the differance and to compare it.

Appendix 2. Use Case Description

| | |
|---|---|
| Use Case Name | Start Game |
| Use Case ID | 01 |
| Scenario | User launches the application. |
| Trigger | User initiates the game. |
| Actor | User |
| Other Participant Actors | None |
| Related Use Cases | None |
| Interested Stakeholders | None |
| Preconditions | Application is installed and functional. |
| Post Conditions | Game interface is displayed. |
| Main scenario | 1. User launches the application.<br>2. Application displays a game interface. |
| Extension | None |

| | |
|---|---|
| Use Case Name | Select Level/Topic |
| Use Case ID | 02 |
| Scenario | User selects a given topic from the list |
| Trigger | User click on the topic |
| Actor | User |
| Other Participant Actors | None |
| Related Use Cases | None |
| Interested Stakeholders | None |
| Preconditions | Game interface is dissed |
| Post Conditions | The selected topic is highlighted. |
| Main scenario | . User selects a topic from the list. <br>2. Application displays corresponding questions |
| Extension | None |

| | |
|---|---|
| Use Case Name | Select Question |
| Use Case ID | 03 |
| Scenario | Users choose the question from the list displayed. |
| Trigger | User select a question |
| Actor | User |
| Other Participant Actors | None |
| Related Use Cases | None |
| Interested Stakeholders | None |
| Preconditions | Topic is selected and questions are displayed. |
| Post Conditions | Selected question i highlighted. |
| Main scenario | 1.User chooses a question from a displayed option. |
| Extension | None |

| | |
|---|---|
| Use Case Name | Submit Answer |
| Use Case ID | 04 |
| Scenario | User provides an answer to a selected question. |
| Trigger | User submit an answer. |
| Actor | User |
| Other Participant Actors | None |
| Related Use Cases | None |
| Interested Stakeholders | None |
| Preconditions | Question is displayed. |
| Post Conditions | User answer is recorded. |
| Main scenario | 1.User provides an answer to the selected question.<br>2. Application records the user's answer. |
| Extension | None |

| | |
|---|---|
| Use Case Name | View Result |
| Use Case ID | 05 |
| Scenario | User complete to answer the question and wants to view the result |
| Trigger | User finishes answering the questions. |
| Actor | User |
| Other Participant Actors | None |
| Related Use Cases | None |
| Interested Stakeholders | None |
| Preconditions | All questions are answered. |
| Post Conditions | Users answer result displayed. |
| Main scenario | 1.User completes answering the questions.<br>2. User wants to view the result.<br>3. Application verifies submitted answers.<br>4. Application displays the user's answers result. |
| Extension | None |

Appendix 3. Approved request letter for conducting a Technical Evaluation in Cavite State University Department of Information Technology

Republic of the Philippines
**CAVITE STATE UNIVERSITY**
**Don Severino delas Alas Campus**
Indang, Cavite

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**
**Department of Information Technology**

May 28, 2024

**DR. WILLIE C. BUCLATIN**
Dean
This College

Dear Dr. Buclatin:

Greetings!

In partial fulfillment of our requirements for our Capstone project, we, the undersigned BSIT students would like to request for your permission to allow us to conduct software evaluation with 10 IT or CS instructors of this College. Our capstone project, "Py – Quest: An Android-Based Game Application for Python Programming Fundamentals" aimed to provide a learning supplemental tool for students to learn Python fundamentals.

During the evaluation, we will demonstrate the mobile application to the respondents and request them to rate our app. Attached is a copy of the software evaluation instrument for your reference.

Rest assured that the data that will be collected will be used for this purpose only and will be treated with utmost confidentiality.

We are hoping for your positive response regarding our request. Thank you so much, and God bless!

Sincerely,

**MICHAEL BLANCO**
Student Researcher

**MARVEL LANDICHO**
Student Researcher

**MIRIAM FAITH MEDINA**
Student Researcher

Recommending Approval:

**ANABELLE J. ALMAREZ**
Research Adviser

**JAY-ARR C. BUHAIN**
Technical Critic

**CHARLOTTE B. CARANDANG**
Chairperson, DIT

Approved:

**WILLIE C. BUCLATIN, PhD, *ASEAN Eng***
Dean

Appendix 4. Technical Evaluation Form

Republic of the Philippines
**CAVITE STATE UNIVERSITY**
Don Severino delas Alas Campus
Indang, Cavite

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**

**Department of Information Technology**

**ISO 25010 EVALUATION INSTRUMENT FOR PY-QUEST: AN ANDROID BASED GAME APPLICATION FOR PYTHON PROGRAMMING FUNDAMENTALS**

**Name of Evaluator:** ______________________ **Position:** ______________

**Instruction:** Please evaluate the software material by using the given scale and placing a checkmark (✓) under the corresponding numerical rating:

| Numerical Rating | Equivalent |
|---|---|
| 5 | Excellent |
| 4 | Very Good |
| 3 | Good |
| 2 | Fair |
| 1 | Poor |

| INDICATORS | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| **A. Functional Suitability** | | | | | |
| 1. Functional completeness (*Degree to which the set of functions covers all the specified tasks and user objectives.*) | | | | | |
| 2. Functional correctness (*Degree to which a product or system provides the correct results with the needed degree of precision.*) | | | | | |
| 3. Functional appropriateness (*Degree to which the functions facilitate the accomplishment of specified tasks and objectives.*) | | | | | |
| **B. Performance Efficiency** | | | | | |
| 1. Time behavior (*Degree to which the response and processing times and throughput rates of a product or system, when performing its functions, meet requirements.*) | | | | | |
| 2. Resource utilization (*Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.*) | | | | | |
| 3. Capacity (*Degree to which the maximum limits of a product or system parameter meet requirements.*) | | | | | |
| **C. Compatibility** | | | | | |
| 1. Co-existence (*Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.*) | | | | | |
| 2. Interoperability (*Degree to which two or more systems, products or components can exchange* | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| information and use the information that has been exchanged.) | | | | | |
| **D. Usability** | | | | | |
| 1. Appropriateness recognizability (*Degree to which users can recognize whether a product or system is appropriate for their needs.*) | | | | | |
| 2. Learnability (*Degree to which a product or system can be used by specified users to achieve specified goals of learning to use the product or system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.*) | | | | | |
| 3. Operability (*Degree to which a product or system has attributes that make it easy to operate and control.*) | | | | | |
| 4. User error protection (*Degree to which a system protects users against making errors.*) | | | | | |
| 5. User interface aesthetics (*Degree to which a user interface enables pleasing and satisfying interaction for the user.*) | | | | | |
| 6. Accessibility (*Degree to which a product or system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.*) | | | | | |
| **E. Reliability** | | | | | |
| 1. Maturity (*Degree to which a system, product or component meets needs for reliability under normal operation.*) | | | | | |
| 2. Availability (*Degree to which a system, product or component is operational and accessible when required for use.*) | | | | | |
| 3. Fault tolerance (*Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.*) | | | | | |
| 4. Recoverability (*Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.*) | | | | | |
| **F. Security** | | | | | |
| 1. Confidentiality (*Degree to which a product or system ensures that data are accessible only to those authorized to have access.*) | | | | | |
| 2. Integrity (*Degree to which a system, product or component prevents unauthorized access to, or modification of, computer programs or data.*) | | | | | |
| 3. Non-repudiation (*Degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.*) | | | | | |
| 4. Accountability (*Degree to which the actions of an entity can be traced uniquely to the entity.*) | | | | | |
| 5. Authenticity (*Degree to which the identity of a subject or resource can be proved to be the one claimed.*) | | | | | |
| **G. Maintainability** | | | | | |
| 1. Modularity (*Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.*) | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| 2. Reusability (*Degree to which an asset can be used in more than one system, or in building other assets.*) | | | | | |
| 3. Analyzability (*Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.*) | | | | | |
| 4. Modifiability (*Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.*) | | | | | |
| 5. Testability (*Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.*) | | | | | |
| **H. Portability** | | | | | |
| 1. Adaptability (*Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.*) | | | | | |
| 2. Installability (*Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.*) | | | | | |
| 3. Replaceability (*Degree to which a product can replace another specified software product for the same purpose in the same environment.*) | | | | | |

**Comments/Suggestions:**

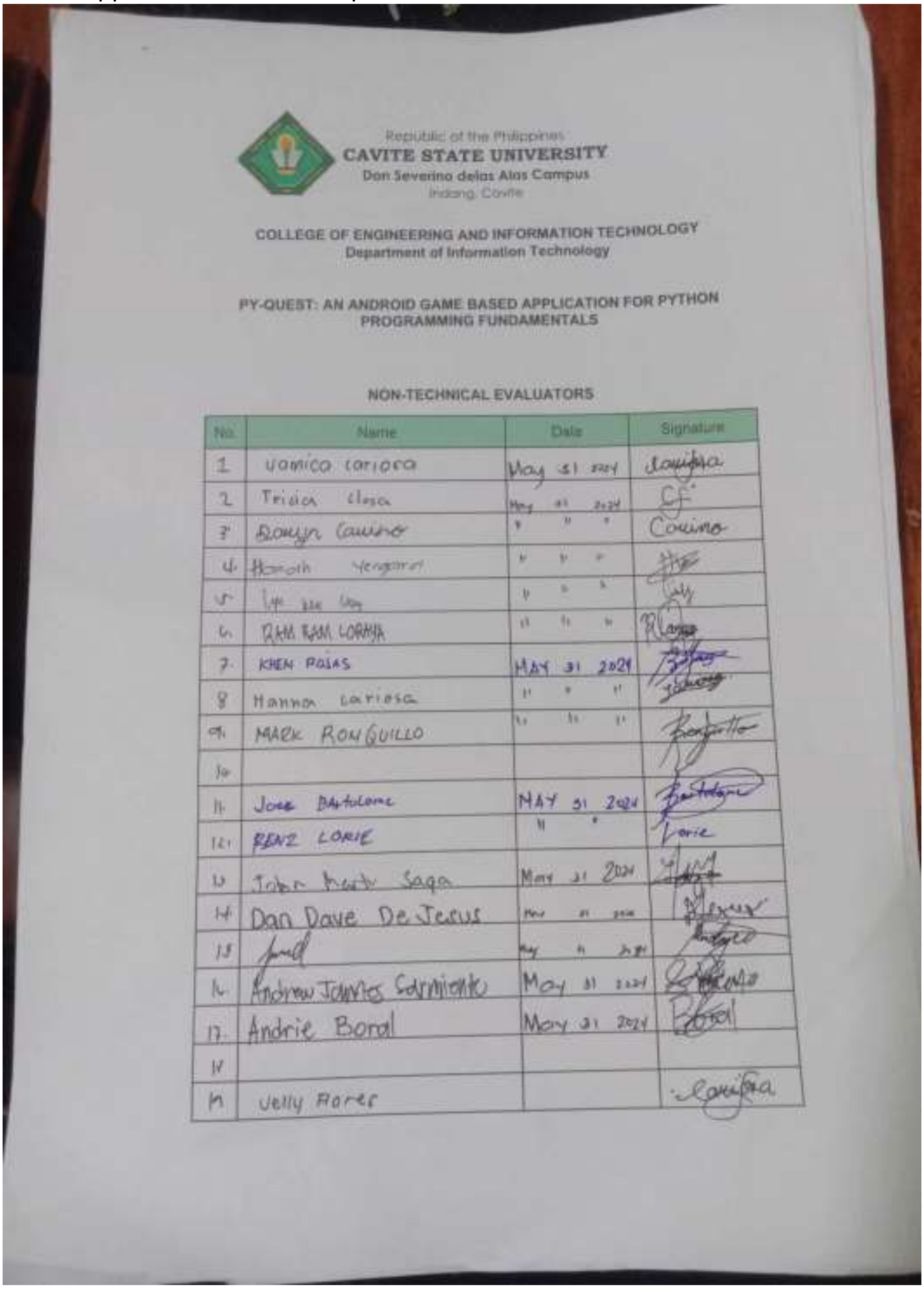
______________________________________________

**NOTE:** *Adopted from ISO/IEC 25010*

______________________

Signature

Appendix 5. Profile and proof of non-technical evaluation



Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino delas Alas Campus
Indang, Cavite

COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY
Department of Information Technology

PY-QUEST: AN ANDROID GAME BASED APPLICATION FOR PYTHON PROGRAMMING FUNDAMENTALS

NON-TECHNICAL EVALUATORS

| No. | Name | Date | Signature |
|---|---|---|---|
| 1 | Yamico Lariosa | May 31 2024 | [illegible] |
| 2 | Tricia Closa | May 31 2024 | [illegible] |
| 3 | Ronyn Cauino | " " " | Cauino |
| 4 | Honorh Yengaran | " " " | [illegible] |
| 5 | [illegible] | " " " | [illegible] |
| 6 | RAM RAM LORAYA | " " " | [illegible] |
| 7. | KHEN POLAS | MAY 31 2024 | [illegible] |
| 8 | Hanna Lariosa | " " " | [illegible] |
| 9. | MARK RONQUILLO | " " " | [illegible] |
| 10 | | | |
| 11. | Jose Bartolome | MAY 31 2024 | [illegible] |
| 12. | RENZ LORIE | " " | Lorie |
| 13 | John Mark Saga | May 31 2024 | [illegible] |
| 14. | Dan Dave De Jesus | May 31 2024 | [illegible] |
| 15 | [illegible] | May 31 2024 | [illegible] |
| 16. | Andrew James Sarmiento | May 31 2024 | [illegible] |
| 17. | Andrie Boral | May 31 2024 | [illegible] |
| 18 | | | |
| 19 | Velly Flores | | [illegible] |

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| | AUÑGON ABEGAIL | | |
| | LEILA AKIM | | |
| | GINA DEVOE REYES | | |
| | UEE1 DOONG | | |
| | [illegible] | | |
| | [illegible] AMALIA | | |
| | JERRY M. ALVAREZ | | |
| | Jheramil G. Onde | | |
| | LOVELY ANGEL LUMANEY | | |
| | Dustine Cascado | | |
| | Jaymar Cabalin | | |
| | KAREN DEVA PENA | | |
| | Kesha Baldon | | |
| | John Lalima | | |
| | Joanna Palmo | | |
| | Jhan Cabanles | | |
| | YESHA FORTUNO | | |
| | [illegible] Rejano | | |
| | Marian Salinas | | |
| | Aprmyr Dino | | |
| | Dre Sicoro | | |
| | Noah Bermudes | | |
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| | GALANZA, IVY A. | | |
| | MARK LORENCE PIANGCO | | |
| | Mark Milan Ballesteros | | |
| | John Dexter D. Opriasa | | |
| | ALEJO J. LAGCO | | |
| | NIÑO NASOL | | |
| | Gian Claire Rosales | | |
| | JAZZY MAE MALANA | | |
| | JENNET CAGARULLA | | |
| | Veronica Jade Quintana | | |
| | Mary Nicole Gutlen | | |
| | Rizalinda Luciniaio | | |
| | Ashwell Merinilla | | |
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| | Joyce Javier | | |
| | Don Dose | | |
| | John Lloyd | | |
| | Rizjo Vargas | | |
| | Melody Landidh | | |
| | CHRG LABCUPIN | | |
| | James Tabangin | | |
| | Pio Vargas | | |
| | Nimfa Dawitoria | | |
| | Shane labadan | | |
| | Joshea Dupu | | |
| | JONATHAN DE LA PEÑA | | |
| | LEA AGUILAR | | |
| | Salvacion Galier | | |
| | Rosemary Nolo | | |
| | Rerto AMANIT | | |
| | Raynor Surban | | |
| | Janiel Tocayo | | |
| | Ryan Kemugat | | |
| | Francis Llorente | | |
| | Carlito Begonia | | |
| | Aurelio Y. Avingo | | |
| | Lovely Jay Asi | | |
| | Mendoza Jay | | |
| | ROMEO TOCAYO | | |
| | Gwen | | |
| | Ma. Cleofas Perena | | |

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| | Casandra Olive | | [illegible] |
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| | GUEL DELA PEÑA | | [illegible] |
| | Jazel Samberi | | Jazel |
| | JEFF ROMERO | | Romero |
| | [illegible] | | [illegible] |
| | JOANNA MARIE | | Joanna M. |
| | Maria Ethalie Nair Tusil | | [illegible] |
| | Somontha Besa | | [illegible] |
| | Kieyan Villatema | | Villatema |
| | Sheena Cutamora | | [illegible] |
| | Belmor Garcia | | [illegible] |
| | Renrch Santos | | [illegible] |
| | Roble Campos | | [illegible] |
| | Gean Españole | | [illegible] |
| | Tina Aires | | [illegible] |
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Republic of the Philippines
**CAVITE STATE UNIVERSITY**
Don Severino delas Alas Campus
Indang, Cavite

COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY
Department of Information Technology

PY-QUEST: AN ANDROID GAME BASED APPLICATION FOR PYTHON PROGRAMMING FUNDAMENTALS

NON-TECHNICAL EVALUATORS

| No. | Name | Date | Signature |
|---|---|---|---|
| | Lance Joatine Bedico | 8/9/24 | |
| | GRAVADOR, SHELLA MARIE C. | 8/9/24 | |
| | Ezariella Claire S. Cabahat | 8/9/24 | |
| | GENE PAUL D. ERNI | 8/9/24 | |
| | Jeremy D. Macapagong | 8/9/24 | |
| | Keith Justin Cuya | 8-9-24 | |
| | John Kent G. Mejorito | 8-9-24 | |
| | Jv Batiancila | 8-9-24 | |
| | ~~[illegible]~~ KHARL EBOL | 8-9-24 | |
| | A. B Belen | 8-9-24 | |
| | Ruel Bojocan | 8-9-24 | |
| | Zhanjoe Christian R. Diño | 8/9/24 | |
| | Añonuevo, Xander B | 8/9/24 | |
| | Omanio, Clarisse | 8/9/24 | |
| | Juliata, Heart | 8/9/24 | |
| | Cabalhug, Angel D. | 8 9 24 | |
| | Rejano, Nanley Maureen B. | Aug 9, 2024 | |
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|---|---|---|---|
| | Fernandez, Melly Jane | Aug 9, 2024 | Melly Jane |
| | Duhan, Mary Grace | Aug 9, 2024 | MGDuhan |
| | Nico Lu Sam Cuta | Aug 9, 2024 | |
| | Cao, Maria Sophia Miles V. | August 9 2024 | |
| | Cabahug, Angel D. | August 9 2024 | |
| | Rejano, Nashley Maureen B | August 9, 2024 | |
| | Macasado Bernie E. | August 9 2024 | |
| | Galarno, John Paul B | August 9 2024 | |
| | Ebin Ezekiel G. | ~~Aug 14~~ 2024 | E.Kie. |
| | Olamosa, Jhunric D. | Aug. 9, 2024 | |
| | Ellaicca Reize S. Santander | Aug 9. 2024 | SANTANDER |
| | Joshua Gerona | Aug 9. 2024 | |
| | Frinz Zyrille Ganal | Aug 9, 2024 | |
| | EARL JUSTINE COSTAS | AUG 9, 2024 | |
| | Justine C. Castillas | Aug 9, 2024 | |
| | Mat Andric T. Caranog | Aug 9, 2024 | |
| | Juroel B. Bagac | Aug 9, 2024 | |
| | Gabriel Mari Bonozo | Aug 9, 2024 | |
| | Jaena, Prince Andrei C. | August 9, 2024 | |
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Appendix 6. Approved request letter for Lesson and Assessment evaluation

Republic of the Philippines
**CAVITE STATE UNIVERSITY**
**Don Severino delas Alas Campus**
Indang, Cavite

**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY**
**Department of Information Technology**

April 4, 2024

**PROF. CHARLOTTE B. CARANDANG**
Chairperson
This Department

Dear Prof. Carandang:

Greetings!

The undersigned are 4th year BSIT students of this university who are currently conducting our research study, respectfully requesting your assistance in validating the correctness of the lessons and assessment about Python. These lessons and assessment, herewith attached, are to be used as content for our research study that aims to develop a mobile application about Python.

As students, we value high-quality and reliable educational resources. We believe it is important to ensure the accuracy of materials provided. Your expertise would greatly help us. Additionally, your validation would improve the credibility and effectiveness of the educational resources offered through our mobile application.

We hope your positive response in this humble matter. Thank you and God bless!

Sincerely,

| | | |
|---|---|---|
| **Miriam Faith Medina**<br>Student Researcher | **Maynel Landicho**<br>Student Researcher | **Michael D. Blanco**<br>Student Researcher |

Recommending Approval:

| | |
|---|---|
| **ANABELLE J. ALMAREZ**<br>Research Adviser | **JAY – ARR C. BUHAIN**<br>Technical Critic |

Approved:

**CHARLOTTE B. CARANDANG**
Chairperson

Appendix 7. Certificate of similarity index using Turnitin

Republic of the Philippines
**CAVITE STATE UNIVERSITY**
**Don Severino delas Alas Campus**
Indang, Cavite

**CERTIFICATE OF SIMILARITY INDEX USING TURNITIN**

This is to certify that the manuscript entitled

**"PY-QUEST: AN ANDROID BASED GAME APPLICATION FOR PYTHIN PROGRAMMING FUNDAMENTALS"**

authored by:

**Michael D. Blanco**
**Mayvel T. Landicho**
**Mirriam Faith D. Medina**

has been subjected to similarity check on July 31, 2024 of 19%.

Processed by:

**GLADYS G. PEREY**
BSIT Unit Research Coordinator

Certified correct by:

**EDWIN R. ARBOLEDA, D. Eng**
College Research Coordinator